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About the UltraSerial10G : Features

Features

The UltraSerial10G hardware has the following features:

- 20 differential drive and receive ports per board
- 40 differential drive with site copy (20 differential drive and 20 differential drive site copy)

- Frequency range of 42Mbps to 10.7Gbps
- 512M (426M available) vectors drive, expect vector memory (VM)
- 1G cycles capture memory (CMEM)
- Receive Align Trigger (used for Clock Data Recovery, 10b Align, and 20-bit Match)
- Pseudo-random binary sequence (PRBS) generator supporting polynomials of 7, 15, 23, and 31
- Jitter insertion
- Jitter measurement
- Eye measurement
- Pin Parametric Measurement Units (PPMU) for DC parametric testing
- Digital loopback that takes differential output from a DUT and drives it back into the DUT receivers
- Two time domains
- Protocol Aware for the PCI Express (PCIe) protocol
- Protocol Aware for sWire protocols, specifically SATA

For UltraSerial10G hardware specifications, see [UltraSerial10G Specifications](#).



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About the UltraSerial10G : Safety Information

Safety Information

When operating an UltraSerial10G instrument, always follow general UltraFLEX test system safety guidelines. See [Test System Safety](#).

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About the UltraSerial10G : UltraSerial10G Software Licensing Requirements

UltraSerial10G Software Licensing Requirements

IG-XL enforces the following node-locked, counted licenses on each UltraSerial10G half board, where two licenses allow you to use all ports on one entire board.

UltraSerial10G Licenses			
License Type	License Name	Enabling Points	Comments
Channels	HSS10G_Channels	1 (enabled)	Counted license, where one license enables any 10 drive and receive ports on any UltraSerial10G board in the test system. You need two licenses to enable all 20 ports on a single UltraSerial10G board.
Data Rate	HSS10G_Speed	In Mbps: 3200, 6400, 8000, and 10000	Counted license, where one license enables the data rate defined by the enabling point for any 10 drive and receive ports on any UltraSerial10G board in the test system. You need two licenses to enable all 20 ports on a single UltraSerial10G board. For

			channels with different speeds on a board, IG-XL uses the license that covers channels with the highest data rate.
Digital Loopback	HSS10G_Loopback	1 (enabled)	Counted license, where one license enables any 10 digital loopback engines on any UltraSerial10G board in the test system. You need two licenses to enable all 20 ports on a single UltraSerial10G board.
PRBS	HSS10G_PRBS	1 (enabled)	Counted license, where one license enables any 10 PRBS engines on any UltraSerial10G board in the test system. You need two licenses to enable all 20 ports on a single UltraSerial10G board.
Jitter Measure	HSS10G_JitterMeasure	1 (enabled)	Counted license, where one license enables jitter measure on any 10 ports on any UltraSerial10G board in the test system. You need two licenses to enable all 20 ports on a single UltraSerial10G board.

Jitter Insertion	HSS10G_JitterInsert	1 (enabled)	Counted license, where one license enables jitter insertion on any 10 ports on any UltraSerial10G board in the test system. You need two licenses to enable all 20 ports on a single UltraSerial10G board.
Protocol Aware	PCleRootComplexEngine	1 (enabled)	Counted license. Each PCIe root complex engine requires one license, up to 20 engines per board.
	PCleEndPointEngine	1 (enabled)	Counted license. Each PCIe end point engine requires one license, up to 20 engines per board.
	sWireEngine	1 (enabled)	Counted license. Each sWire engine requires one license, up to 20 engines per board.

Note that UltraSerial10G license lines start with HSS10G. This license will enable the UltraSerial10G up to 10Gbps.

Online, IG-XL obtains these licenses from the counted license file. Offline, the simulated configuration file defines these counted licenses under the <LICENSE_ENABLERS> section.

In an offline simulated or current configuration file, these licenses appear as follows:

<LICENSE_ENABLERS>

#Off-line enabling levels

#FEATURE	ENABLED	LEVEL	LICENSE COUNT	LEGAL ENABLING POINTS
HSS10G_Channels	1	2	1,0	
HSS10G_Speed	3200	2	10000,8000,6400,3200	
HSS10G_Loopback	1	2	1,0	
HSS10G_PRBS	1	2	1,0	
HSS10G_JitterMeasure	1	2	1,0	
HSS10G_JitterInsert	1	2	1,0	
PCleRootComplexEngine	1	20	1,0	
PCleEndPointEngine	1	20	1,0	

```
sWireEngine      1      20      1,0
</LICENSE_ENABLERS>
```

Handling Offline License Validation Errors

When validating the vector rate license offline using a simulated configuration file with a license enabler section, IG-XL checks for both the correct number of licenses and the legal enabled vector rate. If the test program fails to meet the licensed enable point, IG-XL returns an error. To correct this error, revalidate the test program. You do not need to alter the license enabler section of the simulated configuration file.

Related Topics

- [IG-XL Licensing](#)

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About the UltraSerial10G : Key Terms

Key Terms

10b Align	Stage of the Receive Align Trigger that aligns to the 10 bit boundary of the DUT.
20b Match	Stage of the Receive Align Trigger that aligns to the beginning of the DUT data.
bit error ratio (BER)	The ratio of the number of bits, elements, characters, or blocks incorrectly received to the total number of bits, elements, characters, or blocks sent during a specified time interval. Also called the bit error rate.
bit period	The time width of a databit. Also called unit interval or DUT output frequency.
Clock Data Recovery (CDR)	Stage of the Receive Align Trigger that recovers clock signals that are encoded into pattern data and locates the center of the data eye.
Capture Memory (CMEM)	256M x 1bit memory used to store actual device data.
data dependent jitter (DDj)	Jitter that is correlated to the pattern data such that a change in the pattern data will cause the jitter to change.
differential pair	A group of two single-ended pin components. One single-ended pin represents the true side and the other represents the complement side.
differential pin	A single differential signal (true or complement).
DUT RX	DUT Receiver

DUT TX	DUT Transmitter
History RAM (HRAM)	40K deep memory per UltraSerial10G receive differential pair that captures device data and the state of the pattern generator when a pattern runs.
Instrument Sync-Link (ISL)	A high-speed (2.5 Gbps) serial bus that carries PatGen data between instruments and provides for board-to-board communications through the backplane. PatGen data has a timestamp embedded in the pattern that tells the instrument when to execute a given command or pattern burst to the DUT.
jitter	Variation of edges in time from their ideal position.
jitter insertion	Use of UltraSerial10G hardware to insert a sine wave onto the UltraSerial10G drive pin. The period and amplitude (in UI) of the jitter are user programmable.
LFSR	Linear Feedback Shift Register. A shift register where an input bit is driven by a linear function of the overall shift register value.
Master Oscillator (MOSC)	A 5.35GHz to 10.7GHz clock used to determine the DUT period. DUT period = MOSC period * MOSC Per Vector (MPV).
MOSC Per Vector (MPV)	An integer used to scale the MOSC value.
nondeterministic data (NDD)	A device output that is nondeterministic. NDD situations include uncertain timing and start of data, running disparity, insertion of idles, and out of order data packets.
Pin Parametric Measurement Unit (PPMU)	A DC measuring instrument which can: force a current and measure a voltage, or force a voltage and measure a current (80 per UltraSerial10G board).
Protocol Aware (PA)	A new tester architecture that enables instruments to communicate with standard device protocols without using low-level vector programming. PA allows you to execute direct instructions and emulate real time device interaction at the protocol level.
pseudo random binary sequence (PRBS)	Hardware that generates a sequence of 1's and 0's that exhibit certain randomness and autocorrelation properties. The hardware supports the following sequences: $2^7 - 1$, $2^{15} - 1$, $2^{23} - 1$, and $2^{31} - 1$.
random jitter (Rj)	Jitter that results in a Gaussian distribution of errors in edge position. It is typically expressed as RMS. As more and more samples are taken, the

	probability increases that the upper and lower limit of the variation will increase. As a result, Rj is often characterized as unbounded.
Receive Align Trigger (RAT)	Hardware that performs a series of pattern matches to align to DUT functional data. It consists of three stages: CDR, 10b Align, and 20b Match.
total jitter (Tj)	A jitter measurement that describes a time span around the ideal edge time, beyond which the probability of an edge’s occurrence is less than a specified BER.
UltraSerial10G Driver	Hardware that drives differential data to the DUT Receiver (RX).
UltraSerial10G Receiver	Hardware that receives differential data from the DUT Transmitter (TX).
unit interval (UI)	The time width of a single data bit. (Also called bit interval)
Vector Memory (VM)	A 256M x 2 bit memory used to store programmed drive and receive vector data (0, 1, H, L, X).

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UltraSerial10G Debug Displays

UltraSerial10G Debug Displays

The UltraSerial10G Serial Channel Debug Display is an interactive, window-based tool you can use to view the state of the UltraSerial10G instrument and make changes to the instrument for debugging.

This section describes features specific to the UltraSerial10G debug display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

This section describes the following topics:

- Starting the Serial Channel Debug Display
- Serial Channel Debug Display Interface
- PPMU Debug Display

For more information, see:

- Jitter Tool
- Serial Capture Tool

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UltraSerial10G Debug Displays : Starting the Serial Channel Debug Display

Starting the Serial Channel Debug Display

To launch the Serial Channel Debug Display, first validate an UltraSerial10G test program. Then, on the IG-XL Tools tab, in the Launch group, select Debug Displays>Serial Channel. This opens the display for active sites only.

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For more information on starting debug displays, see Starting Debug Displays in the Test Debug Environment (TDE) section.

For information on fields used in the Serial Channel debug display, see Serial Channel Debug Display Interface.

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UltraSerial10G Debug Displays : Serial Channel Debug Display Interface

Serial Channel Debug Display Interface

The display shows one UltraSerial10G differential drive or receive pair at a time. The Pin Name menu allows you to select which differential pair to view. The display reflects your choice of UltraSerial10G drive or receive pins.

Differential Drive Channels

The following figure shows the Serial Channel display with a drive channel selected.

Serial Channel Debug Display, Drive Channel Selected

Drive Timing parameters are as follows:

Init State	The initial state of the driver. Valid values are: Lo, Hi.
Start State	The start state of the driver. Valid values are: Lo, Hi, None.
Start Delay	Delay offset for the drive pin. Range is: 0 to 16383 x MOSC Period.
MOSC Period	Period of the MOSC, which is determined from the programmed MPV and DUT Period. This is a read-only value and cannot be modified in this display. This is a read-only value and cannot be modified in this display.
MPV	Number of MOSC per vector. Valid values are: 1, 2, 4, 8, 16, 32, 64, and 128.
DUT Period	The period of the DUT. Valid range is: 1 x MOSC Period to 128 x MOSC Period. This range is calculated from the minimum MPV value of 1 to the

	maximum MPV value of 128.
MPV0/MPV1	MOSC per vector for the currently selected differential channel pair. Valid values are: 1, 2, 4, 8, 16, 32, 64, and 128. These options are only available in Protocol Aware (PA) channel mode. When running PCIe PA, IG-XL uses these selectors to switch on-the-fly between two speeds, 2.5Gbps and 5.0Gbps, while connecting to a PCIe device. The UltraSerial10G always starts on 2.5Gbps, and IG-XL can increase the speed to 5.0Gbps.

Drive Pin Electronics parameters are as follows:

Pin Name	The name of the UltraSerial10G drive pin.
Vid	The differential input voltage. Valid spec range is: +100mV to +1.8V.
Vicm	The common mode voltage, which is programmed when the DC Coupling Relay is closed. Valid range is: 0V to +1.8V.
PE/PPMU Enabled Selectors	Selects which resource to apply to the DUT. Click a radio button to choose the enabled connection: • PE Enabled: Channel connected to the Pin Electronics • PPMU Enabled: Channel connected to the PPMU • Not Enabled: Channel not connected to either the Pin Electronics or the PPMU Note that you cannot connect the pin electronics and PPMU at the same time.
PPMU	See PPMU Debug Display.
Levels Definition	This button brings up a window which shows a graphical representation of the various drive levels. See Drive Levels Definition.
Channel Mode	This read-only field shows the channel mode. Values are: Serial, Serial Drive-only, Protocol Aware, and Loopback.
Jitter Insert	This read-only field shows whether or not the pin is enabled for jitter insertion. Values are: Enabled or Disabled.

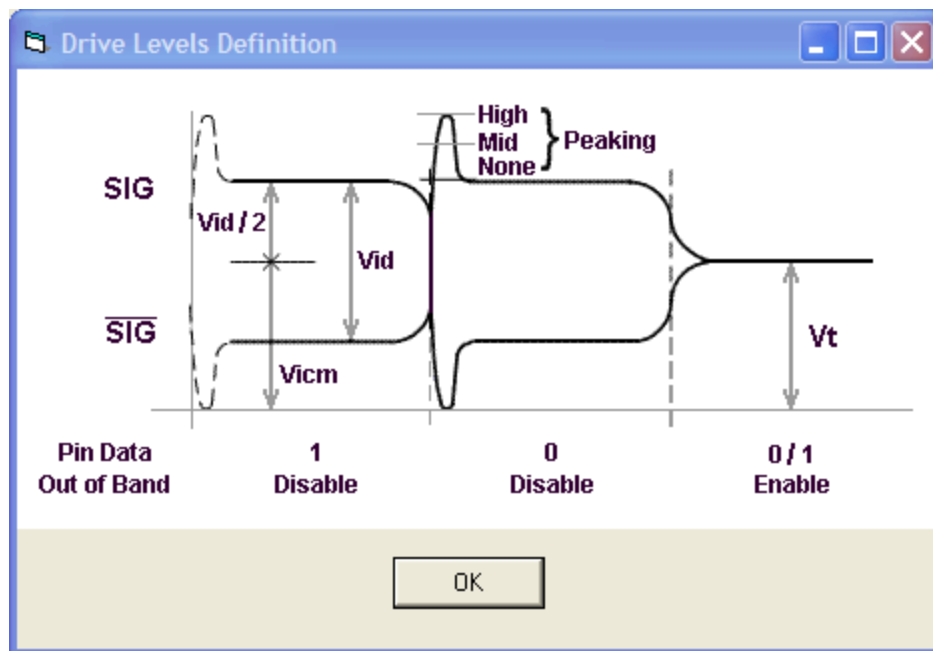
Peaking

The peaking mode of the driver. Valid values are:

- **None**: No peaking applied. Default.
- **Mid**: Mid level peaking applied.
- **High**: High level peaking applied.

✓ Note:

The Serial Channel Display allows you to program a value for the Vid that is outside the spec range. The programmable range is +100mV to +1.8V. A value outside the spec range is not guaranteed to be accurate.

**Drive Levels Definition****Differential Receive Channels**

The following figure shows the Serial Channel display with a receive channel selected.

Serial Channel Debug Display, Receive Channel Selected

Receive Timing parameters are as follows:

CDR Mode	The clock data recovery mode of the Receive Align Trigger. The supported modes are: Train & Track and Disabled. Train & Hold mode is not supported.
Start Delay	Delay offset for the receive pin. Range is: 0 to 16383 x MOSC Period.
MOSC Period	Period of the MOSC, which is determined from the programmed MPV and DUT Period. This is a read-only value and cannot be modified in this display. This is a read-only value and cannot be modified in this display.
MPV	Number of MOSC per vector. Valid values are: 1, 2, 4, 8, 16, 32, 64, and 128.

DUT Period	The period of the DUT. Valid range is: 1 x MOSC Period to 128 x MOSC Period. This range is calculated from the minimum MPV value of 1 to the maximum MPV value of 128.
MPV0/MPV1	MOSC per vector for the currently selected differential channel pair. These options are only available in Protocol Aware (PA) channel mode. When running PCIe PA, IG-XL uses these selectors to switch on-the-fly between two speeds, 2.5Gbps and 5.0Gbps, while connecting to a PCIe device. The UltraSerial10G always starts on 2.5Gbps, and IG-XL can increase the speed to 5.0Gbps.

The Receive Align Trigger parameters are as follows:

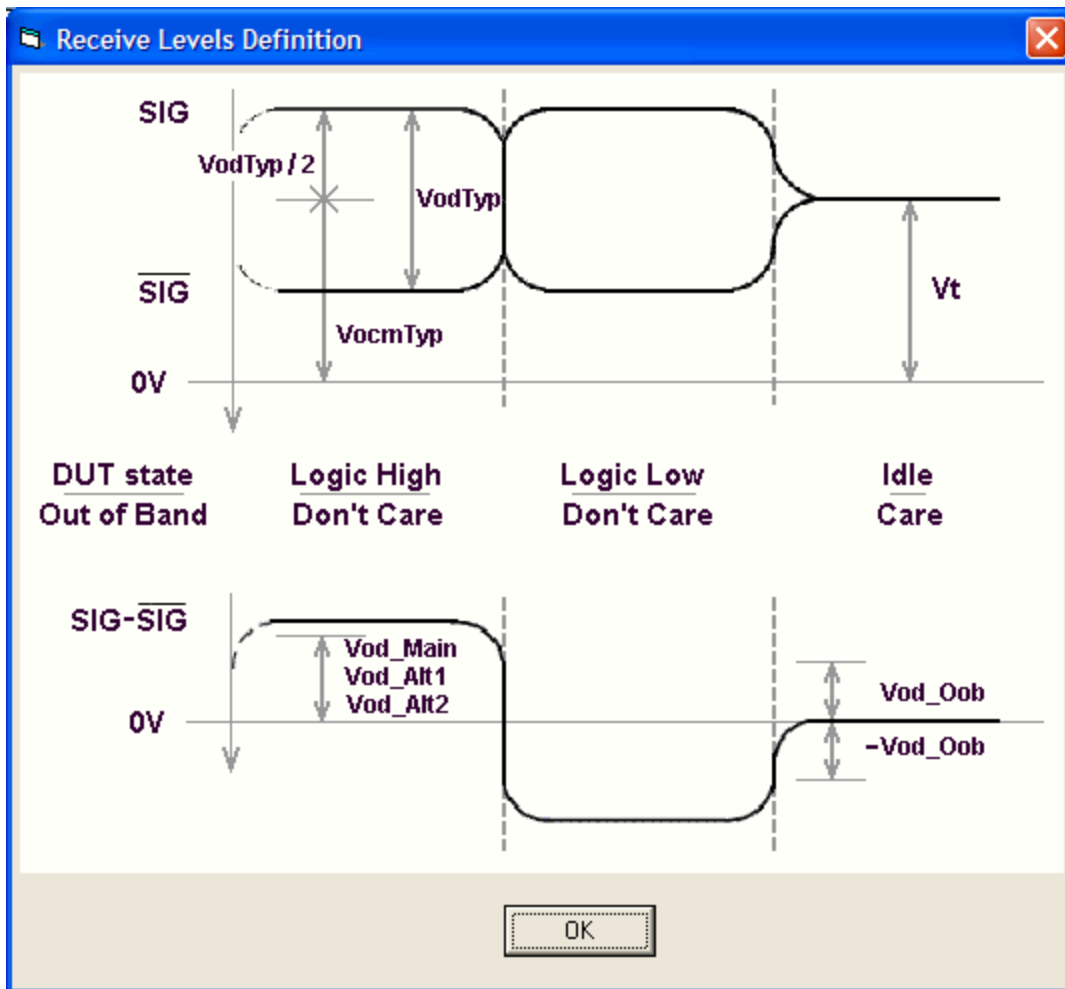
Mode	The mode of the Receive Align Trigger. Valid values are: CDR Lock, 10b, and 20b.
10b Disparity Enabled	If this box is checked, the 10b symbol match will occur if the symbol is found regardless of positive or negative disparity. If this box is not checked, the 10b symbol match will only occur if the symbol is found matching the programmed disparity. This box is only enabled if the Mode is 10b.
20b Disparity Enabled	If this box is checked, the 20b symbol match will occur if the symbol is found regardless of positive or negative disparity. If this box is not checked, the 20b symbol match will only occur if the symbol is found matching the programmed disparity. This box is only enabled if the Mode is 20b.
Timeout Enabled	If this box is checked, the PatGen will time out if the Receive Align Trigger does not occur in the amount of time specified by the Timeout parameter.
Timeout	The amount of time to wait for the Receive Align Trigger to occur.
Alignment Symbol	The 10b alignment symbol.
Match Status	The actual state of the Receive Align Trigger. Valid values are: No CDR Lock, CDR Lock, 10b, and 20b. This is a read-only value.
Timeout Status	Whether or not Receive Align Trigger timeout occurred. Valid values are: No Timeout and Timed Out. This is a read-only value.

RAT Timeout	The time it took for Receive Align Trigger to occur from serial PatGen start. This is a read-only value.
-------------	--

The Receive Pin Electronics parameters are as follows:

Pin Name	The name of the UltraSerial10G receive pin.
Vod_Main	For functional testing, this level is the standard setting which is typically set to 0.0V. For jitter testing, this level is the mid Vod level which is typically set to 50% of the output voltage swing. Valid values: -0.8V to +0.8V
Vod_Alt1	For six-point eye and rise time/fall time testing, this level is the minimum Vod level which is typically set to 20% of the output voltage swing. Valid values: -0.8V to +0.8V
Vod_Alt2	For six-point eye and rise time/fall time testing, this level is the maximum Vod level which is typically set to 80% of the output voltage swing. Valid values: -0.8V to +0.8V
Vod_Oob	The out of band detection threshold for the receive channel. Valid values: 25mV to 100mV.
Vt	The termination voltage. Valid range is: 0V to +1.7V.
PE/PPMU Enabled Selectors	Selects which resource to apply to the DUT. Click a radio button to choose the enabled connection: • PE Enabled: Channel connected to the Pin Electronics • PPMU Enabled: Channel connected to the PPMU • Not Enabled: Channel not connected to either the Pin Electronics or the PPMU Note that you cannot connect the pin electronics and PPMU at the same time.
PPMU	See PPMU Debug Display below.

Levels Definition	This button brings up a window which shows a graphical representation of the various receive levels. See Receive Levels Definition .
Channel Mode	This read-only field shows the channel mode. Values are: Serial, Serial Drive-only, Protocol Aware, and Loopback.
Peaking	The peaking mode of the driver. Valid values are: • None: No peaking applied. Default. • Mid: Mid level peaking applied. • High: High level peaking applied.



Receive Levels Definition

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UltraSerial10G Debug Displays : PPMU Debug Display

PPMU Debug Display

The PPMU Debug Display allows you to modify the PPMU settings for an UltraSerial10G pin. The PPMU for a UltraSerial10G pin is different from an HSD pin in the following ways:

- [UltraSerial10G pins support the following](#) Force Current Range values: 200uA, 2mA, and 30mA.
- [UltraSerial10G pins support the following](#) Measure Current Range values: 200uA, 2mA, and 30mA.
- [UltraSerial10G pins do not support programmable PPMU voltage clamps](#) (Clamps VHi and Clamps VLo). These field are read-only.

For more information, see [PPMU Debug Display](#).

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UltraSerial10G Examples

UltraSerial10G Examples
For more information, see:

- Serial (VBT Syntax Reference)
- UltraSerial10G Programming

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UltraSerial10G Programming

UltraSerial10G Programming

This section describes how to program an UltraSerial10G to source a waveform to a device and includes these topics:

- [Simulated Configuration Files](#)
- [UltraSerial10G Compatibility](#)
- [Programming Overview](#)
- [UltraSerial10G DataTool Programming](#)
- [UltraSerial10G Pattern Control](#)
- [UltraSerial10G Functional Test Programming](#)
- [Capturing and Processing Data](#)
- [UltraSerial10G Jitter and Eye Measurement Test Programming](#)
- [Job Execution](#)
- [Using UltraSerial10G in Concurrent Testing](#)

For more information, see:

- [Serial](#) (VBT Syntax Reference)

- [UltraSerial10G Examples](#)

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UltraSerial10G Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the UltraSerial10G offline by forcing IG-XL to map the UltraSerial10G. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the UltraSerial10G board in slot 3, as the HSS-10G type:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]  Type              idprom (type serial rev company)

3.0      HSS-10G      974-245-40 00000000 0000-A 0000
          HSS-10G      956-128-02 00000000 0000-A 0000
          PowerBoard   956-462-10 00000000 0000-A 0000
          ChanBoard0    956-160-10 00000000 0000-A 0000
          ChanBoard1    956-160-10 00000000 0000-A 0000

4.0      HSD-M        979-219-00 00000000 0000-A 0000
```

	HSD-M	956-361-00 0000000 0000-A 0000
12.0	HexVS	974-294-00 0000000 0000-A 0000
	HexVS	956-388-00 0000000 0000-A 0000
21.0	VSM	974-296-00 0000000 0000-A 0000
	VSM	956-361-00 0000000 0000-A 0000
24.0	SupportBoard	974-220-12 0000000 0000-A 0000
	SupportBoard	956-354-00 0000000 0000-A 0000
62.0	DistributionBoard	956-355-00 0000000 0000-A 0000
	DistributionBoard	956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the [UltraSerial10G offline](#), the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the UltraSerial10G online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see [Tester Configuration](#).

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UltraSerial10G Programming : UltraSerial10G Compatibility

UltraSerial10G Compatibility

This section describes differences between the UltraSerial10G and the SB6G serial instruments. The UltraSerial10G has significantly higher performance and a higher channel density than the SB6G. You can use the UltraSerial10G and the SB6G in the same test system. The following topics are available:

- [SB6G and UltraSerial10G Physical Description](#)
- [SB6G and UltraSerial10G Feature Comparison](#)
- [SB6G to UltraSerial10G Migration Considerations](#)
- [SB6G to UltraSerial10G Test Program Conversion](#)
- [Using SB6G and UltraSerial10G in the Same Job](#)

For more information, see:

- [SB6G Specifications](#)
- [UltraSerial10G Specifications](#)

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UltraSerial10G Programming : UltraSerial10G Compatibility : SB6G and UltraSerial10G Physical Description

SB6G and UltraSerial10G Physical Description

The SB6G is a high-level assembly that includes a power board component, an instrument board component, and a DIB launch assembly. The DIB assembly includes cabling up to one DIB interposer block that is aligned to the UltraFLEX test head’s sidewalk 2. A liquid cooling assembly for the instrument board occupies one universal card slot in the test head.

The UltraSerial10G combines two PCBs, each of which supports 10 differential pairs, back-to-back with a cooling assembly between them. It uses timing and pattern generation architecture similar to the SB6G, which accounts for its high degree of compatibility in function and performance. The UltraSerial10G DIB assembly includes cabling for up to three DIB interposer blocks that are aligned to the UltraFLEX test head’s sidewalks 2, 3, and 4.

Due to differences in pin electronics timing architecture and PPMUs on the UltraSerial10G and SB6G, use guidelines in this section to make a smooth transition from the SB6G to the UltraSerial10G instrument.

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UltraSerial10G Programming : UltraSerial10G Compatibility : SB6G and UltraSerial10G Feature Comparison

SB6G and UltraSerial10G Feature Comparison

The table below compares major features of the SB6G and UltraSerial10G, showing differences that can affect functionality and performance.

SB6G and UltraSerial10G Feature Comparison		
Feature	SB6G	UltraSerial10G
General Features		
Min to Max Data Rates	33Mbps to 6.4Gbps	42Mbps to 10.7Gbps
Differential Ports	8	20
Site Copy Mode	N/A	Up to 40 RX ports
Time Domains	1	2
Pin Electronics and Levels		
AC/DC Coupling	Supported	DC coupling only
Peaking on Driver and Comparator	No	Yes
Out of band (OOB) Signaling	Yes	Yes
Vid Range	50mV to 1.2V	100mV to 1.6V
Vicm	-0.9V to 3.5V	50mV to 1.75V
Vod_Main/Alt1/Alt2	-0.9V to 0.9V	-0.8V to 0.8V

Vt	-1.0V to 3.6V	0V to 1.7V
Timing		
Maximum data rate using VM	6.4Gbps	8.0Gbps
Maximum data rate using PRBS	6.4Gbps	10.7Gbps
Maximum data rate using loopback	N/A	10.7Gbps
Maximum data rate using jitter insertion and measurement	6.4Gbps	10.7Gbps
Timing FPGA mode	Base Serial	Base serial, loopback, site copy, and PCIe Protocol Aware (PA)
MOSC range	8.55GHz to 12.8GHz	5.35GHz to 10.7GHz
Start delay range	0 to 1.9s	0 to 1.5s
MOSC period (start delay resolution)	78ps to 117ps	93ps to 187ps
MOSC per vector (MPV) values	2, 3, 4, 5, 6, 7, 8, 9, 11, 16, 22, 32, 44, 64, 88, 128, 176, and 256	1, 2, 4, 8, 16, 32, 64, and 128
Vector Memory (VM) Support		
Drive/Expect pattern depth	256M	512M (426M available)
Drive Midband (X)	Per 32 bits, Modulo-32	Per 10 bits, Modulo-10
Compare Masking	Per bit	Per 10 bits, Modulo-10
Minimum pattern size	1024	2560 or 40 (if last pattern module in pattern set)
Pattern modulo size	1024	2560 or 40 (if last pattern)
Capture Memory (CMEM) Support		
Capture depth	256M cycles	1G cycles
Resolution	Per bit	Per 10 bits
PRBS Hardware		
PRBS polynomials	15, 23, and 31	7, 15, 23, and 31
Fail Counter	16K	16K

Cycle Counter	2^49	2^44
Reseed	No	Yes
Digital Loopback		
Instrument loopback	No	Yes
Drive level programming	N/A	Yes
Jitter insertion	N/A	Yes
Capture DUT output	N/A	Yes
Receive Align Trigger (RAT)		
CDR Support	Yes	Yes
10b align	Yes	Yes
10b disparity	No	Yes
20b match + disparity	Yes	Yes
CDR Training Interval	<30k UI (UI = 1 DUT period)	80s, nominal
Protocol Aware (PA)		
Signature receive	Yes	No
PCIe Generations 1 and 2	No	Yes
PPMU		
Current range	200A, 2mA, 50mA	200A, 2mA, 30mA
Voltage range	~-1V to 3.6V	-1V to 1.5V
Simultaneous connection to pin electronics (PE)	Yes	Yes, but you must use specific connection rules. See PPMU.
Licensing		
Licenses	One license for all features (HSS_Speed)	Multiple licenses, one for each feature (HSS10G_Speed, HSS10G_Channels, HSS10G_Loopback, HSS10G_PRBS, HSS10G_JitterMeasure, HSS10G_JitterInsert, PCIeRootComplexEngine,

		PCleEndPointEngine, and sWire)
Data rate enabling points (Mbps)	3200, 6400	3200, 6400, 8000, 10000
Port enabling	One license enables all ports on one instrument board.	Each channel and data rate (speed) license enables any 10 drive and receive ports on any US10G board in the test system. Each digital loopback, PRBS, jitter measure, and jitter insertion license enables any 10 engines on any US10G board in the test system. Each PCIe root complex, PCIe end point, and sWire engine requires one license, up to 20 engines per board.
DIB Support		
Sidewalk 2	8 ports	8 ports
Sidewalk 3	None	8 ports
Sidewalk 4	None	4 ports
DGS signal	At pad D204	Not used

For more information, see:

- SB6G Specifications
- UltraSerial10G Specifications

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UltraSerial10G Programming : [UltraSerial10G Compatibility](#) : SB6G to UltraSerial10G Migration Considerations

SB6G to UltraSerial10G Migration Considerations

This section describes how differences between the SB6G and UltraSerial10G affect test program migration to the UltraSerial10G and provides recommendations for resolving compatibility issues. It is divided into these topics:

- [Serial Ports, Data Rates, and Site Copy Mode](#)
- [Pin Electronics and Levels](#)
- [Serial Timing](#)
- [Vector Memory Support](#)
- [Capture Memory Support](#)
- [PRBS Hardware](#)
- [Digital Loopback](#)
- [Receive Align Trigger](#)
- [Protocol Aware \(PA\)](#)

- PPMU

- Licensing

- DIB Support

Serial Ports, Data Rates, and Site Copy Mode

SB6G only supports one time domain and eight ports. UltraSerial10G can have up to two time domains, and each time domain can contain up to 10 ports on each half board for a total of 20 ports on each instrument board. Note that the UltraSerial10G ports are not divided so that the first time domain includes the first 10 ports and second time domain includes the last 10 ports. Instead, the UltraSerial10G time domains include these ports:

- First Time Domain (0) Ports: 4, 5, 6, 7, 8, 9, 10, 11, 14, 15

- Second Time Domain (1) Ports: 0, 1, 2, 3, 12, 13, 16, 17, 18, 19

UltraSerial10G also runs at a higher data rate and has a higher channel density than the SB6G. This allows testing at higher data rates and higher site counts without using relays for resource sharing and impacting test time with site serial loops.

Finally, UltraSerial10G supports a drive-only site copy mode that allows you to reuse all receive channels as drive channels on a different site. These channels become site copies of the primary drive channels within the same timing group. Basically, this mode changes TX ports to RX ports so that compare pins become drive pins. Each TX pin drives the same level, timing, and pattern data as its RX drive pair pin. This mode is useful for testing more sites in parallel in applications, like HDMI receivers, where you do not require the UltraSerial10G receive pins. For more information, see [Site Copy and Receive Channel Reuse](#) and [Site Copy Guidelines](#).

Pin Electronics and Levels

UltraSerial10G has several differences from the SB6G related to pin electronics and levels programming.

First, UltraSerial10G does not support an AC coupling option, like the SB6G, where you can disable DC coupling and VT termination on receive pins by setting the Pin Levels sheet's DCCoupling and VtEnable parameters to Disable. On UltraSerial10G, DC coupling is always enabled for all pins. An UltraSerial10G Pin Levels sheet does not include the SB6G DCCoupling or VtEnable parameters.

When porting SB6G test programs to UltraSerial10G, the Pin Levels sheet retains these parameters for compatibility, but IG-XL returns a validation error if you set the DCCoupling or VtEnable value to Disable on the UltraSerial10G.

Second, UltraSerial10G provides both driver and comparator peaking to compensate for losses in the transmission line. Using the [Pin Levels sheet](#) Peaking parameter, you can program driver and comparator peaking separately using these options: None, Mid, and High. SB6G does not support peaking levels.

Finally, you can program an out of band (OOB) Vod detection threshold level (Vod_Oob) on the UltraSerial10G, which is not available on SB6G. This comparator threshold level allows you to detect OOB levels for certain SerDes tests.

Pin Levels

Pin/Group	Seq.	Parameter	Value
RX_ALL		Vid	0.1
RX_ALL		Vicm	0
RX_ALL		Peaking	None
TX_ALL		Vod_Main	0
TX_ALL		Vod_Alt1	0
TX_ALL		Vod_Alt2	0
TX_ALL		Vod_Oob	0.065
TX_ALL		Vt	0
TX_ALL		Peaking	None
TX_ALL		VodTyp	0
TX_ALL		VocmTyp	0

UltraSerial10G New Pin Levels Parameters

If a device requires AC coupling, then you must include this coupling option on the UltraSerial10G DIB, usually with a relay so that you can switch it on and off. SB6G uses a 100nF capacitor for AC coupling.

You must also check that the UltraSerial10G driver can drive the necessary levels and swings based on the current level ranges in a SB6G test program. You may need to program UltraSerial10G peaking settings to minimize transmission loss and achieve better accuracy, yield, and test results.

Serial Timing

The primary difference in timing between the SB6G and UltraSerial10G is the MOSC range and period:

- SB6G: 8.55GHz to 12.8GHz with a period of 78ps to 117ps
- UltraSerial10G: 5.35GHz to 10.7GHz with a period of 93ps to 187ps.

The valid values for MOSC per vector (MPV) are also different for the instruments:

- SB6G: 2, 3, 4, 5, 6, 7, 8, 9, 11, 16, 22, 32, 44, 64, 88, 128, 176, and 256
- UltraSerial10G: 1, 2, 4, 8, 16, 32, 64, and 128

Since you get different resolutions from the different MOSC ranges and MPV values, the start delay range is also different between these instruments. For UltraSerial10G, the maximum start delay is 8K multiplied by 187ps which equates to ~1.5s. For the SB6G, it is 16K multiplied by 117ps, which equates to ~1.9s.

To ensure timing compatibility between instruments, remove any MOSC values programmed in the Serial Timing sheet's Cycle Master Period column. Since the MOSC range and MPV values are different between these instruments, drive and receive delay resolutions are also different. You can either update these values or allow IG-XL to program them to the next nearest possible value. Finally, you should update or remove any MOSC and MPV programming in the VBT section of the test program.

UltraSerial10G also includes a field on the Serial Timing sheet called Max Period Error. This value is the maximum difference allowed between the value specified in the Cycle DUT Period column and the nearest supported hardware period. Use this field to specify the maximum acceptable difference between the desired period and the period that IG-XL calculates. If this difference is exceeded, then IG-XL returns a validation error.

Finally, UltraSerial10G supports several additional FPGA timing modes besides the base serial mode used with SB6G. These include digital loopback, site copy, and PCIe Protocol Aware (PA). When using these modes, make sure that you follow channel selection guidelines for UltraSerial10G timing groups so that your DIB design works correctly. UltraSerial10G has 10 timing groups, five on each half instrument board. Each timing group controls two pin electronics (PE) blocks. Each PE has one TX and one RX port. Both PE blocks in each timing group must operate in only one of the four FPGA modes.

For more information on channel selection, see the UltraSerial10G DIB Design Guidelines.

Vector Memory Support

UltraSerial10G has a larger vector memory size than the SB6G, and you must update the following items when using the vector memory:

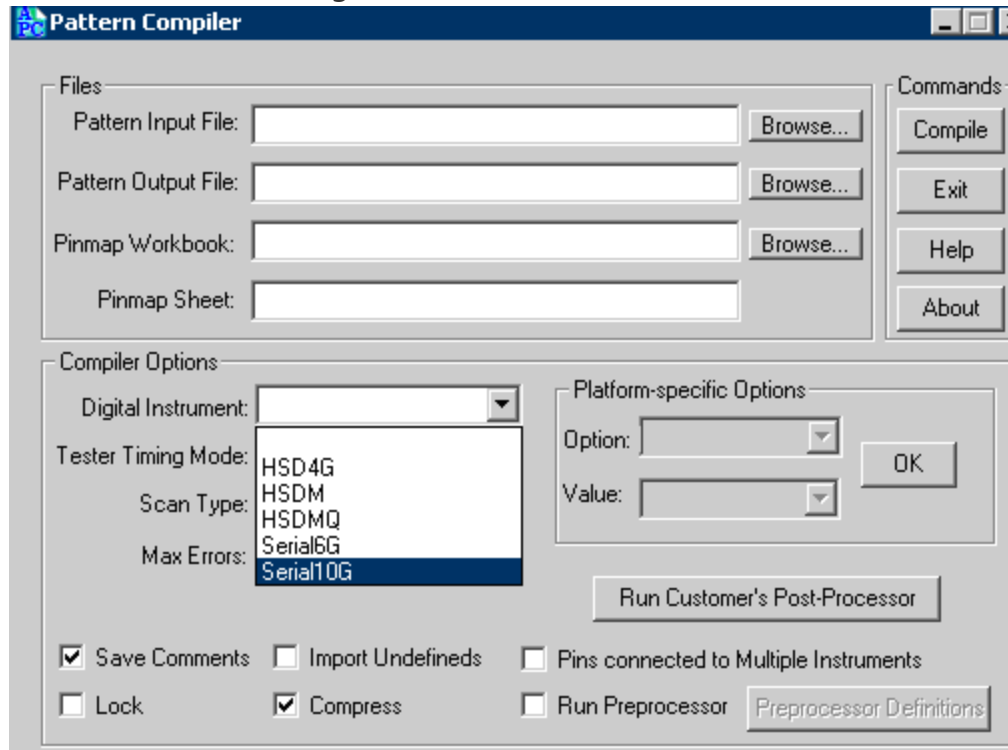
- Pattern file's digital instrument header
- Pattern compiler's timing mode
- Pattern length modulo requirements
- Masking bits in serial patterns
- Out of band (OOB) requirements

Digital Instrument Header

When transitioning from an SB6G pattern, change the digital instrument headers from SB6G syntax (digital_inst = Serial6G;) to UltraSerial10G syntax (digital_inst = Serial10G;). If you start the serial pattern from an HSD pattern, then you also must change the headers in that pattern.

Pattern Compiler

When compiling UltraSerial10G patterns, select the Serial10G option in the Tester Timing Mode field, as shown in the figure below.



Pattern Compiler Serial10G Tester Timing Mode Option

Pattern Length Requirements

You must also adjust the pattern length between SB6G and UltraSerial10G patterns. On SB6G, the pattern length must be modulo 1024 vectors. On UltraSerial10G, it must be modulo 2560 vectors, unless it is a pattern set with only one pattern or the last pattern module in a pattern set, in which case the pattern length can be modulo 40 vectors.

For SerDes patterns that use the 8b10b encoding scheme, the pattern size must be modulo 10. Typically, SB6G patterns have a minimum pattern size of 5120 (1024 multiplied by 5) to meet the 10-bit boundary requirement. You can convert most of these patterns by changing the header to Serial10G since a modulo size of 2560 is compatible with the 5120 modulo size (2560 multiplied by 2). In cases where the SB6G patterns do not use a 10-bit boundary length, check if the pattern set has only one pattern or if the pattern module is the last in the pattern set. If either of these cases is true, then you can use a modulo 40 vector length to meet any boundary restrictions. If the pattern cannot use the modulo 40 vector length, then you must pad the pattern to meet the modulo 2560 vector restriction.

Note that you must also update the operand in the Serial Pattern Sets sheet for patterns that use the PRBS opcode. Note also that UltraSerial10G does not support the NoData opcode.

Masking Bits

Masking bits in serial patterns is also different between these instruments. On SB6G, you can mask on a per-bit basis, but on UltraSerial10G, masking is based on a block of 10 bits and on a 10-bit boundary within the pattern.

Most SerDes patterns are not affected by this restriction, but you may encounter a problem if you run low-speed patterns like scan that do not have a 10-bit boundary and require masking control on a per-bit basis. For these patterns, you can either mask more bits than desired with the possibility of losing some coverage or expand the pattern by 10X, which allows you to mask on a per-bit basis.

Expanding the pattern by 10X, however, has potential drawbacks. First, depending on the number of scan patterns, pattern expansion can quickly consume the vector memory space. Second, you must be able to visualize the 10X expansion when debugging with PatternTool, which will show 10 continuous failures for one actual failure.

OOB Requirements

Finally, out of band (OOB) requirements are different between SB6G and UltraSerial10G. On UltraSerial10G, the OOB programming is per 10 bits and on a 10-bit boundary. On SB6G, it is per 32 bits and on a 32-bit boundary. Since you program most SerDes patterns on a 10-bit boundary, SB6G patterns typically follow the UltraSerial10G restriction. If they do not, you must pad the OOB patterns to meet the modulo 10 requirement.

Capture Memory Support

UltraSerial10G CMEM has a larger capture depth (1G cycles) than SB6G (256M cycles). The capture resolution is also different between instruments. (SB6G is per bit, and UltraSerial10G is per 10 bits.) When running above 8Gbps on the UltraSerial10G, you can only capture CMEM data using one lane at a time per timing group.

PRBS Hardware

UltraSerial10G PRBS hardware is the same as the SB6G, except that UltraSerial10G adds a PRBS 7 polynomial and reseed feature, which allows you to use PRBS 7 hardware instead of vector memory patterns. The cycle counter size is also different between the two instruments, but this should not affect most programs. If this is an issue, simply reduce the run time of the PRBS sequence.

Note, however, that you must make changes in your VBT module when programming PRBS hardware for these instruments. For SB6G, you program PRBS algorithms, bit error rate (BER), and signatures using the VBT node, TheHdw.Serial.Pins.Pattnerns.PRBS. For UltraSerial10G, however, IG-XL no longer supports this PRBS node to program these PRBS elements or return their values in an immediate window. Instead, you must use the Serial Pattern Sets sheet to program the PRBS algorithm and to see the current PRBS seed value from the PRBS generators. To program the BER, use the VBT node, TheHdw.Serial.Pins.Patgen.BitErrorRate(WaitTime), which returns the bit error

rate and takes an optional wait time parameter that is the time to wait before taking the BER measurement.

Digital Loopback

UltraSerial10G allows you to conduct digital loopback. This option provides the capability to loop back device output into its receiver. Typically, you must rely on DIB relays to achieve this capability, but this feature eliminates the need for these relays. This feature allows you to perform more exhaustive testing with level control and jitter insertion. It also provides better debug capability since you can capture device output during loopback testing. Use the Loopback or Loopback_oob values in the Serial Pattern Sets sheet's Receive Patterns Opcode column to set up digital loopback.

Receive Align Trigger

UltraSerial10G has the same receive align trigger (RAT) architecture as the SB6G. When porting an SB6G test program to UltraSerial10G, the only modification you need to make is to adjust the time allowed for the clock data recovery (CDR) training interval (also known as the CDR lock time). This CDR training interval is the time between when the DUT starts sending data and when the CDR circuit starts tracking the actual data that you want to compare or capture. For SB6G allow a minimum of 30k UI, where UI = 1 DUT period; for UltraSerial10G allow a minimum of 80s. IG-XL includes several updates to the receive align trigger VBT to support UltraSerial10G:

- `TheHdw.Serial.Pins.Patgen.ReceiveAlign.Disparity(Disparity)`

This node adds a disparity filter for 10b alignment, which helps eliminate the need for rebursting a pattern multiple times to get the 10b alignment. To enable this 10b align disparity filter, use `tlSerialReceiveAlignDisparity10b`. You can also enable a 20b match disparity filter or both the 10b align and 20b match disparity filters using this node.

- `TheHdw.Serial.Pins.Patgen.ReceiveAlign.TriggerMode`

This node hides the `tlSerialReceiveAlignStart` trigger mode for the UltraSerial10G, although it is still supported for SB6G. On UltraSerial10G, use the `tlSerialReceiveAlignCDR` trigger mode to set the RAT to start the expect pattern execution for clock data recovery. You can still use the `tlSerialReceiveAlign10b` and `tlSerialReceiveAlign10b20b` trigger modes on both instruments.

Protocol Aware (PA)

Protocol Aware (PA) is tester architecture that enables instruments to communicate with standard device protocols without using low-level vector programming. PA allows you to execute direct instructions and emulate real time device interaction at the protocol level. The UltraSerial10G instrument supports PA to test PCIe controllers for the following PCIe specifications:

- PCIe Generation 1 at 2.5Gbps

- [PCIe Generation 2 at 5.0Gbps](#)

For more information, see [PCIe Protocol Aware](#).

It also supports PA for sWire engines for testing of high-speed serial bus interfaces, such as SATA.

For more information, see [sWire Protocol Aware](#).

Note that UltraSerial10G does not support the Signature Receive option that is available on the SB6G.

PPMU

UltraSerial10G and SB6G PPMU voltage and current ranges differ slightly, so you must modify SB6G programs that use the 50mA range so that they use the UltraSerial10G's 30mA range instead. You must also check that the SB6G PPMU voltage range is no greater than 1.5V when porting it to the UltraSerial10G.

Simultaneous Pin Electronics and PPMU Connections

On the SB6G you can connect the pin electronics (PE) and PPMU at the same time without using any special programming or connection sequence. On the UltraSerial10G, however, you cannot connect the PE and the PPMU at the same time without following specific connection rules.

Moreover, IG-XL does not notify you when it automatically disconnects one resource when you connect the other. Note that this behavior applies to V8.10.10 and earlier.

To connect the PE and PPMU at the same time on UltraSerial10G, follow these rules:

- If the PPMU is connected using a current range smaller than 30mA (either 2mA or 200A), then you can connect to the PE, and IG-XL allows both to be connected at the same time.

Note that if the PPMU debug display is open for the serial pin while you try to make a simultaneous connection in this way, IG-XL may disconnect the PPMU and connect the PE. This known issue will be fixed in V8.20.00.

- If the PE and PPMU are connected at the same time, IG-XL automatically disconnects the PE when the PPMU changes to the 30mA range.

- If the PPMU is connected in 30mA range, then connecting the PE does nothing, and IG-XL leaves the PPMU connected.

- If only the PE is connected and you connect the PPMU using any current range, then IG-XL automatically disconnects the PE and connects the PPMU.

- In all versions of IG-XL, you must use the following VBT syntax to connect both the PE and PPMU at the same time as long as the PPMU current range is not 30mA:

Call TheHdw.PPMU.AllowPPMUFuncRelayConnection(True)

Licensing

SB6G uses a single license (HSS_Speed) that supports two data rates (3200Mbps and 6400Mbps) and includes all other features (jitter insertion, jitter measure, PRBS, and so on) for each instrument board.

UltraSerial10G, however, uses eight licenses to support all of its features. Each channels (HSS10G_Channels) and data rate (HSS10G_Speed) license enables any 10 drive and receive ports on any instrument board in the test system. UltraSerial10G supports the following data rate enabling points: 3200, 6400, 8000, and 10000Mbps. Each of the following licenses enables any 10 ports on any UltraSerial10G board in the test system:

- Digital loopback (HSS10G_Loopback)

- PRBS (HSS10G_PRBS)

- Jitter measure (HSS10G_JitterMeasure)

- Jitter insertion (HSS10G_JitterInsert)

Each of the following UltraSerial10G Protocol Aware licenses supports one engine, up to 20 engines per board.

- PCIe root complex (PCleRootComplexEngine)

- PCIe end point (PCleEndPointEngine)

- sWire for testing high-speed serial bus interfaces such as SATA (sWire)

Note that this UltraSerial10G licensing structure is in place for V8.00.01 Patch 6 and for V8.10.10 and later releases. However, UltraSerial10G licensing was handled differently in V8.00.00 and V8.10.00. For information on licensing in these versions, contact the Teradyne applications support team.

DIB Support

SB6G DIBs use the same interposer, so they are compatible with UltraSerial10G. When designing new DIBs for UltraSerial10G, follow the pinout, channel selection, and wiring guidelines in the UltraSerial10G DIB Design Guidelines.

The only difference between the SB6G and UltraSerial10G signal layout is that pad location D204 is a device ground sense (DGS) connection on SB6G and a no connect (NC) signal on UltraSerial10G. So, when designing an UltraSerial10G DIB that is compatible with SB6G, you must ground the NC signal at pad location D204.

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UltraSerial10G Programming : UltraSerial10G Compatibility : SB6G to UltraSerial10G Test Program Conversion

SB6G to UltraSerial10G Test Program Conversion

To convert an SB6G test program to use the UltraSerial10G, you must first move to IG-XL V8.00.00 or later. Earlier versions of IG-XL do not support the UltraSerial10G instrument.

The following procedure provides a guideline when converting SB6G test programs to run on a test system that uses UltraSerial10G:

1. [Modify Channel Map](#)

2. [Update Serial Timing](#)

3. [Modify Pin Levels](#)

4. [Convert Patterns](#)

5. [Update VBT Calls](#)

Modify Channel Map

On the [Channel Map](#) sheet you must modify the Tester Channel Type column from Serial to Serial10G. If designing a new UltraSerial10G DIB, assign UltraSerial10G pins to the proper resources and specify the tester channel type. Channel numbers range from 0 to 19 for the drive and receive pins.

Update Serial Timing

You may need to make the following updates to the [Serial Timing](#) sheet:

- Remove the MOSC programmed value in the Cycle Master Period column since both the MOSC and MPV values for the UltraSerial10G and SB6G are different.
- Adjust the start delay value to use a different MOSC resolution.
- Modify any VBT code that programs the MOSC or MPV.

Modify Pin Levels

When updating pin levels for UltraSerial10G, disable the AC coupling option since it is not supported for UltraSerial10G. Also review and change any voltage ranges from SB6G that are incompatible with the UltraSerial10G. Finally, program UltraSerial10G level features, if required. Review this list of Pin Levels sheet parameters to make sure your pin levels work for UltraSerial10G:

DCCoupling	Remove or program to Enable.
VtEnable	Remove or program to Enable.
Vid, Vicm, Vod_Main, Vod_Alt1, Vod_Alt2, and Vt	Make sure they are programmed within the UltraSerial10G specification.
Vod_oob	Use this UltraSerial10G parameter to set an out of band (OOB) Vod detection threshold level for OOB voltage testing on serial buses like Serial ATA.
Peaking	Use this UltraSerial10G parameter to set the peaking mode (none, mid or high) on the driver and comparator to compensate for skin effect loss.

Convert Patterns

Update all SB6G pattern digital instrument headers as described above and ensure that the pattern length meets the UltraSerial10G modulo size requirements. Once you update the pattern, recompile it using the Serial10G option in the Tester Timing Mode field in the Pattern Compiler.

Update VBT Calls

Apart from making changes to the PRBS and receive align trigger VBT code, as described above, you must also update the capture configuration type to capture data and enable MoveLink for the serial jitter and eye tests. To do this, replace the following VBT code:

```
TheHdw.Serial.Pins(Serial_Pins).Patterns.CaptureMemory.CaptureConfig =  
    tISerialCaptureConfigMove  
With:
```

TheHdw.Serial.Pins(Serial_Pins).Patterns.CaptureMemory.CaptureConfig =
tlSerialCaptureConfigSamplerMove

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UltraSerial10G Programming : UltraSerial10G Compatibility : Using SB6G and UltraSerial10G in the Same Job

Using SB6G and UltraSerial10G in the Same Job

When using the SB6G and UltraSerial10G instruments in the same test program, you can include both SB6G and UltraSerial10G entries in the following DataTool sheets:

- | | |
|---|---------------------|
| • | Pin Map |
| • | Channel Map |
| • | Pin Levels |
| • | Serial Timing |
| • | Serial Pattern Sets |

You cannot include both SB6G and UltraSerial10G entries in the same Jitter sheet. To insert jitter on one instrument and measure jitter on the other instrument, you must use VBT syntax to program the instrument used for jitter insert, and use the Jitter sheet to program the instrument used for jitter measure.

SB6G and UltraSerial10G channels must be in separate pin groups and cannot be programmed in the same VBT statements. You cannot include SB6G and UltraSerial10G pins in the same pattern module. You can start patterns separately for each instrument using the following VBT syntax to call each pattern:

```
TheHdw.Serial.Patterns(patName).Start
```

You can also start these SB6G and UltraSerial10G patterns together using this VBT syntax:
TheHdw.Serial.Patterns(SB6GpatName & "+" & US10GpatName).Start

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UltraSerial10G Programming : Programming Overview

Programming Overview

To develop a test program that uses the UltraSerial10G instrument, use the following worksheets:

- | | |
|---|--|
| • | Pin Map |
| • | Channel Map |
| • | DC Specs |
| • | AC Specs |
| • | Pin Levels |
| • | Time Sets (Basic) or Time Sets and Edge Sets |
| • | Serial Timing (UltraSerial10G specific parameters) |
| • | Serial Pattern Sets (UltraSerial10G specific parameters) |
| • | Jitter (UltraSerial10G specific parameters) |
| • | Test Instances |

•	Flow Table
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Programming of several sheets (focused primarily on UltraSerial10G specific parameters) is described in this section. Other sheets are programmed in the standard fashion explained in the DataTool section.

After setting up the worksheets, you can create UltraSerial10G functional, jitter measure, jitter insertion, digital loopback, and parametric tests. To program UltraSerial10G functional tests, you must create a UltraSerial10G pattern file and, in most cases, an HSD pattern file (HSD1000, UltraPin800, or UltraPin1600) and execute them using VBT language. To program a UltraSerial10G parametric test, create an instance of the PPMU template. You use the Flow Table to execute UltraSerial10G functional, jitter measure, jitter insertion, digital loopback, and parametric tests. For information on VBT syntax for UltraSerial10G, see [Serial](#) in the VBT Syntax Reference.

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UltraSerial10G Programming : UltraSerial10G DataTool Programming

UltraSerial10G DataTool Programming

This section describes how to program several DataTool worksheets to handle basic UltraSerial10G parameters. It includes the following topics:

- Programming the Pin Map
- Programming the Channel Map
- Programming Pin Levels
- Programming the Serial Timing
- Programming Serial Pattern Sets
- Programming UltraSerial10G Serial Jitter
- Programming UltraSerial10G Test Instances

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[UltraSerial10G Programming](#) : [UltraSerial10G DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. You must add the DUT pins that will use UltraSerial10G resources to a Pin Map sheet.

Pin Map

Group Name	Pin Name	Type
	RX0P	Input
	RX0N	Input
	RX1P	Input
	RX1N	Input
	RX2P	Input
	RX2N	Input
	TX0P	Output
	TX0N	Output
	TX1P	Output
	TX1N	Output
	TX2P	Output
	TX2N	Output
	pin_a	I/O
	pin_b	I/O
RX0	RX0P	Differential
RX1	RX1P	Differential
RX2	RX2P	Differential
TX0	TX0P	Differential
TX1	TX1P	Differential
TX2	TX2P	Differential
TX_ALL	TX0	Differential
RX_ALL	RX0	Differential
RX_TX_ALL	RX_ALL	Differential
hsd_all	pin_a	I/O

UltraSerial10G Pin Map

Each UltraSerial10G pin is either a single-ended input or a single-ended output. For UltraSerial10G, you must create a differential pair pin group. Each differential pair consists of two single-ended pins. One single-ended pin represents the true side, and the other single-ended pin represents the complement side. In addition, you can create differential pin groups which contain differential pairs and other differential pin groups. (Differential pin groups cannot contain single-ended pins.)

UltraSerial10G Pin Map Parameters

Parameter	Description
Group Name	The pin group name for a differential pair or differential group.

Pin Name	The name of a UltraSerial10G single-ended pin. For pin groups, either the name of a UltraSerial10G single-ended pin, the name of a UltraSerial10G differential pair, or the name of a UltraSerial10G differential group.
Type	Selects a type for each pin name or pin group from the Type list. For UltraSerial10G, the options are: • Input: UltraSerial10G input pins, where "input" refers to the DUT pins, so these pins are connected to the input pins of the DUT and send a signal to the DUT. • Output: UltraSerial10G output pins, where "output" refers to the DUT pins, so these pins are connected to the output pins of the DUT and receive a signal from the DUT. • Differential: Use this type for pin groups.

For more information, see [Pin Map Sheet](#) in the DataTool section.

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UltraSerial10G Programming : [UltraSerial10G DataTool Programming](#) : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. You specify how individual DUT pins are connected to UltraSerial10G channels as shown below.

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comment
RX0P		Serial10G	0.drvinfos0	
RX0N		Serial10G	0.drvneg0	
RX1P		Serial10G	0.drvinfos1	
RX1N		Serial10G	0.drvneg1	
RX2P		Serial10G	0.drvinfos2	
RX2N		Serial10G	0.drvneg2	
TX0P		Serial10G	0.rcvinfos0	
TX0N		Serial10G	0.rcvneg0	
TX1P		Serial10G	0.rcvinfos1	
TX1N		Serial10G	0.rcvneg1	
TX2P		Serial10G	0.rcvinfos2	
TX2N		Serial10G	0.rcvneg2	
pin_a		I/O	4.ch50	
pin_b		I/O	4.ch51	

UltraSerial10G Channel Map

Each UltraSerial10G instrument has 40 drive channels (20 true and 20 complement) and 40 receive channels (20 true and 20 complement). The signal naming conventions are:

- <slot #>.drvpos<channel 0-19> (drive channels, true side)
- <slot #>.drvneg<channel 0-19> (drive channels, complement side)
- <slot #>.rcvpos<channel 0-19> (receive channels, true side)
- <slot #>.rcvneg<channel 0-19> (receive channels, complement side)

Note that all channels on sidewalk 2 are compatible with the SB6G DIB pinout. You must make the channel assignments for all single-ended pins. The Channel Map sheet returns an error if you try to assign a channel to a differential pair. Both the true and the complement sides of a differential pair are assigned a channel connection on the Channel Map sheet.

UltraSerial10G Channel Map Parameters

Parameter	Description
Pin Name	The name of a UltraSerial10G true or complement pin.
Type	Each UltraSerial10 pin name can be one of these types: - Serial10G: UltraSerial10G channel type. - Serial10GDrvOnly: UltraSerial10G drive-only channel type.
Site	The UltraSerial10G channel.

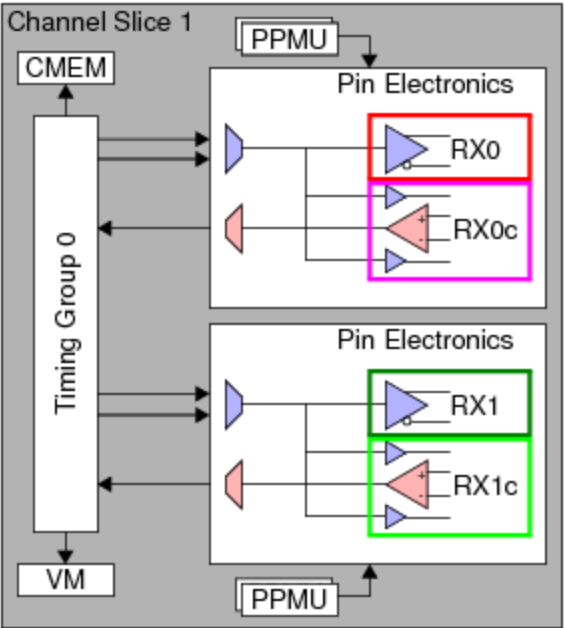
For more information, see Channel Map Sheet in the DataTool section.

Site Copy and Receive Channel Reuse

The UltraSerial10G site copy feature allows you to reuse all of the receive channels as drive channels on a different site. These channels become site copies of the primary drive channels within the same timing group.

When using site copy, make sure to connect the primary channel pair and related site copy channel pair to the same DUT pins on different sites. For example, if you connect drvpos0 and drvneg0 to PinA_pos and PinA_neg on site 0, then you should connect drvpos0c and drvneg0c to PinA_pos and PinA_neg on site 1.

The figure below shows the how a channel map programs the drive-only site copy for one timing group across two sites.



Channel Map				
DIB ID:		View Mode: Signal		
Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
Rx0_T		Serial10GDrvOnly	11.drvpos0	11.drvpos0c
Rx0_C		Serial10GDrvOnly	11.drvneg0	11.drvneg0c
Rx1_T		Serial10GDrvOnly	11.drvpos1	11.drvpos1c
Rx1_C		Serial10GDrvOnly	11.drvneg1	11.drvneg1c

UltraSerial10G Drive-only Site Copy Channel Map Programming

The table below lists the correct channel assignments for site copy applications arranged by sidewalk (SW). Note that each timing group has one differential pair of drive channels and one differential pair of receive channels.

UltraSerial10G Site Copy Channel Selection Assignments

SW	Timing Group	Drive DIB Location	Drive Channels	Receive DIB Location	Receive Channels (Site Copy Disabled)	Renamed Drive Channels (Site Copy Enabled)	Time Domain
2	0	F205	drvpos0	D205	rcvpos0	drvpos0c	1
2	0	E205	drvneg0	D206	rcvneg0	drvneg0c	1
2	0	E206	drvpos1	C206	rcvpos1	drvpos1c	1
2	0	F206	drvneg1	C205	rcvneg1	drvneg1c	1
2	1	B205	drvpos2	B203	rcvpos2	drvpos2c	1
2	1	A205	drvneg2	A203	rcvneg2	drvneg2c	1
2	1	A206	drvpos3	A204	rcvpos3	drvpos3c	1
2	1	B206	drvneg3	B204	rcvneg3	drvneg3c	1
2	2	F201	drvpos4	F203	rcvpos4	drvpos4c	0
2	2	F202	drvneg4	E203	rcvneg4	drvneg4c	0

2	2	E202	drvpos5	E204	rcvpos5	drvpos5c	0
2	2	E201	drvneg5	F204	rcvneg5	drvneg5c	0
2	3	B201	drvpos6	D201	rcvpos6	drvpos6c	0
2	3	B202	drvneg6	D202	rcvneg6	drvneg6c	0
2	3	A202	drvpos7	C202	rcvpos7	drvpos7c	0
2	3	A201	drvneg7	C201	rcvneg7	drvneg7c	0
3	4	F305	drvpos8	D305	rcvpos8	drvpos8c	0
3	4	E305	drvneg8	D306	rcvneg8	drvneg8c	0
3	4	E306	drvpos9	C306	rcvpos9	drvpos9c	0
3	4	F306	drvneg9	C305	rcvneg9	drvneg9c	0
3	5	B305	drvpos10	B303	rcvpos10	drvpos10c	0
3	5	A305	drvneg10	A303	rcvneg10	drvneg10c	0
3	5	A306	drvpos11	A304	rcvpos11	drvpos11c	0
3	5	B306	drvneg11	B304	rcvneg11	drvneg11c	0
3	6	F301	drvpos12	F303	rcvpos12	drvpos12c	1
3	6	F302	drvneg12	E303	rcvneg12	drvneg12c	1
3	6	E302	drvpos13	E304	rcvpos13	drvpos13c	1
3	6	E301	drvneg13	F304	rcvneg13	drvneg13c	1
3	7	B301	drvpos14	D301	rcvpos14	drvpos14c	0
3	7	B302	drvneg14	D302	rcvneg14	drvneg14c	0
3	7	A302	drvpos15	C302	rcvpos15	drvpos15c	0
3	7	A301	drvneg15	C301	rcvneg15	drvneg15c	0
4	8	F403	drvpos16	E404	rcvpos16	drvpos16c	1
4	8	E403	drvneg16	F404	rcvneg16	drvneg16c	1
4	8	C406	drvpos17	D405	rcvpos17	drvpos17c	1
4	8	C405	drvneg17	D406	rcvneg17	drvneg17c	1
4	9	B403	drvpos18	A404	rcvpos18	drvpos18c	1
4	9	A403	drvneg18	B404	rcvneg18	drvneg18c	1
4	9	C402	drvpos19	D401	rcvpos19	drvpos19c	1

4	9	C401	drvneg19	D402	rcvneg19	drvneg19c	1
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For more information on this feature, see [Drive-only Site Copy](#).

			
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[UltraSerial10G Programming](#) : [UltraSerial10G DataTool Programming](#) : Programming Pin Levels

Programming Pin Levels

UltraSerial10G channel levels are programmed using a Pin Levels sheet. The Pin Levels sheet defines values for input and output voltages and current loads to be applied to the DUT.

Pin Levels

Pin/Group	Seq.	Parameter	Value
RX_ALL		Vid	0.1
RX_ALL		Vicm	0
RX_ALL		Peaking	None
TX_ALL		Vod_Main	0
TX_ALL		Vod_Alt1	0
TX_ALL		Vod_Alt2	0
TX_ALL		Vod_Oob	0.065
TX_ALL		Vt	0
TX_ALL		Peaking	None
TX_ALL		VodTyp	0
TX_ALL		VocmTyp	0
hsd_all		Vil	0
hsd_all		Vih	1
hsd_all		Vol	0.5
hsd_all		Voh	0.5
hsd_all		VCmpMain	0
hsd_all		Vt	0
hsd_all		Vcl	-1
hsd_all		Vch	6
hsd_all		VoutLoTyp	0
hsd_all		VoutHiTyp	0
hsd_all		DriverMode	Largeswing-HiZ

UltraSerial10G Pin Levels Sheet

For the UltraSerial10G instrument, you must assign a pin levels parameter for each pin name or pin group. The following table shows the UltraSerial10G columns and pin level Parameter and Value fields. If a pin does not have levels specified, a validation error occurs.

Pin/Group	The name of a UltraSerial10G differential pair. IG-XL automatically creates parameters based on the pin group type (which is differential in this case).
Seq.	A value to control the sequence in which the electronic levels are applied to the DUT. This entry is optional.
Parameter	When you enter a UltraSerial10G differential pair in the Pin/Group column, parameters for pins of that type are automatically inserted in this field.

UltraSerial10G drive pin levels:

- **Vid**: The differential input voltage (assuming an open connection). Valid Values: +100mV to +1.8V
- **Vicm**: The common mode voltage. Valid Values: -0V to +1.8V
- **Peaking**: The voltage levels setting used to compensate for skin effect loss on the drive signal.

UltraSerial10G receive pin levels:

- **Vod_Main**: Threshold voltage output reference level for the differential comparator. For functional testing, this level is the standard setting, which is typically set to 0.0V. For jitter testing, this level is the mid Vod level which is typically set to 50% of the output voltage swing. Valid values: -0.8V to +0.8V
- **Vod_Alt1**: For six-point eye and rise time/fall time testing, this level is the minimum Vod level which is typically set to 20% of the output voltage swing. See also [Programming Vod_Alt and Vt Levels](#). Valid values: -0.8V to +0.8V
- **Vod_Alt2**: For six-point eye rise time/fall time testing, this level is the maximum Vod level which is typically set to 80% of the output voltage swing. See also [Programming Vod_Alt and Vt Levels](#). Valid values: -0.8V to +0.8V
- **Vod_Oob**: The out of band (OOB) Vod detection threshold level.
- **Vt**: The termination input voltage level. See also [Programming Vod_Alt and Vt Levels](#). Valid values: 0V to +1.7V
- **Peaking**: The voltage levels setting used to compensate for skin effect loss on the comparator signal.

Value

Enter the value for each parameter as either a numeric value or a formula. If you enter a number, you can use a formula to include engineering units: for example, $=200\text{mV}$.

For the DrvPeaking and CmpPeaking parameters, select one of these values from a pull down menu:

- **None**: No peaking applied. Default.
- **Mid**: Mid level peaking applied.
- **High**: High level peaking applied.

For more information, see [Pin Levels Sheet](#) in the DataTool section.

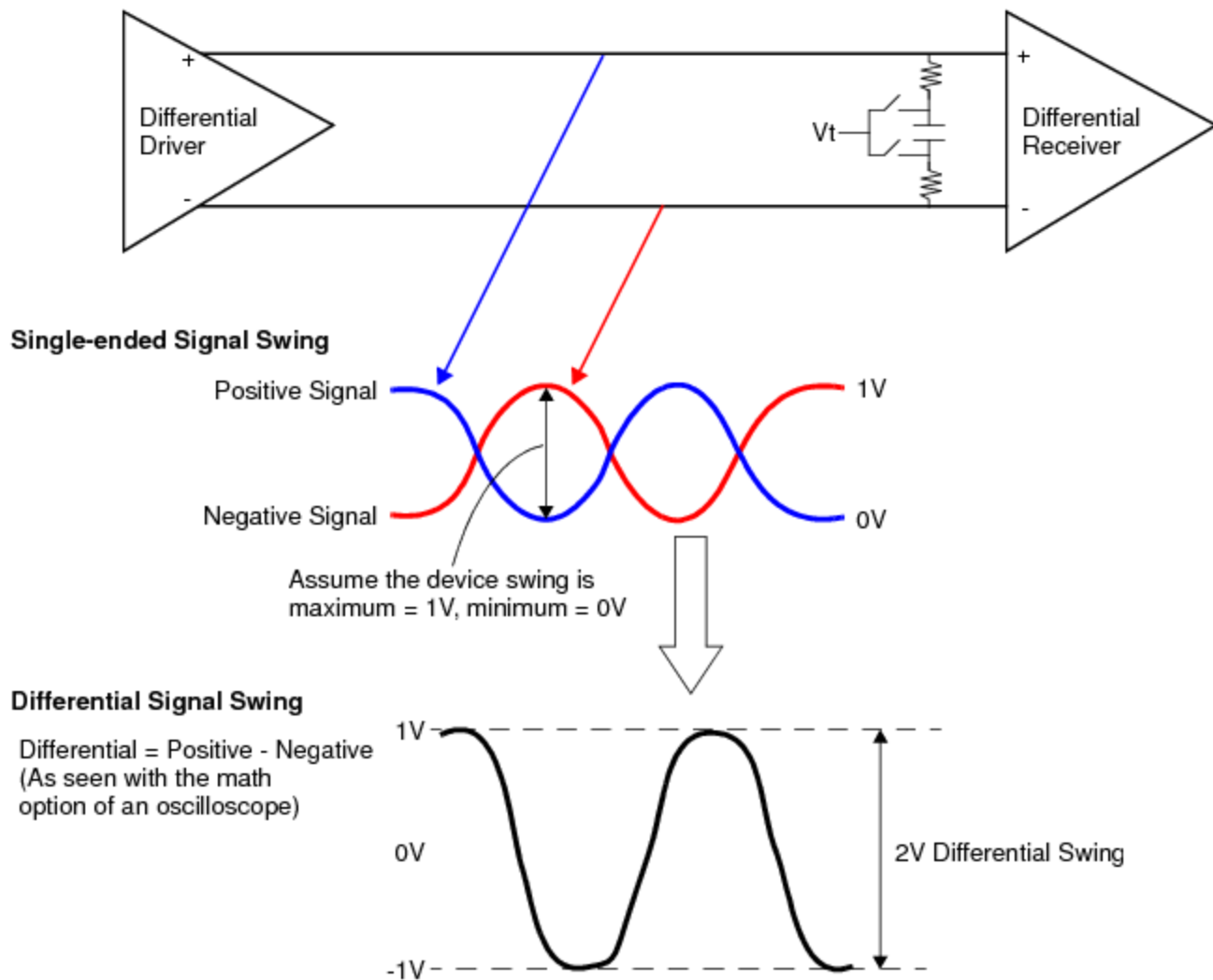
Programming Vod_Alt and Vt Levels

Program device Vod levels for the receive (TX) pins within the limits specified by the following inequality:

$$\text{Vod_Alt1} \leq \text{Vod_Main} \leq \text{Vod_Alt2}$$

Generally, the Alt levels are set to 20% and 80% values with respect to the differential swing seen by the receiver (relative to no offset). This can be made clear with an example.

Assuming a 2V differential swing, as shown in the figure, the 20% and 80% levels are as follows: $(2.0\text{V} * (0.8-0.2)/2)$ implies $\text{Vod_Alt1} = -0.6$ and $\text{Vod_Alt2} = 0.6$.



Computing the Vod_Alt Levels

The V_t parameter is set according to the signal-ended swing values. Consequently, the value should be set to the middle of the single-ended swing. In the example above, this is a setting of 0.5V.

Incremental Levels

The UltraSerial10G instrument supports incremental levels, which allows you to change levels and driver mode for an individual pin or a pin group that has already been defined on the Pin Levels sheet. IG-XL applies the last set of specified levels for a pin or pin group when adding a new pin or pin group, applying and sequencing levels, characterizing pins, and determining specification dependencies. With this flexible programming model, you can reassign new pin levels to existing pins on the sheet while preserving previous pin level setups. You can always revert to previous levels settings and behavior by removing the most recent pin level change on the sheet. For more information, see [Incremental Levels Mode](#).

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UltraSerial10G Programming : [UltraSerial10G DataTool Programming](#) : Programming the Serial Timing

Programming the Serial Timing
Use a Serial Timing sheet to set the period and CDR mode for UltraSerial10G differential pairs.

Serial Timing

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Max Period Error:

Timing Set	Pin/Group	Cycle		Clock Recovery	Start Delays	
		Master Period	DUT Period	Mode	Drive Delay	Receive Delay
serial_timing	RX_ALL		125.E-12		0	
serial_timing	TX_ALL		125.E-12	Train&Track		0

Serial Timing Sheet
For more information, see [Serial Timing Sheet](#) in the DataTool section.
Program the Serial Timing sheet by filling in these parameters:

Max Period Error	Enter a maximum period error value for any UltraSerial10G pin group listed in the Timing Set column. This value is the maximum difference allowed between the value specified in the Cycle DUT Period column and the nearest supported hardware period. Default value is infinite.
Timing Set	Enter a unique set name for the given grouping of timing information.
Pin/Group	Enter a differential pair or differential group to which the timing information applies.
Cycle Master Period	This field is optional. It corresponds to the board level MOSC. If you do not enter a value, it is automatically calculated from the Cycle DUT

Period. Only program this field to override the MOSC value.

Normally, you leave this field blank unless there is a reason to set the MOSC to a specific value. The DUT period programmed is the result of MOSC and MOSC Per Vector (MPV). Available MPVs are listed in [UltraSerial10G Timing](#). Specifying a master period in addition to the DUT period can result in the DUT period being different than desired because only discrete MPVs are available.

All UltraSerial10G driver edges are placed at the effective DUT period. In a few cases, the programmed DUT period may be different from the specified DUT period. One case is when you set the Master Period column on the Serial Timing sheet. In this case, IG-XL uses the Master Period value to set the MOSC, and then programs the closest DUT period by selecting the closest MPV. Another case is when you specify two different DUT periods on the same UltraSerial10G board. In this case, IG-XL selects the MOSC (if not specified in the Master Period field of the Serial Timing sheet) based on the smallest DUT period. IG-XL derives the other DUT period on the board by selecting an MPV closest to the desired DUT period for that MOSC.

For either of these cases, use the [Serial Channel Debug Display](#) or [VBT language](#) (see `TheHdw.Serial.Pins.Timing.Period.Value`) to read back the actual DUT period programmed to confirm it is programmed to be within an acceptable tolerance.

Cycle DUT Period	The period. If more than one DUT period is specified on an UltraSerial10G board, the DUT periods programmed may be different than those shown on the Serial Timing sheet. See UltraSerial10G Timing .
Clock Recovery Mode	The mode of the CDR. This field is required for pin groups that program receive pins. Valid CDR modes are: • Train&Track: Trains the CDR circuit by finding the transition of the incoming data and then placing the strobe at the center of the data eye. Then the CDR circuit starts tracking, where it continually monitors and adjusts to any data movement within the CDR circuitry bandwidth. The UltraSerial10G receive channel requires a training interval of 80s (nominal) to allow the CDR to find the center of the data eye.

- **Disabled:** The CDR circuit is disabled (OFF). In this state the strobe position is programmed using the Start Delays Receive Delay column.

Start Delays Drive Delay

A delay offset for Drive (input) channels on the UltraSerial10G instrument, which will delay when the UltraSerial10G begins driving data to the DUT.

In ISL mode, the user delay occurs after the UltraSerial10G instrument receives the ISL start command.

Specify units of time. IG-XL rounds up to the closest MOSC tick resolution (MOSC period). The maximum drive delay is 1.5s. (This is the MOSC period multiplied by 8K, so taking the slowest MOSC period of 187ps * 8K equals approximately 1.5s. When running the fastest MOSC rate, the maximum delay is smaller.) The default Drive Delay value is 0.

Start Delays Receive Delay

A delay offset for Receive (output) channels on the UltraSerial10G instrument, which will delay when the UltraSerial10G begins receiving data from the DUT.

In ISL mode, the user delay occurs after the UltraSerial10G instrument receives the ISL start command.

If the CDR circuit is Train&Track mode, the user delay occurs before the 80s CDR training interval.

Specify units of time. IG-XL rounds up to the closest MOSC tick resolution (MOSC period). The maximum receive delay is 1.5s. The default Receive Delay value is 0.

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UltraSerial10G Programming : UltraSerial10G DataTool Programming : Programming Serial Pattern Sets

Programming Serial Pattern Sets

All UltraSerial10G pattern execution is defined by a pattern set or a series of pattern sets that you create on the Serial Pattern Sets sheet.

Serial Pattern Sets

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Pattern Set	Pins		Drive Patterns			Receive Patterns				
	Time Domain	Pin/Group	Drive Module	Opcode	Operand	Receive Module	Opcode	Operand	Level Control	Capture
hw_prbs7		RX_ALL	0X00000001	PRBS_7	Inf					
hw_prbs15		RX_ALL	0X00000001	PRBS_15	Inf					
hw_prbs23		RX_ALL	0X00000001	PRBS_23	Inf					
hw_prbs31		RX_ALL	0X00000001	PRBS_31	Inf					
hw_prbs7_all		RX_TX_ALL	0X00000001	PRBS_7	Inf		PRBS_7	5120		
clk_pattern		RX_TX_ALL	clk_drv.pat	Repeat	Inf	PCle_comp_rcv.pat				
pcie_comp		RX_ALL	pcie_comp_scope.pat	Repeat	Inf					
pcie_loopback		RX_TX_ALL	PCle_training_drv.pat	Repeat	500	PCle_comp_rcv.pat				
pcie_loopback		RX_TX_ALL	PCle_comp_drv.pat							
single_shot		RX_TX_ALL	SingleShot_drv.pat			PCle_comp_rcv.pat				
drv_only		RX_ALL	PCle_training_drv.pat	Repeat	500					
drv_only		RX_ALL	PCle_comp_drv.pat	Repeat	Inf					
rcv_only		TX_ALL				PCle_comp_rcv.pat				Capture
loopback		TX_ALL					Loopback	38400		Capture
loopback_oob		TX_ALL					Loopback_Oob	Inf		NoCapture

Serial Pattern Sets Sheet

Program the Serial Pattern Sets sheet by filling in these parameters:

Pattern Set	The unique name for each pattern set. A single serial pattern set represents a single pattern run by the UltraSerial10G PatGen.
Time Domains	Not used.
Pin/Group	The name of the serial pin group.

Drive Patterns Drive Module

The UltraSerial10G drive pattern. The options are:

- **<File>**: The pattern file to be executed. If a pattern file contains multiple modules, then all modules in the file are executed.
- **<File>:<Module>**: The pattern module to be executed.
- **<0XXXXXXXX>**: The drive PRBS seed value (in hex format), which is only valid with an opcode field entry of PRBS.
- **[Empty]**: No pattern will be executed.

Drive Patterns Opcode

The opcode for pattern execution. Entries for this field are:

- **[Empty]**: Executes the pattern and moves to the next entry. Default.
- **Repeat**: Repeat the pattern a specified number of times (finite) or forever (infinite). Infinite repeat repeats only the last module in a pattern file. You cannot use the Repeat opcode on modulo 40 patterns.
- **PRBS_31**: Executes a PRBS pattern using the 31-bit PRBS hardware.
- **PRBS_23**: Executes a PRBS pattern using the 23-bit PRBS hardware.
- **PRBS_15**: Executes a PRBS pattern using the 15-bit PRBS hardware.
- **PRBS_7**: Executes a PRBS pattern using the 7-bit PRBS hardware.
- **NoData**: Allocates a temporary module of unknown pattern data. Not supported on UltraSerial10G.

Drive Patterns Operand

The execution behavior of an opcode. Some opcodes require an operand, and other opcodes do not. Entries for this field are:

- [\[Empty\]](#): Is equivalent to Nop. Default.
- [Number](#): For the Repeat opcode, Number is the number of times to repeat the drive module pattern. The drive pattern will be executed Number + 1 times.

For the PRBS opcodes, Number represents the length of the PRBS pattern (modulo 2560 or 40).

For the NoData opcode, Number represents the number of UltraSerial10G cycles to allocate (modulo 2560 or 40); the maximum value is 256Mb.

For a pattern set with only one pattern or the last module in a pattern set, you can use modulo 40. You cannot use the Repeat opcode on modulo 40 patterns.

- [Inf](#): Repeats the drive module pattern an infinite number of times. When using the Inf operand, the entry must be the last in the pattern set.

Receive Patterns Receive Module

The UltraSerial10G receive pattern. The options are:

- [<File>](#): The pattern file to be executed. If a pattern file contains multiple modules, then all modules in the file are executed.
- [<File>:<Module>](#): The pattern module to be executed.
- [\[Empty\]](#): No pattern will be executed.

Receive Patterns Opcode

The opcode for pattern execution. Entries for this field are:

- [\[Empty\]](#): Executes the pattern and moves to the next entry. Default.
- [Repeat](#): Repeat the pattern a specified number of times (finite) or forever (infinite). Infinite repeat repeats only the last module in a

pattern file. You cannot use the Repeat opcode on modulo 40 patterns.

- [PRBS_31](#): Executes a PRBS pattern using the 31-bit PRBS hardware.
- [PRBS_23](#): Executes a PRBS pattern using the 23-bit PRBS hardware.
- [PRBS_15](#): Executes a PRBS pattern using the 15-bit PRBS hardware.
- [PRBS_7](#): Executes a PRBS pattern using the 7-bit PRBS hardware.
- [NoData](#): Allocates a temporary module of unknown pattern data. Not supported on UltraSerial10G.

Receive Patterns Operand

The execution behavior of an opcode. Some opcodes require an operand; other opcodes do not require an operand. Entries for this field include:

- [\[Empty\]](#): Is equivalent to Nop. Default.
- [Number](#): For the Repeat opcode, Number is the number of times to repeat the drive module pattern. The drive pattern will be executed Number + 1 times.

For the PRBS opcodes, Number represents the length of the PRBS pattern (modulo 2560 or 40). The operand range is 2560 to 1,342,177,280.

For the NoData opcode, Number represents the number of UltraSerial10G cycles to allocate (modulo 2560 or 40); the maximum value is 256 Mbits.

For a pattern set with only one pattern or the last module in a pattern set, you can use modulo 40. You cannot use the Repeat opcode on modulo 40 patterns.

- [Inf](#): Repeats the drive module pattern an infinite number of times. When using the Inf operand, the entry must be the last in the pattern set.

Receive Patterns Level Control

The Vod level setting to choose during pattern execution. Entries for this field include:

- [\[Empty\]](#): The default value of Vod_Main is used.
- [Vod_Main](#): Selects main voltage as the receive level of the pattern. Vod_Main is programmed on the Pin Levels sheet. Not supported on UltraSerial10G.
- [Vod_Alt1](#): Selects alternate voltage 1 as the receive level of the pattern. Vod_Alt1 is programmed on the Pin Levels sheet. Not supported on UltraSerial10G.
- [Vod_Alt2](#): Selects alternate voltage 2 as the receive level of the pattern. Vod_Alt2 is programmed on the Pin Levels sheet.

Receive Patterns Capture

Specifies whether to capture data in Capture Memory for the pattern. Entries for this field include:

- [Capture](#): Captures data in Capture Memory
- [NoCapture](#): Does not capture data in Capture Memory

The Capture parameter is only used when the Capture Memory is set to Capture Mode in VBT language. When Capture Memory is set to Mask Mode, this parameter is ignored.

Note:

You cannot mix PRBS programming, loopback programming, and pattern file/module programming for the same pins in the same pattern set.

Pin Validation Rules

UltraSerial10G pins and pin groups must be consistent when specified in the Serial Pattern Sets sheet and in pattern files. Listed below are the definitions of the different pin designations used in the validation rules.

- Pattern set pins: Pin groups listed in the Pin/Group column of the Serial Pattern Sets sheet.
- Pattern file pins: Pin groups listed in the UltraSerial10G pattern files.
- Pattern set pins group: All instances of a single pattern set pin within a pattern set.
- Pattern run pins: Pins that will be run either by a databus start or ISL command in the primary PatGen.

The validation rules are listed in the table.

Pin Validation Rules

Rule	Error Type
1. For any two different pattern set pins within a set, no pins may overlap.	Validation
2. Within a pattern set pins group, all pins in a drive or receive pattern file must have the same pattern file pins.	Run-time
3. All pattern file pins in a pattern set pins group must be members of the pattern set pins.	Run-time
4. When loading concurrent pattern sets, all pattern set pins among any two sets cannot have any overlapping pins.	Run-time

Run-time

5. You cannot reuse pattern set pins for different groups of pins in pattern files within a pattern set (see rule 3).

Run-time

6. All pattern files listed in the Drive Module Serial Pattern Sets sheet column must contain only drive pins. Likewise, all pattern files listed in the Receive Module Serial Pattern Sets sheet column must contain only receive pins.

Run-time

7. Pattern run pins must exist 1) in the pattern file pins from the last loaded pattern set, or 2) if the UltraSerial10G pattern set is not using files (i.e., PRBS, NoData), then the pattern run pins are checked against the pattern set pins.

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UltraSerial10G Programming : UltraSerial10G DataTool Programming : Programming UltraSerial10G Serial Jitter

Programming UltraSerial10G Serial Jitter

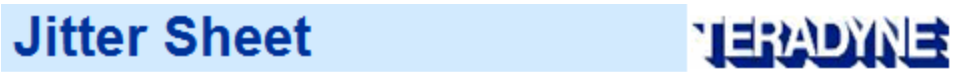
Jitter is a measure of the variation of edges from their ideal positions. Use the Jitter sheet to set up jitter insertion on UltraSerial10G drive pins and jitter measurement on UltraSerial10G receive pins. This section is divided into the following topics:

- Jitter Insertion
- Jitter Measurement

See also UltraSerial10G Jitter and Eye Measurement Test Programming.

Jitter Insertion

The UltraSerial10G can insert jitter on the UltraSerial10G drive pins using VM, PRBS, or loopback. The jitter insertion waveform is a sine wave. You can program the period and amplitude of the jitter. The maximum jitter that can be inserted is a function of data rate and jitter frequency. You program serial jitter insertion for UltraSerial10G drive pins using a Jitter sheet. On a given board, jitter will be present at all or none of the drive pins.



		Insertion		
Jitter Set	Pin/Group	Mode	Period	Amplitude in UI
jitter_insertion	RX_ALL	Added	10.E-06	300.E-03

Jitter Sheet (Insertion Section)

To insert jitter on UltraSerial10G drive pins, program these sheet parameters:

Jitter Sheet Insertion Parameters

Parameter	Description
Jitter Set	A unique set name for the given grouping of serial jitter information.
Pin/Group	The differential pair or differential group to which the serial jitter information applies. This field must contain a pin of type input when specifying a jitter insertion mode.
Mode	The jitter insertion mode which determines how the jitter insertion amplitude is interpreted. Entries for this field include: Added: The jitter seen at the device will equal the specified amplitude (in UI) plus the jitter inherent to the tester. Therefore, this jitter may be more than the specified amplitude. Disabled: The jitter insertion circuitry is disabled.
Period	The period of the jitter insertion sine wave. The range is 10ns (1/100 MHz) to 10s (1/100 kHz).
Amplitude in UI	The amplitude of the jitter insertion sine wave, which is a function of the currently selected DUT UI and the jitter insertion frequency. The maximum amplitude that can be inserted is a function of data rate and jitter frequency. See Jitter Insertion in the UltraSerial10G specifications.

For more information, see Jitter Insertion in the UltraSerial10G specifications.

Jitter Measurement

The UltraSerial10G uses the parameters provided on a Jitter sheet to determine walking strobe programming for a given jitter or eye measurement setup.

Jitter Sheet		TERADYNE								
		Insertion			Measurement					
Jitter Set	Pin/Group	Mode	Period	Amplitude in UI	Sample Type	Target Resolution	Target Pattern Reps	Bits Per Pattern	Min Pattern Skips	Bit Period
jitter_measure	TX_ALL	Disabled			One Level	500.E-15	4	127		
					Disabled One Level Three Level Custom One Level Custom Three Level					

Jitter Sheet (Measurement Section)

To measure jitter on UltraSerial10G receive pins, program these sheet parameters:

Jitter Sheet Measurement Parameters

Parameter	Description
Jitter Set	A unique set name for the given grouping of serial jitter information.
Pin/Group	The differential pair or differential group to which the serial jitter information applies. This field must contain a pin of type output when specifying a jitter sample type (measurement mode).
Sample Type	The jitter and eye measurement modes. Entries for this field include: • Disabled: No jitter or eye measurement is performed on this pin or pin group. • One Level: A jitter measurement is to be performed on this pin or pin group. • Three Level: An eye measurement is to be performed on this pin or pin group. • Custom One Level: A custom jitter measurement is to be performed on this pin or pin group. • Custom Three Level: A custom eye measurement is to be performed on this pin or pin group.
Target Resolution	The effective sampling frequency. The minimum value for this field is 50 fs. See Guidelines for Jitter Parameters. Using a smaller resolution increases the number of samples taken. This provides a more accurate result but increases the measurement and analysis time. Using a larger resolution decreases the number of samples taken. This provides a less accurate result but decreases the measurement and analysis time. Follow this guideline: Target resolution should be no more than 1/6 the expected measured RMS (Rj) value. For example, if you expect a minimum of 6ps Rj RMS out of the DUT, then set the target resolution to $<6\text{ps}/6$, which $<1\text{ps}$.

Target Pattern Reps	The number of times to sample the repeating pattern from the DUT. Using a higher number of reps increases the number of samples taken. This provides a more accurate result but increases the measurement and analysis time. Using a smaller number of reps decreases the number of samples taken. This provides a less accurate result but decreases the measurement and analysis time. See Guidelines for Jitter Parameters.
Bits per Pattern	The length of the repeating pattern being sampled, in an integer number of bits.
Min Pattern Skips	The minimum number of pattern lengths to skip between subsequent effective samples. See How Pattern Skips Affect a Measurement.
Bit Period	The actual DUT output period of the pattern to be measured. UltraSerial10G does not use this column since the period value is extracted from the Serial Timing sheet.

Guidelines for Jitter Parameters

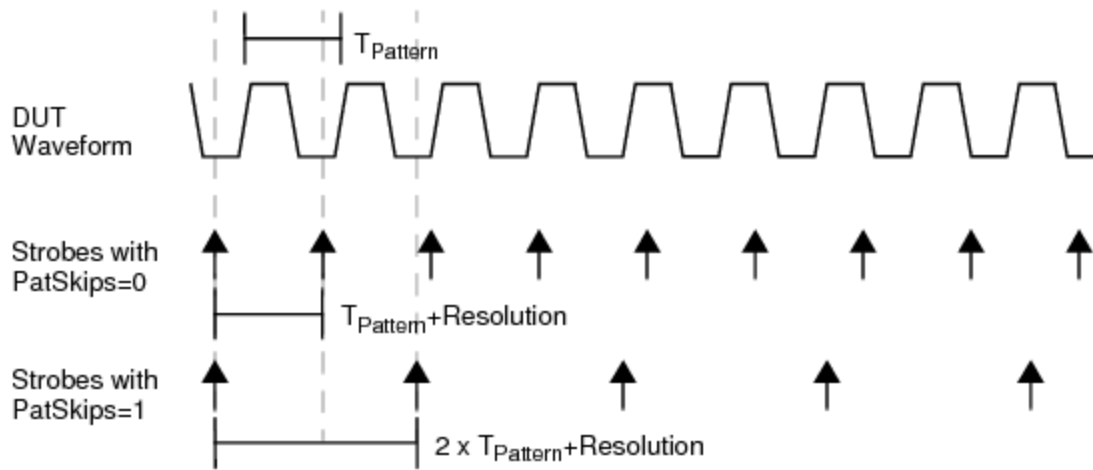
Use the following guidelines for setting jitter parameters used for making jitter and jitter eye measurements. These guidelines have proven helpful in achieving good jitter measurement results.

- Target resolution should be no more than 1/6 the expected measured RMS (Rj) value.
- Target pattern reps for measure jitter and eye tests:

Target reps = 250/number of bits per pattern (rounded-up)

How Pattern Skips Affect a Measurement

Pattern skips provide control over the bandwidth of jitter and eye measurements. This is done by modifying the time required to acquire certain information about the DUT waveform, while maintaining the number of samples and target resolution of the measurement. The diagram illustrates how the strobing is modified for two different settings of pattern skips.



Modified Strobes for Different Settings of Pattern Skips (resolution typically finer than shown)

The frequency response of jitter and eye measurements is like that of a high-pass filter. Jitter with harmonic content below F_c (the cutoff frequency) is ignored by the measurement, much like a serial receiver will track jitter below its specified F_c .

If PatternSkips = 0 for a short pattern, such as that of a clock, the F_c of the measurement may be so high that little or no jitter is measured. Consequently, pattern skips are most often used for short patterns.

Pattern skips have only a small effect on DDj measurements.

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UltraSerial10G Programming : UltraSerial10G DataTool Programming : Programming UltraSerial10G Test Instances

Programming UltraSerial10G Test Instances

You can use the instance editor to create and edit Functional_T, PinPMU_T, and other custom VBT test instances. A test instance defines a test that is applied to the DUT when the test program runs. You can create as many test instances as you want. The Test Instances sheet lists all the test instances for a test program.

To create or edit a Functional_T or PinPMU_T test instance in the instance editor, see [Creating and Editing Test Instances](#).

Once you have created a test instance, program the Functional_T test instance by filling in the parameter fields. For a reference to the Functional_T test parameters, see [Functional VBT Test](#). For a reference to the PinPMU_T test parameters, see [Pin PMU VBT Test](#).

To create custom VBT test, see [Creating VBT Tests](#) in the VBT Syntax Reference.

You specify the timing sets to use for a test on the Test Instances sheet, which include serial timing and serial jitter for UltraSerial10G programming.

Test Instances

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Test Procedure				DC Specs		AC Specs		Sheet Parameters				
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
RXTX_func	VBT	RXTX_Func		func_spec	nom	3125Mbps	nom	Func TSB,Serial Timing		Levels		

Test Instances Sheet

Instance Editor [VBT Test 'RXTX_Func']

Instance: RXTX_func

Parameters | Timing and Levels | Limits

Name	Value	Comment
☐ hsd_pattern	.\patterns\hsd_drive_TD_receive_RD_serdes.PAT	
serial_patset	RXTX_func_patset	
tx_pins	TX	
ten_bit_only_flag	False	
ten_bit_symbol	K28.5+	
wait	0.01	

VBT Instance Editor Parameters Tab

Instance Editor [VBT Test 'RXTX_Func']

Instance: RXTX_func

Parameters | Timing and Levels | Limits

Time Set: Func TSB AC Selector: nom

Edge Set: AC Category: 3125Mbps

Pin Levels: Levels DC Selector: nom

Mixed-Sig: DC Category: func_spec

Overlay: Serial Timing: Serial Timing

Jitter:

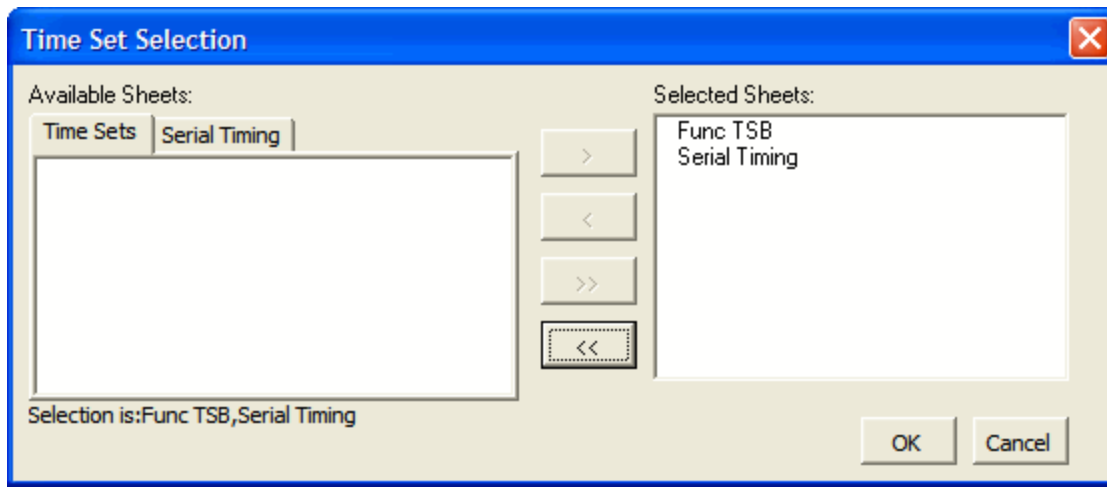
Time Domains: ☐

Time Domain

Add Remove

VBT Instance Editor Timing and Levels Tab

The Time Sets column specifies a comma-separated list of time sets to use for this test, including UltraSerial10G and other digital instruments. Valid time sets include Time Set, Serial Timing, and Serial Jitter. To select serial timing and serial jitter, right click in the Time Sets column and select Edit. This opens the Time Set Selection window. The Time Set Selection window allows you to select Serial Timing and Jitter sheets to use for this test.



Time Set Selection Window

			
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UltraSerial10G Programming : UltraSerial10G Pattern Control

UltraSerial10G Pattern Control

Patterns contain the data that digital channels send to the DUT. All patterns in a burst must contain the same list of instruments.

This section describes how to create UltraSerial10G test patterns in an ASCII text file using digital pattern microcodes. This section includes the following topics:

- | | |
|---|---|
| • | Header Statements |
| • | Pattern Module Statements |
| • | Controlling the UltraSerial10G in HSD Pattern Files |
| • | Pattern Execution |

UltraSerial10G pattern file structure is similar to HSD pattern file structure but has some additional syntax and rules:

- | | |
|---|--|
| • | Drive and receive pin data states must exist in different pattern files. See also Pin Validation Rules for other pin related restrictions. |
| • | Each pattern module must contain a minimum of 2560 vectors and a maximum of 32M vectors. |

- Each pattern module must be a multiple of 2560 vectors in length. For a pattern set with only one pattern or the last module in the pattern set, the multiple can be 40 vectors.
- Opcodes are not supported in the pattern.
- Patterns cannot contain time sets.

The UltraSerial10G uses the Serial10G compiler option on the standard pattern compiler GUI.

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UltraSerial10G Programming : [UltraSerial10G Pattern Control](#) : Header Statements

Header Statements

The [UltraSerial10G](#) pattern must define the digital instrument type for the pattern compiler using the `digital_inst` statement:

```
digital_inst = Serial10G;
```

The `Serial10G` keyword identifies the pins as `UltraSerial10G` pins.

You must specify the [UltraSerial10G](#) pattern compiler options when compiling for the `UltraSerial10G` instrument.

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UltraSerial10G Programming : UltraSerial10G Pattern Control : Pattern Module Statements

Pattern Module Statements

You can only program vector pin data as differential pin pairs (not single-ended pins) in a UltraSerial10G pattern. Define vector pin data using the `vm_vector` statement:

`vm_vector UltraSerial10GPattern (UltraSerial10G_PINS)`

Where UltraSerial10G_PINS can be an UltraSerial10G pin group or a comma-separated list of UltraSerial10G pin groups.

Pattern Data States

The UltraSerial10G instrument supports drive and receive pin data states. It uses the data states listed below.

UltraSerial10G Drive Pattern Data States

Pin State	Drive
0	Low
1	High
X	Midband

When an UltraSerial10G drive pin is programmed to midband (X), both single-ended pins of the differential pair drive the same midband level. Drive midband must start on a modulo 10 boundary and occur in blocks of 10. If drive midband occurs in a pattern module, any one pin cannot have 10 consecutive zeros starting from a mod-10 boundary. If any of these rules are broken, the pattern will not compile.

UltraSerial10G Receive Pattern Data States

Pin State	Expect
L	Low
H	High
X	Don't Care

When an UltraSerial10G compare pin is programmed to don't care (X), compare masking must start on a modulo 10 cycle boundary and occur in blocks of 10 cycles. If any cycle in a block of 10 cycles has an X in it, then IG-XL masks that full block of 10 cycles.

Pattern Labels

UltraSerial10G supports two types of vector labels: local labels and global labels. Local labels are used as landmarks inside a pattern and are not visible to IG-XL. Global labels are used to modify vector memory at a particular label and are visible to IG-XL. The UltraSerial10G cannot be started at an offset into a pattern module.

```
PATTERN_LABEL:    > .H .L ;
```

```
global PATTERN_LABEL: > .H .L ;
```

Where PATTERN_LABEL is a user defined label name.

Example UltraSerial10G Drive Pattern File

```
pinmap_workbook = ".\example.xls";
```

```
digital_inst = Serial10G;
```

```
vm_vector rx_mod (RX)
```

```
{
global rx_start:
> .0 ;
> .1 ;
> .0 ;
> .1 ;
> .1 ;
> .0 ;
> .0 ;
> .1 ;
> .0 ;
> .1 ;
...
> .0 ; // Must be a multiple of 2560 or 40
}
```

Example UltraSerial10G Receive Pattern File

```
pinmap_workbook = ".\example.xls";
digital_inst = Serial10G;

vm_vector tx_mod (TX)
{
global tx_start:
> .L ;
> .H ;
> .L ;
> .H ;
> .H ;
> .L ;
> .L ;
> .H ;
> .L ;
> .H ;
...
> .L ; // Must be a multiple of 2560 or 40
}
```

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UltraSerial10G Programming : UltraSerial10G Pattern Control : Controlling the UltraSerial10G in HSD Pattern Files

Controlling the UltraSerial10G in HSD Pattern Files

When ISL mode is used, the HSD PatGen sends ISL commands to the UltraSerial10G instrument to start or stop it. The ISL command is sent to the UltraSerial10G instrument, and the pins begin pattern execution approximately 4.0s after the command is issued. Multiple ISL commands must be executed a minimum of 3.0s apart.

For information on programming HSD pattern files, see:

- HSD1000 Test Pattern Programming
- UltraPin800 Test Pattern Programming

Instrument Statement

The HSD pattern must define UltraSerial10G differential pins to which the ISL commands will be routed. UltraSerial10G pins are defined using an instruments statement:

```
instruments = {  
  (UltraSerial10G_PINS):Serial10G;  
}
```

Where UltraSerial10G_PINS can be a UltraSerial10G pin group or a comma-separated list of UltraSerial10G pin groups. The Serial10G keyword identifies the pins as UltraSerial10G pins.

For more information, see Instruments Statement in the Pattern Language section.

UltraSerial10G Pattern Microcodes

The UltraSerial10G instrument can receive two ISL commands: START and STOP.

The following table lists ISL microcodes that are valid for HSD vectors.

ISL Microcodes

Microcode	Description
(UltraSerial10G_PINS):Serial10G = START	Start executing the UltraSerial10G pattern data for the specified UltraSerial10G drive and receive pins.
(UltraSerial10G_PINS):Serial10G = STOP	Stop executing the UltraSerial10G pattern data for the specified UltraSerial10G drive and receive pins.

Example: Controlling PatGen Pattern File

The following example shows an HSD1000 pattern that sends a START and STOP microcode to a UltraSerial10G instrument.

```
digital_inst = HSDM;
pinmap_workbook = "example.xls";
opcode_mode = dual;
import tset tset_hsd;

instruments = { ( TX, RX):Serial10G; }

vm_vector (          $tset  TDA          ...  RD  controls)
{
/* Start UltraSerial10G drive and receive */
    > tset_hsd 0011001100  ...  .X  .0;
((TX, RX):Serial10G = Start) > tset_hsd 0011001100  ...  .X  .0;
...
/* Stop UltraSerial10G drive and receive */
    > tset_hsd 0011001100  ...  .X  .0;
((TX, RX):Serial10G = Stop) > tset_hsd 0011001100  ...  .X  .0;
...
halt          > tset_hsd 0010011001  ...  .X  .0;
}
```

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UltraSerial10G Programming : UltraSerial10G Pattern Control : Pattern Execution

Pattern Execution

You can start UltraSerial10G pattern sets in two ways:

- | |
|--|
| <ul style="list-style-type: none">Synchronously using HSD PatGen with ISL commands |
| <ul style="list-style-type: none">Asynchronously using UltraSerial10G PatGen start |

When ISL commands are used, the HSD pattern starts the UltraSerial10G pattern. When UltraSerial10G PatGen start is used, the UltraSerial10G pattern is executed independent of the HSD pattern.

Each pattern set represents a single burst. Patterns are executed in the order in which they appear in the pattern set. When the UltraSerial10G PatGen starts, there are no intermediate stops. This implies that you must start the UltraSerial10G and wait for it to complete its execution.

For more information on creating interpose functions to start the UltraSerial10G pattern sets in a functional test, see [UltraSerial10G Functional Test Programming](#).

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UltraSerial10G Programming : UltraSerial10G Functional Test Programming

UltraSerial10G Functional Test Programming

To run an UltraSerial10G functional test, you can create an interpose function that sets up the UltraSerial10G instrument. This interpose function is then called from the functional template that executes the HSD pattern. Alternatively, you can create a VBT function that executes the HSD and UltraSerial10G patterns.

This section describes how to create an interpose function that runs an UltraSerial10G pattern based on the two ways of starting UltraSerial10G pattern sets:

- | | |
|--|---|
| | <ul style="list-style-type: none">• Pattern Execution Using ISL Start |
| | <ul style="list-style-type: none">• Pattern Execution Using UltraSerial10G PatGen Start |
| | <ul style="list-style-type: none">• Pass/Fail Determination for PatGen Start |

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UltraSerial10G Programming : UltraSerial10G Functional Test Programming : Pattern Execution Using ISL Start

Pattern Execution Using ISL Start
Use the following steps to run a UltraSerial10G pattern using ISL start:

1. Apply and connect HSD and UltraSerial10G levels and timing:

TheHdw.Digital.ApplyLevelsTiming

2. Set up receive align trigger for UltraSerial10G receive pins:

TheHdw.Serial.Pins(serial_rcv_pins).Patgen.ReceiveAlign.TriggerMode _
 (TriggerMode, 10bSymbol)
TheHdw.Serial.Pins(serial_rcv_pins).Patgen.ReceiveAlign.TimeoutEnable = _
 True
TheHdw.Serial.Pins(serial_rcv_pins).Patgen.ReceiveAlign.Timeout = 1*msec

3. Load HSD patterns:

TheHdw.Patterns(pattern).Load

4. Load UltraSerial10G patterns:

TheHdw.Serial.Patterns(serial_patset).Load
The serial .Load statement must be called before each HSD pattern burst to set up the correct UltraSerial10G pattern set that will be executed from the HSD ISL start microcode.

5. [Run the HSD pattern \(which issues the ISL command\) to start the UltraSerial10G pattern and test pass fail status for both HSD and UltraSerial10G patterns:](#)

[TheHdw.Patterns\(pattern\).Test\(SetPassFail, StopOnFirstFail\)](#)

Example

Public Function UltraSerial10G_func(hsd_pat As String, patset As String, TX_pins _
As String, RAT_timeout As Double, symbol_10b As String)

On Error GoTo ErrorHandler

' Apply HSDM & UltraSerial10G Levels and Timing

TheHdw.Digital.ApplyLevelsTiming True, True, True, tlPowered

' Set the RAT mode to 10b20b

TheHdw.Serial.Pins(TX_pins).Patgen.ReceiveAlign.TriggerMode tlSerialReceiveAlign10b20b,
symbol_10b

' Set RAT to timeout

TheHdw.Serial.Pins(TX_pins).Patgen.ReceiveAlign.TimeoutEnable = True

TheHdw.Serial.Pins(TX_pins).Patgen.ReceiveAlign.TimeOut = RAT_timeout

' Load HSDM pattern

TheHdw.Patterns(hsd_pat).Load

' Load UltraSerial10G patterns

TheHdw.Serial.Patterns(patset).Load

' Start HSD pattern that bursts UltraSerial10G using ISL and determines

' pass/fail result

TheHdw.Patterns(hsd_pat).test(pfAlways)

UltraSerial10G_func = tl_SUCCESS

Exit Function

ErrorHandler:

Call TheExec.LogErrorMessage("Unexpected error " & Err.Number & _
" in VBT function UltraSerial10G_func." & vbCrLf & Err.Description)

UltraSerial10G_func = TL_ERROR

End Function

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UltraSerial10G Programming : UltraSerial10G Functional Test Programming : Pattern Execution
Using UltraSerial10G PatGen Start

Pattern Execution Using UltraSerial10G PatGen Start
Use the following steps to run an UltraSerial10G pattern using PatGen start:

1. Apply and connect HSD and UltraSerial10G levels and timing:

TheHdw.Digital.ApplyLevelsTiming

2. Set up receive align trigger for UltraSerial10G receive pins:

TheHdw.Serial.Pins(serial_rcv_pins).Patgen.ReceiveAlign.TriggerMode _
 (TriggerMode, 10bSymbol)
TheHdw.Serial.Pins(serial_rcv_pins).Patgen.ReceiveAlign.TimeoutEnable = _
 True
TheHdw.Serial.Pins(serial_rcv_pins).Patgen.ReceiveAlign.Timeout = 1*msec

3. Load HSD patterns:

TheHdw.Patterns(pattern).Load

4. Load UltraSerial10G patterns:

TheHdw.Serial.Patterns(serial_patset).Load

5. Start the HSD pattern:

[TheHdw.Patterns\(pattern\).Start](#)

6. [Start the UltraSerial10G pattern:](#)

[TheHdw.Serial.Patterns\(serial_patset\).Start](#)

7. [Wait for the UltraSerial10G pattern to finish:](#)

[TheHdw.Serial.Patgen.HaltWait](#)

8. [Wait for the HSD pattern to finish:](#)

[TheHdw.Digital.Patgen.HaltWait](#)

[UltraSerial10G PatGen Start Example](#)

Public Function UltraSerial10G_func(hsd_pat As String, patset As String, tx_pins As String, RAT_timeout As Double, _ symbol_10b As String)

On Error GoTo ErrorHandler

' Apply HSD & UltraSerial10G Levels and Timing

TheHdw.Digital.ApplyLevelsTiming True, True, True, tlPowered

' Set the RAT mode to 10b20b

TheHdw.Serial.Pins(tx_pins).Patgen.ReceiveAlign.TriggerMode tlSerialReceiveAlign10b20b, symbol_10b

' Set RAT to timeout

TheHdw.Serial.Pins(tx_pins).Patgen.ReceiveAlign.TimeoutEnable = True

TheHdw.Serial.Pins(tx_pins).Patgen.ReceiveAlign.TimeOut = RAT_timeout

' Load HSD Pattern

TheHdw.Patterns(hsd_pat).Load

' Load UltraSerial10G Patterns

TheHdw.Serial.Patterns(patset).Load

' Start HSD Pattern

TheHdw.Patterns(hsd_pat).Test

' Start UltraSerial10G Pattern

TheHdw.Serial.Patterns(patset).Start

UltraSerial10G_func = tl_SUCCESS

Exit Function

ErrorHandler:

Call TheExec.LogErrorMessage("Unexpected error " & Err.Number & _
" in VBT function UltraSerial10G_func." & vbCrLf & Err.Description)

UltraSerial10G_func = TL_ERROR

End Function

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UltraSerial10G Programming : UltraSerial10G Functional Test Programming : Pass/Fail
Determination for PatGen Start

Pass/Fail Determination for PatGen Start
If the PatGen start command is used to execute the pattern, use the following statements to determine if the UltraSerial10G and HSD patterns passed or failed and log the results:

- Test if the UltraSerial10G pattern passed:

TheHdw.Serial.Patgen.PatternBurstPassed(site)

- Test if the HSD pattern passed:

TheHdw.Digital.Patgen.PatternBurstPassed(site)

- Log the functional result:

TheExec.Datalog.WriteFunctionalResult(SiteNum, TestNum, result)

- Set the site result:

TheExec.Sites.Site(site).TestResult
See UltraSerial10G PatGen Start Example.

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UltraSerial10G Programming : Capturing and Processing Data

Capturing and Processing Data

The UltraSerial10G can capture data into Capture Memory and automatically move the captured data to a DSP computer for real-time processing.

To move captured data to the DSP computer, you must create a DspWave object that points to the UltraSerial10G Capture Memory DSP Wave. For more information, see [TheHdw.Serial.Pins.Patterns.CaptureMemory.DSPWave](#).

In addition, you must program the Capture Memory mode to Move the data to the DSP computer. See [TheHdw.Serial.Pins.Patterns.CaptureMemory.CaptureConfig](#) for details. Both of these steps must be done before the pattern is executed.

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UltraSerial10G Programming : Capturing and Processing Data : DSP Example

DSP Example

The following code shows the VBT needed to set up capture memory, run the UltraSerial10G pattern, and use the DSP computer to process the captured data.

Capture Memory Setup

The DSPSetup function is used to create a pointer to the DSP wave and configure capture memory to move the captured data to the DSP computer.

Public Function DSPSetup(testName As String, PDEWave As String, Capture_pin _
As String)

On Error GoTo ErrorHandler

' Create a pointer to the DSPWave

Dim DSPWaves As New PinListData

DSPWaves = TheHdw.Serial.Pins(Capture_pin).Patterns.CaptureMemory.DSPWave

' Bind the DSPWave object to the PDEWave variable

thevars(PDEWave) = DSPWaves.Pins(0)

' Configure the capture memory to enable MoveLink.

TheHdw.Serial.Pins(Capture_pin).Patterns.CaptureMemory.CaptureConfig = _
tlSerialCaptureConfigMove

DSPSetup = tl_SUCCESS

Exit Function

ErrorHandler:

Call WriteMessage(testName & ": " & "Unexpected error " & Err.number & " in _
VBT function DSPSetup." & vbCrLf & Err.Description)

DSPSetup = tl_ERROR

End Function

UltraSerial10G Pattern Execution

The DSP_Test function is used to execute the UltraSerial10G pattern. After execution of this function, the data is stored in capture memory.

Public Function DSP_Test(HSDPattern As String, RecPattern As String, _
Capture_pin As String, ten_bit_a As String)

On Error GoTo ErrorHandler

' Connect all pins and Apply Levels and Timing to HSD and UltraSerial10G

TheHdw.Digital.ApplyLevelsTiming True, True, True, tlPowered

' Set the 10b symbol

Call TheHdw.Serial.Pins(Capture_pin).Patgen.ReceiveAlign. _

TriggerMode(tlSerialReceiveAlign10b20b, ten_bit_a)

' Set RAT to timeout after 1 ms

TheHdw.Serial.Pins(Capture_pin).Patgen.ReceiveAlign.TimeoutEnable = True

TheHdw.Serial.Pins(Capture_pin).Patgen.ReceiveAlign.Timeout = 0.01

' Load HSD pattern

TheHdw.Patterns(HSDPattern).Load

' Load UltraSerial10G pattern set

TheHdw.Serial.Patterns(RecPattern).Load

' Start HSD pattern that has an ISL command to start UltraSerial10G

TheHdw.Patterns(HSDPattern).Start

' Wait for Digital PG Burst

TheHdw.Digital.Patgen.HaltWait

' Wait for Serial PG Burst

TheHdw.Serial.Patgen.HaltWait

DSP_Test = tl_SUCCESS

Exit Function

ErrorHandler:

Call WriteMessage("Unexpected error " & Err.number & " in VBT function _
DSP_Test." & vbCrLf & Err.Description)

DSP_Test = tl_ERROR

End Function

DSP Processing

The Functional_DSP_test function is used to count the number of "1's" in the captured data.

✓ Note:

DSP procedures must be saved in VBT modules that begin with the prefix DSP.

Public Function Functional_DSP_test(wave As DSPWave, ByRef N1 As Long) As Long

' This function counts number of 1s in the capture data.

Dim DspSize As Long

Dim index As Long

```
Dim i As Long
```

```
Dim counter As Long
```

```
Dim Binary_array() As Integer
```

```
counter = 0
```

```
' Find out how many elements of 32 bit wide data are in the DSPWave object.
```

```
DspSize = wave.SampleSize
```

```
' Resize the data array to support the number of samples.
```

```
ReDim Binary_array(DspSize * 32)
```

```
' Count the number of 1s in the data.
```

```
For index = 0 To DspSize - 1
```

```
    For i = 31 To 0 Step -1
```

```
        If i <> 31 Then
```

```
            Binary_array(index * 32 + i) = IIf(wave.Element(index) And (2 ^ _  
                i), 1, 0)
```

```
            counter = counter + IIf(wave.Element(index) And (2 ^ i), 1, 0)
```

```
        Else
```

```
            Binary_array(index * 32 + i) = IIf(wave.Element(index) And _  
                &H80000000, 1, 0)
```

```
            counter = counter + IIf(wave.Element(index) And &H80000000, 1, 0)
```

```
        End If
```

```
    Next i
```

```
Next index
```

```
N1 = counter
```

```
Functional_DSP_test = tl_SUCCESS
```

```
End Function
```

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming

UltraSerial10G Jitter and Eye Measurement Test Programming

Jitter and eye measurement tests are similar in many respects to functional tests. As with functional tests, they require levels, timing, and pattern information be loaded before execution. However, jitter and eye measurement tests differ in that samples are taken of the DUT output for later DSP analysis, instead of the hardware immediately comparing test results to predefined vectors in expect memory. Only after DSP analysis are results compared to predefined limits.

This section is divided into these topics:

- | | |
|---|---|
| • | Jitter Measurement Concepts |
| • | Eye Measurement Concepts |
| • | Jitter and Eye Measurement Programming Procedure |
| • | Jitter or Eye Measurement Test Pattern and DataTool Sheet Setup |
| • | Jitter or Eye Measurement Test VBT Setup |
| • | Jitter or Eye Measurement Test DSP Setup |
| • | Jitter or Eye Measurement Test Instance and Limits Setup |
| • | Debugging Jitter or Eye Measurement Tests |

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Jitter Measurement Concepts

Jitter Measurement Concepts

Jitter measurements are performed while the DUT produces a looping pattern. Sampling is done using undersampling techniques. To perform serial undersampling, the pattern executes at a certain frequency while an asynchronous sample clock takes samples at different points in the pattern. The sample clock is referred to as the walking strobe (WS) clock since the serial undersampling technique mimics a strobe walking across the pattern.

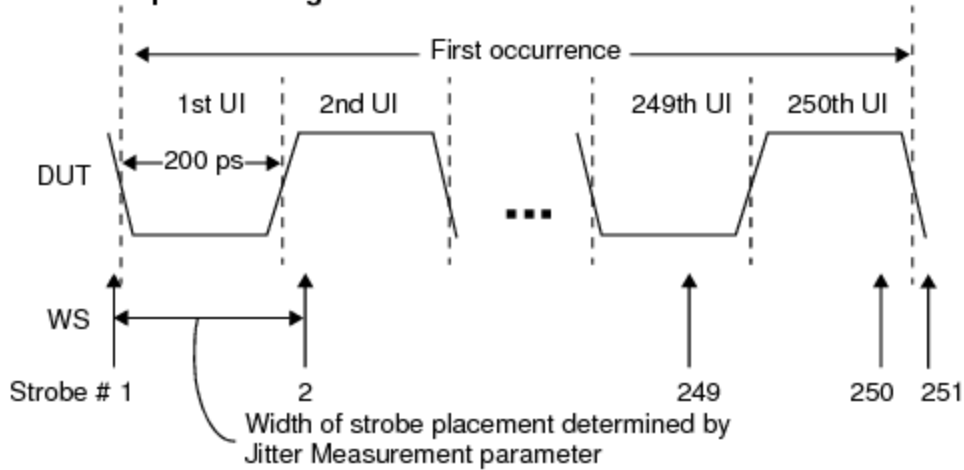
To set up undersampling, use the following parameters:

Bit Period or Unit Interval (UI)	The data rate.
Jitter Measure Pins	Which pins to perform the measurement.
Bits Per Pattern	Number of bits in the repeating pattern.
Desired Effective Sampling Resolution	Desired time between samples (once reordered).

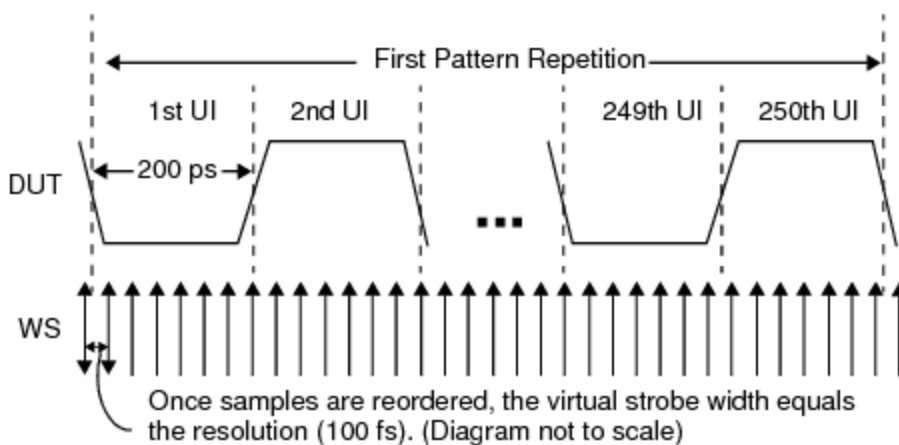
As an example, assume the DUT frequency is 5Gbps (bit period or UI of 200 ps), bits per pattern is 250, target sampling resolution is 100 fs, and a pattern of repeating 1s and 0s (10101...). The WS frequency would be calculated so that as many samples as possible could be taken for each repeat of the pattern. By the time the sample strobe reached the first bit again, it would be offset by 100 fs.

The following figure shows (in three steps) how undersampling data is captured for analysis. In this case, the WS frequency would be the bit period plus a little more for the resolution (to make it walk).

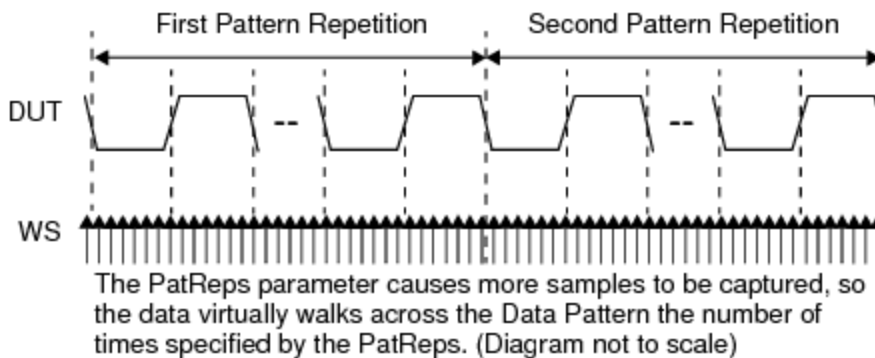
1. Data capture during the burst



2. Data capture reorder in the DSP



3. Effect of pattern repetition (PatReps) on the measurement



[Jitter Undersampling Process on UltraSerial10G](#)



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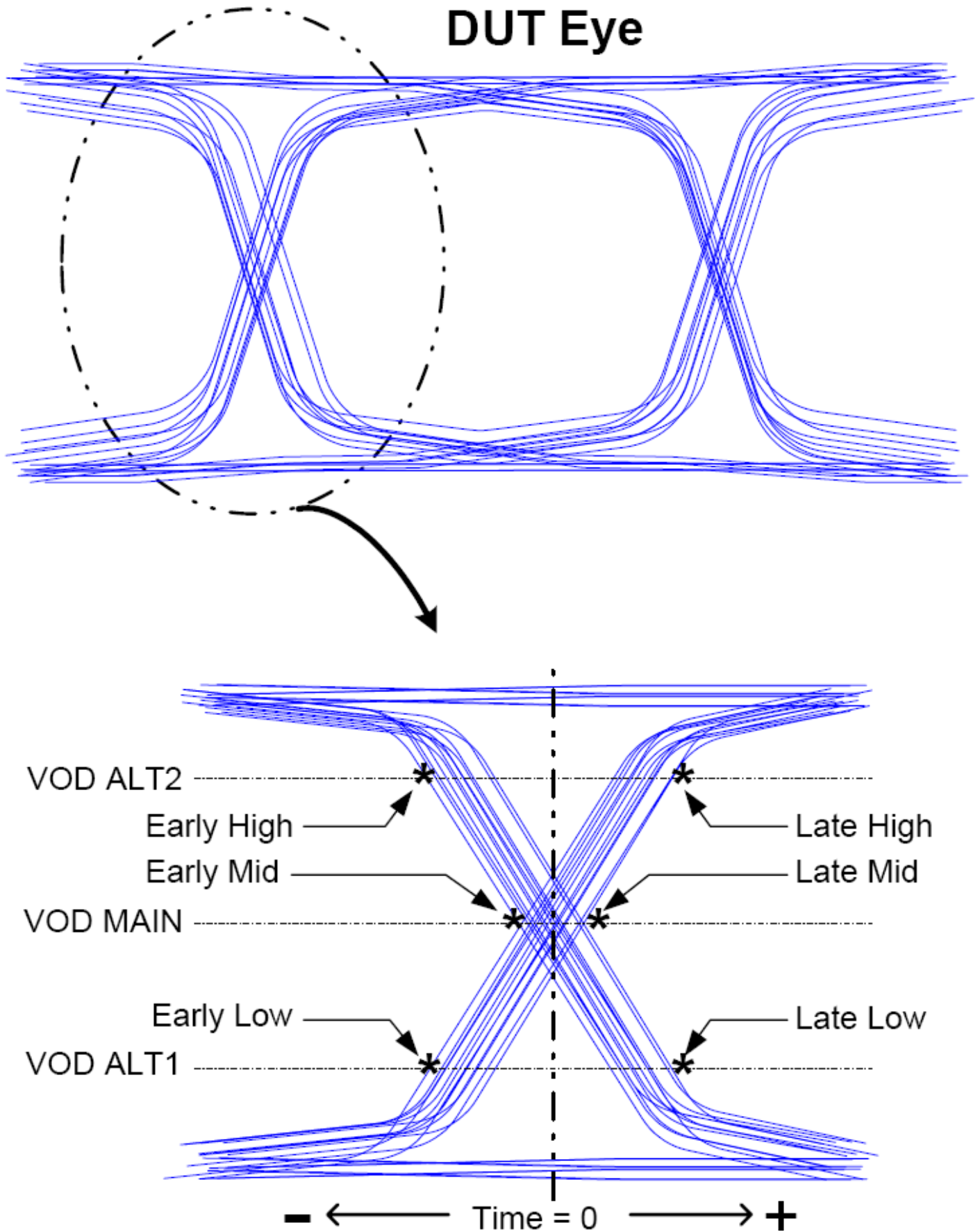
	
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[UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Eye Measurement Concepts](#)

Eye Measurement Concepts

A eye measurement test is similar to a jitter measurement, except that samples are captured at three different vod levels. DSP analysis determines the earliest and latest time that each edge in the pattern occurred at each vod level and returns six points corresponding to the overall earliest and latest positions with the edges superimposed on each other. The six points can be used to compare the DUT output to four point and six point templates.

DUT Eye



Six Eye Measurement Points (Three Earliest and Three Latest)

The eye analysis data is not returned in the standard hexagon format, usually thought of as determining the opening of the eye. The points returned are normalized to zero time at the

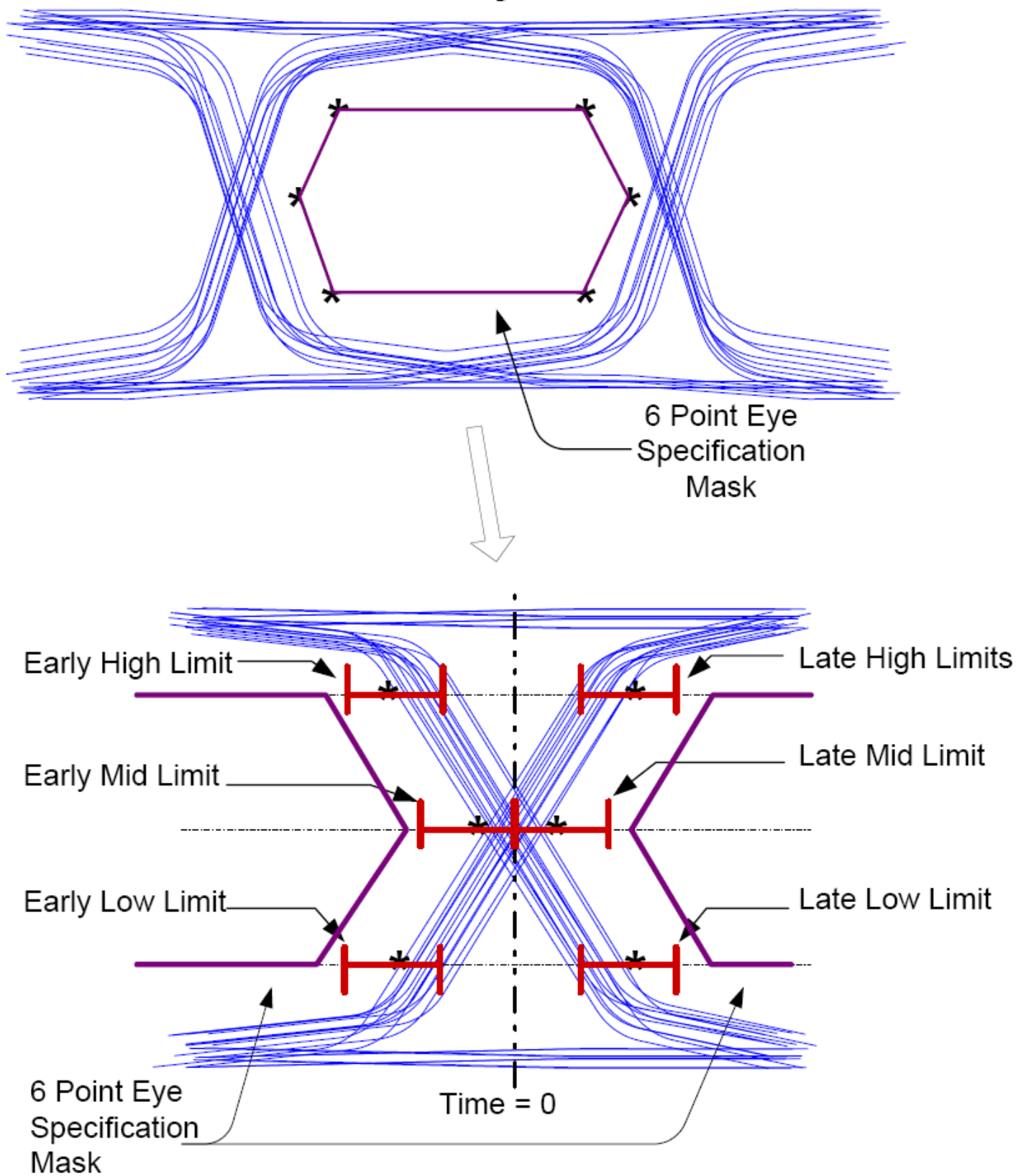
transition region as opposed to the open part of the eye. The reason for this format is to allow multiple eye analysis results to be overlaid to determine the effects different patterns and frequencies have on the jitter and slew rate without adjusting for the unit interval (UI).

Six Point Eye Measurement

A six point eye is the standard hexagonal mask to validate the opening off an eye. To set the job limits for a hexagonal eye, do the following:

1. Translate the right side of the hexagonal mask to the left (earlier in time) by subtracting the unit interval (UI) from the values.
2. Normalize the midpoint values about zero.
3. Set the limits to avoid the specification mask, while leaving room for required design margin or instrument accuracy limits.

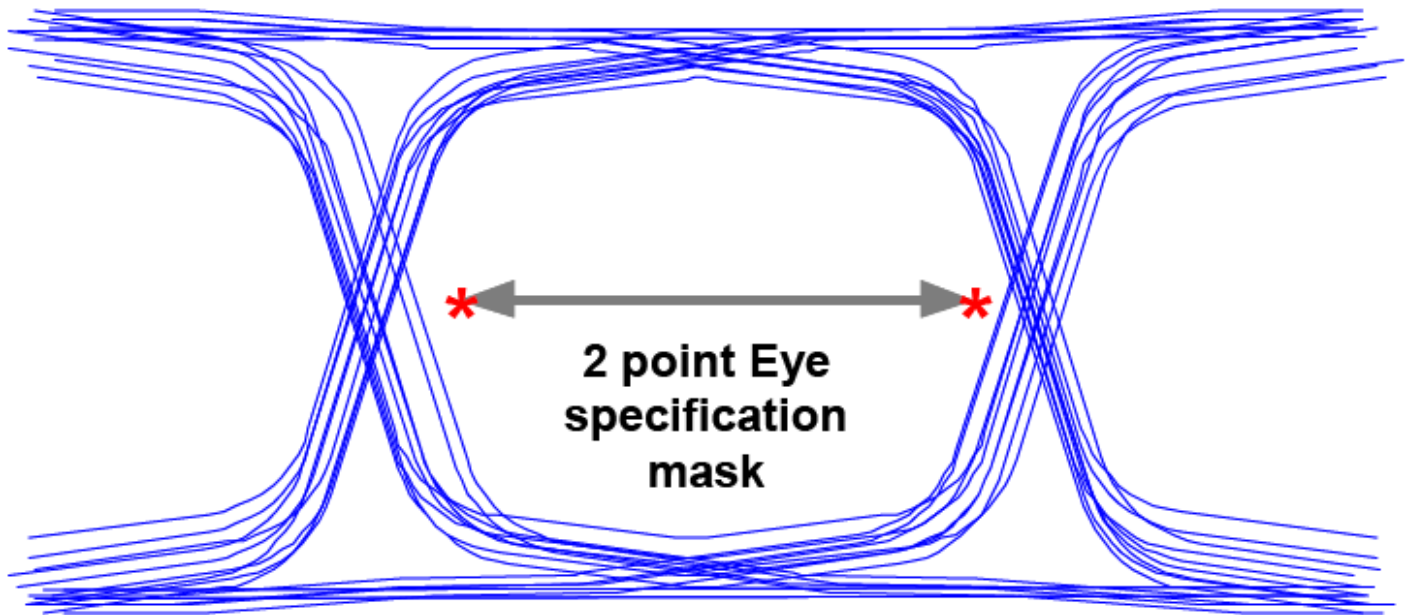
DUT Eye



Six Point Eye Limits Setup

Two Point Eye Measurement

A two point eye is the same setup as the six point eye, except the early and late midpoints are used in the limits table. A two point eye determines the total jitter or the width of the eye.



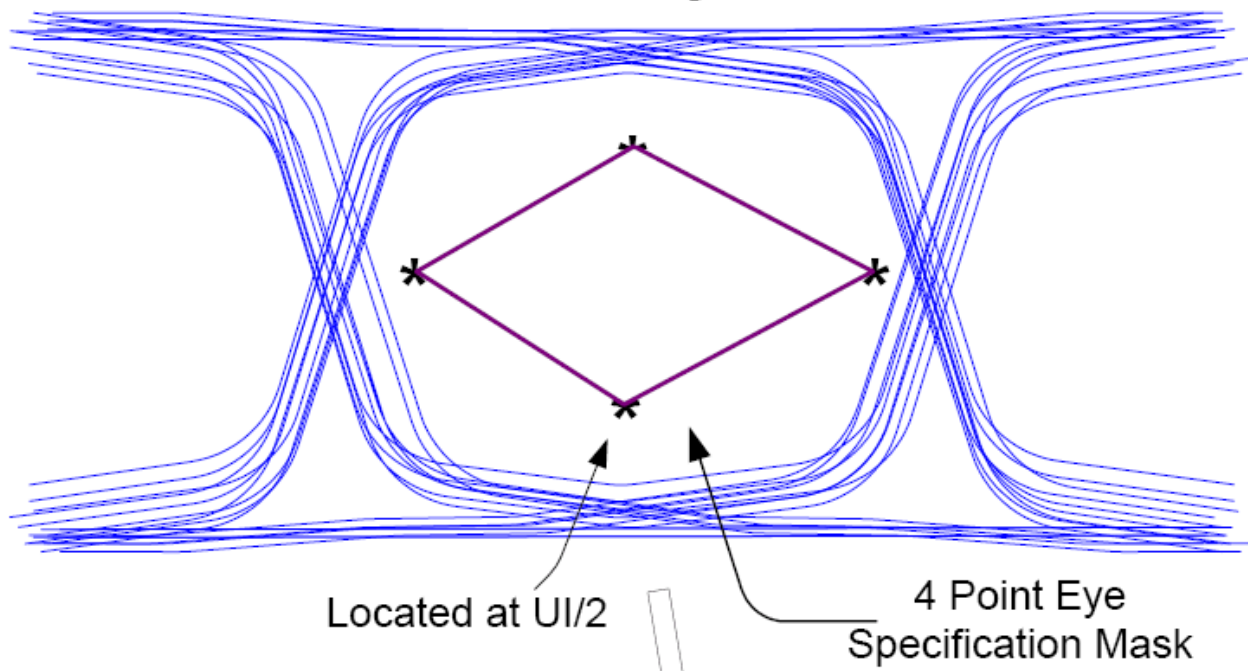
Two Point Eye Measurement

Four Point Eye Measurement

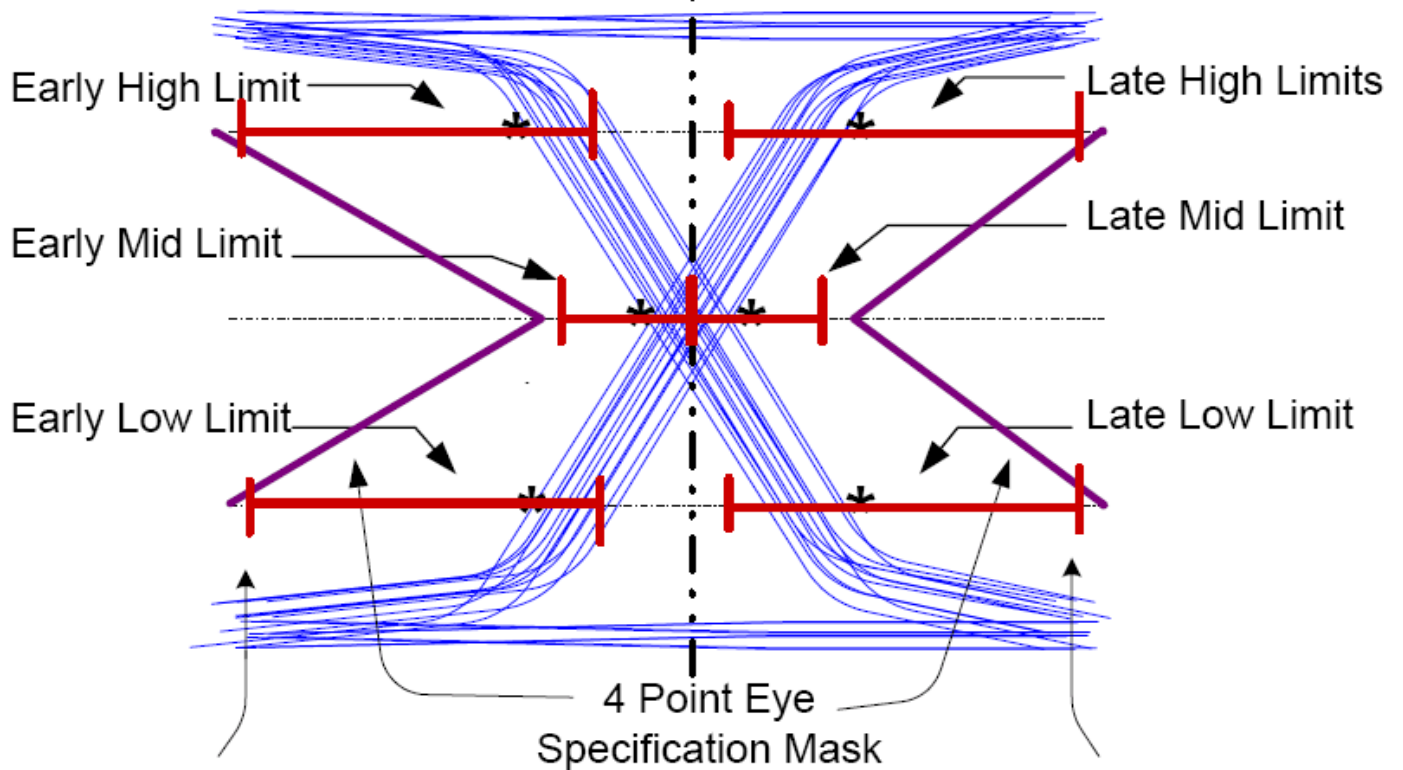
A four point eye mask is in the shape of a diamond. A four point measurement is used to determine the width of the eye and to verify that the eye has not closed at the top or bottom. To set up four point eye measurements, all six points returned from the eye analysis are required. The only difference between this and the six point eye is that the limits for the high/low points must be adjusted as follows:

- Low for Early High/Low = $-(UI/2 - \text{Resolution})$
- High for Late High/Low = $+(UI/2 - \text{Resolution})$

DUT Eye



Time = 0



Lower Limit= $-(UI/2 - \text{Resolution})$

Upper Limit= $UI/2 - \text{Resolution}$

Four Point Eye Limits Setup

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Jitter and Eye Measurement Programming Procedure

Jitter and Eye Measurement Programming Procedure

To perform a jitter or eye measurement test, follow these steps:

- | | |
|----|--|
| 1. | Create a pattern and fill out the required DataTool sheets (see Jitter or Eye Measurement Test Pattern and DataTool Sheet Setup). |
| 2. | Create a VBT function to set up and perform the measurement (see Jitter or Eye Measurement Test VBT Setup). |
| 3. | Create a DSP function to perform the jitter measurement, analyze the sampled data, and check results (see Jitter or Eye Measurement Test DSP Setup). |
| 4. | Create test instances to call the functions and add them to the flow table (see Jitter or Eye Measurement Test Instance and Limits Setup). |
| 5. | Debug the jitter or eye measurement test (see Debugging Jitter or Eye Measurement Tests). |

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Jitter or Eye Measurement Test Pattern and DataTool Sheet Setup

Jitter or Eye Measurement Test Pattern and DataTool Sheet Setup

Follow these steps to set up a jitter or eye measurement test pattern and DataTool sheets:

- | | |
|----|---|
| 1. | Create a pin map (see Programming the Pin Map). |
| 2. | Create a channel map (see Programming the Channel Map). |
| 3. | Create a pattern that causes the desired bit pattern to be driven out of the DUT until the jitter or eye measurement is complete. |

Since the methods for doing this vary from device to device, they are not described here. Note that measurement times can vary greatly depending on the type of measurement and the setup parameters used. To avoid sampling problems, make sure the desired bit stream is looping indefinitely while sampling is done.

- | | |
|----|--|
| 4. | Program device Vod levels for UltraSerial10G pins (see Programming Pin Levels). |
| 5. | Create a Jitter sheet for each measurement (jitter and eye) since only one jitter set is allowed per sheet (see Programming UltraSerial10G Serial Jitter). |

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Jitter or Eye Measurement Test VBT Setup

Jitter or Eye Measurement Test VBT Setup
Follow these steps to set up a VBT function for a jitter or eye measurement test:

1. Apply and connect HSD and UltraSerial10G levels and timing:

TheHdw.Digital.ApplyLevelsTiming(...)

2. Load HSD patterns:

TheHdw.Patterns(hsdm_pattern).Load

3. Load serial patterns.

Although a receive pattern is not required to run jitter or eye measure tests, you must configure the UltraSerial10G to run long enough to capture the required number of samples. IG-XL creates a dummy receive pattern that sets up the UltraSerial10G correctly. You must load this pattern:
' Retrieve software generated jitter pattern for capture
JitterPattern = TheHdw.Serial.Jitter.Pattern
TheHdw.Serial.Patterns(JitterPattern).Load

4. Create pointers to DSP Waves and bind them to the pins being measured. (See Capturing and Processing Data.)

Dim DspWave_AllPins As New PinListData
DspWave_AllPins = TheHdw.Serial.Pins(JitterPins).Patterns.CaptureMemory _

.DSPWave

5. Set up the RAT to start immediately and disable the RAT timeout and CDR:

```
TheHdw.Serial.Pins(JitterPins).Timing.CDRMode = tlSerialCDRDisable
```

```
Call TheHdw.Serial.Pins(JitterPins).Patgen.ReceiveAlign.TriggerMode _
```

```
(tlSerialReceiveAlignStart)
```

```
TheHdw.Serial.Pins(JitterPins).Patgen.ReceiveAlign.TimeoutEnable = False
```

6. Configure the capture memory to move captured data to the DSP hardware:

```
TheHdw.Serial.Pins(JitterPins).Patterns.CaptureMemory.CaptureConfig =
```

```
tlSerialCaptureConfigSamplerMove
```

7. Start the HSD (if necessary) and serial jitter patterns.

The HSD patterns and the serial patterns are started asynchronously. The jitter measurement starts when the jitter measure serial pattern starts.

```
TheHdw.Patterns(hsdm_pattern).Start
```

```
TheHdw.Serial.Patterns(JitterPattern).Start
```

☑ Note:

Make sure that you do not create a race condition by starting the jitter measure before the pattern is repeating. Doing this causes the algorithm to fail.

8. Wait for UltraSerial10G to complete capture and halt any infinitely looping drive patterns.

```
TheHdw.Serial.Pins(JitterPins).Patgen.HaltWait
```

```
TheHdw.Digital.Patgen.Halt
```

9. After jitter or eye test is complete, execute the following statement to disable the jitter mode.

```
TheHdw.Serial.Pins(JitterPins).Patterns.CaptureMemory.CaptureConfig = _
```

```
tlSerialCaptureConfigNone
```



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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Jitter or Eye Measurement Test DSP Setup

Jitter or Eye Measurement Test DSP Setup
Follow these steps to set up a DSP function for a jitter or eye measurement test:

1. [Create a DSP function that runs the VBT code created in Jitter or Eye Measurement Test VBT Setup.](#)

You can find examples of a [Jitter DSP Function](#) and a [Jitter Eye Measurement DSP Function](#) below.

2. [Execute the DSP analysis.](#)

The analysis is run on the DSP hardware if the test element Run Mode is set to Automatic. If the Run Mode is set to Host-Debug, then the analysis is run in software on the host computer.

Jitter DSP Function

The jitter DSP function requires the test program contain a VBT module name that begins with DSP. This VBT module should contain a function that executes the jitter or eye measurement analysis. The function call for performing a jitter measurement analysis looks like this:

[SerialMeasureJitter\(\)](#)

The SerialMeasureJitter function requires that you send it a DSP wave to analyze. This is done by attaching a DSP wave to the measure jitter function call:

[Dim](#) wave1 As New DSPWave

Call wave1.SerialMeasureJitter(Rj, Peak2PeakDDj, MeasuredUI, dspStatus)

The SerialMeasureJitter function returns four results:

Rj	Holds the measured Rj value. This can be any variable name you create, but it must be defined as type Double.
Peak2PeakDDj	Holds the measured peak-to-peak DDj value. This can be any variable name you create, but it must be defined as type Double.
MeasuredUI	Holds the estimated measured UI value. This can be any variable name you create but, must be defined as type Double.
dspStatus	The DSP analysis routine returns a DSP status of zero if it can measure Rj, DDj, and estimated UI. Otherwise, the DSP status bit is set to a nonzero value. This can be any variable name you create, but it must be defined as type Long.

For more information on these parameters, see [SerialMeasureJitter](#) and [Jitter Measurement Concepts](#).

Here is a more complete VBT code example for jitter measurement:

```
Public Function JitterMeas(ByVal Wave As DSPWave, DDj As Double, _
    Rj As Double, measUI As Double, DspStatus As Long) As Long
```

```
Call Wave.SerialMeasureJitter(Rj, DDj, measUI, DspStatus)
```

```
' Export the file (only works in debug mode).
```

```
' Call TheHdw.Serial.Jitter.FileExport(Wave, ".\JM_Jitter.txt")
```

```
End Function
```

Jitter Eye Measurement DSP Function

The function call for performing a jitter eye width measurement analysis looks like this:

```
SerialMeasureEye()
```

The [SerialMeasureEye](#) function requires you to send it a DSP wave to analyze. Note that this DSP wave must contain sampled data for all three Vod levels. Attaching a DSP wave to the measure eye function call is the same as attaching one to the measure jitter function call:

```
Dim wave1 As New DSPWave
```

```
Call wave1.SerialMeasureEye(Rj, Peak2PeakDDj, MeasuredUI, Risetime, Falltime, EarlyLow, LateLow,
EarlyMid, LateMid, EarlyHigh, LateHigh, dspStatus)
```

The [SerialMeasureEye](#) routine returns the following results:

Rj	Holds the measured Rj value. This can be any variable name you create, but it must be defined as type Double.
----	---

Peak2PeakDDj	Holds the measured peak to peak DDj value. This can be any variable name you create, but it must be defined as type Double.
MeasuredUI	Holds the estimated measured UI value. This can be any variable name you create, but it must be defined as type Double.
Risetime	Holds the result for the measured rise time. This can be any variable name you create, but it must be defined as type Double.
Falltime	Holds the result for the measured fall time. This can be any variable name you create, but it must be defined as type Double.
EarlyLow	Holds the result for the earliest edge time measured at Vod_Alt1. This can be any variable name you create, but it must be defined as type Double.
LateLow	Holds the result for the latest edge time measured at Vod_Alt1. This can be any variable name you create, but it must be defined as type Double.
EarlyMid	Holds the result for the earliest edge time measured at Vod_Main. This can be any variable name you create, but it must be defined as type Double.
LateMid	Holds the result for the latest edge time measured at Vod_Main. This can be any variable name you create, but it must be defined as type Double.
EarlyHigh	Holds the result for the earliest edge time measured at Vod_Alt2. This can be any variable name you create, but it must be defined as type Double.
LateHigh	Holds the result for the latest edge time measured at Vod_Alt2. This can be any variable name you create, but it must be defined as type Double.
dspStatus	The DSP analysis function returns a DSP status of zero if it can measure Rj, DDj and estimated UI. Otherwise, the DSP status bit is set to a nonzero value. This can be any variable name you create, but it must be defined as type Long.

For more information on these parameters, see [SerialMeasureEye](#) and [Eye Measurement Concepts](#). Here is a more complete example of a jitter eye width measurement.

```
Public Function EyeMeas(ByVal Wave As DSPWave, DDj As Double, Rj As Double, _
    Tj As Double, measUI As Double, Tr As Double, Tf As Double, _
    Eye20 As Double, Eye50 As Double, Eye80 As Double, DspStatus As Long) _
    As Long
```

Dim Eye20E As Double
Dim Eye20L As Double
Dim Eye50E As Double
Dim Eye50L As Double
Dim Eye80E As Double
Dim Eye80L As Double

Call Wave.SerialMeasureEye(Rj, DDj, measUI, Tr, Tf, Eye20E, Eye20L, Eye50E, _
Eye50L, Eye80E, Eye80L, DspStatus)

Eye20 = measUI - Abs((Eye20E - Eye20L))
Eye50 = measUI - Abs((Eye50E - Eye50L))
Eye80 = measUI - Abs((Eye80E - Eye80L))

Tj = DDj + Rj * 14 ' Change multiplication factor as needed.

' Export the file (only works in host debug mode).
' Call TheHdw.Serial.Jitter.FileExport(Wave, ".\Eye_Jitter.txt")

End Function

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming : Jitter or Eye Measurement Test Instance and Limits Setup

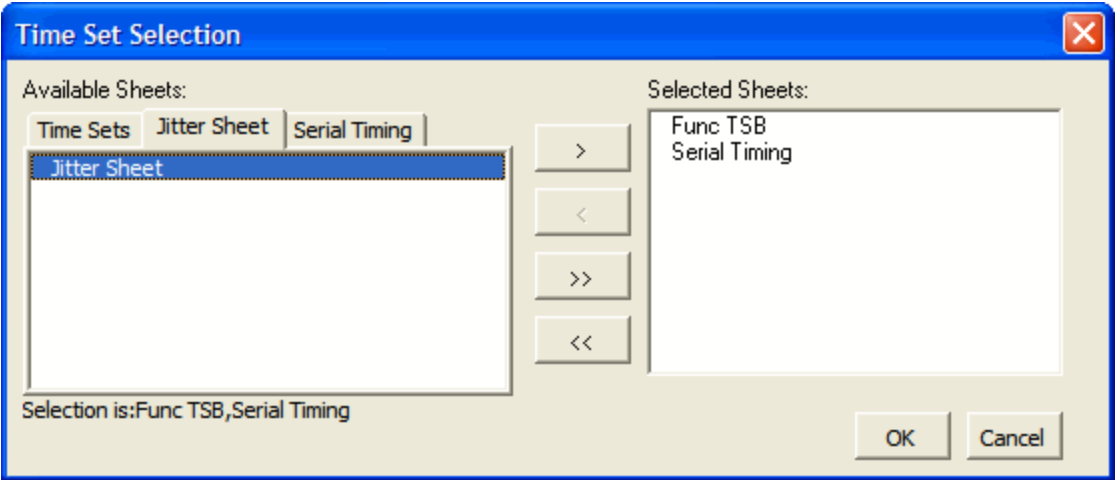
Jitter or Eye Measurement Test Instance and Limits Setup
The final step in the procedure is to set up the test instances and limits:

1.

Create a test instance that uses the VBT and DSP functions. (See also Programming UltraSerial10G Test Instances.)
2.

Specify the Jitter sheet by right-clicking in the Time Sets column of the Test Instances sheet and selecting Edit.

This opens the TimeSet Selection window. This shows a box with multiple timing set tabs. Select the Serial Jitter tab and select which Jitter sheet to use:



Selecting a Jitter Time Set

3. [Add the test instance to the flow table. When using the DSP hardware, you have the advantage of using the Test-defer-limits flow opcode:](#)

Flow Table



		Gate			Command				Limits	
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim
					Test	Func_2.5G_PClk_Comp	Func	200		
					Test	BER_PRBS31_Test	BER	201		
					Test	SigGen_Test	SigGen	202		
					Test-defer-limits	JM_2p5G_Clk	JM_Clk	203		
					Use-Limit	JM_2p5G_Clk	DDj	204	0	2E-11
					Use-Limit	JM_2p5G_Clk	Rj	205	0	2E-11
					Test-defer-limits	Eye_2p5G_Clk	Eye Clk	215		
					Use-Limit	Eye_2p5G_Clk	DDj	216		
					Use-Limit	Eye_2p5G_Clk	RJ	217		
					Use-Limit	Eye_2p5G_Clk	Tj	218		
					Use-Limit	Eye_2p5G_Clk	Eye20	219		
					Use-Limit	Eye_2p5G_Clk	Eye50	220		
					Use-Limit	Eye_2p5G_Clk	Eye80	221		
					Use-Limit	Eye_2p5G_Clk	Tr	222		
					Use-Limit	Eye_2p5G_Clk	Tf	223		
					limits-all					
					set-device					

UltraSerial10G Flow Table

The Test-defer-limits opcode allows the DSP processing to occur in the background while the test flow continues on to the next test.

For more information, see [Test Limits in Flow](#).

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UltraSerial10G Programming : UltraSerial10G Jitter and Eye Measurement Test Programming :
Debugging Jitter or Eye Measurement Tests

Debugging Jitter or Eye Measurement Tests

Jitter Tool

You can use Jitter Tool to view and debug jitter and eye measurement results. To run Jitter Tool, you must first call a function that exports a file from a DSP function that has analyzed the sample jitter measurement data. To save the results of a jitter test to a file, use this VBT method in a DSP function:

TheHdw.Serial.Jitter.FileExport(<DspWave>, <FileName>)

This method writes out a single DspWave per site. For more information on this method, see FileExport. Once the file has been created, open Jitter Tool from the main IG-XL menu.

For more information, see Jitter Tool.

DSP Status Fail Codes

In UltraSerial10G jitter measurement, the jitter or eye algorithm may fail to process the data. When the jitter or eye algorithm fails, a nonzero fail code is returned in the DSP_STATUS variable. The same fail code is also returned in the Rj, DDj, and eye variables returned from the algorithm.

The following topics describe common fail codes.

Groups Too Wide

Fail Code	-58
Description	The apparent edges for one pattern edge spread over 1 UI in time (edge times of these apparent edges spread over 1 UI). This error normally is caused by missing edges at region boundary or too much edge variation due to extensive jitter/noise.

Possible Causes

- Large jitter/PJ component
- Wrong pattern period
- Pattern reps too low

Groups Too Close

Fail Code	-59
Description	Two groups of apparent edges corresponding to two pattern edges of opposite polarity are within UI/2 in time and are at risk of overlapping. This error (two pattern edges within UI/2) is frequently caused by extensive jitter/noise.
Possible Causes	• Wrong pattern period • Wrong bit per pattern • High DDJ and PJ

Too Few Edges

Fail Code	-60
Description	Less than two apparent edges were extracted from sampling data. (Any pattern should have at least two bits, so a minimum of two edges must be detected.) This error is typically when problems occur with extracting edges.
Possible Causes	• Too much jitter • Resolution is too coarse

- Output stuck at all high or all low
- Captured data is not toggling

Data Too Coarse

Fail Code	-61
Description	A setup problem: the minimal requirements for samples per UI have not been met. At least 100 samples per UI are required (UI/resolution < minimal sample number).
Possible Causes	• Resolution too large

Bad Edge

Fail Code	-64
Description	Found a bad edge data in what should be good data. Bad edges are typically high glitch, low glitch, or all unstable (no edge can be detected at all). This is usually related to job setup.
Possible Causes	• Wrong pattern period • Wrong bit per pattern • Data is glitching or is not stable

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UltraSerial10G Programming : Job Execution

Job Execution

Use the Flow Table to execute UltraSerial10G functional and parametric tests. Before the first test instance is executed, the UltraSerial10G hardware is initialized to a known state, which means that the databus is initialized and registers are programmed to an initial value. After the job finishes execution, the UltraSerial10G hardware is restored.

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UltraSerial10G Programming : Using UltraSerial10G in Concurrent Testing

Using UltraSerial10G in Concurrent Testing

To support concurrent testing, you can assign each half board of an UltraSerial10G to a unique time domain. Concurrent features for UltraSerial10G include:

- You can configure up to two time domains total for each UltraSerial10G board (one for each half instrument board).

Note that the UltraSerial10G ports are not divided so that the first time domain includes the first 10 ports and second time domain includes the last 10 ports. Instead, the UltraSerial10G time domains include these ports:

– First time domain (0) ports: 4, 5, 6, 7, 8, 9, 10, 11, 14, 15

– Second time domain (1) ports: 0, 1, 2, 3, 12, 13, 16, 17, 18, 19

- Each UltraSerial10G half board can be controlled by unique time domains simultaneously using Sync-Link. Features include:

– Independent access to Sync-Link

– Independent timing programming

– Independent synchronization with other UltraSerial10G instruments or with digital timing

- 5/16/26, 5:32 PM
- Combined Documentation
- Independent control by digital pattern

–

Independent capture memory and triggers

•

Half boards can be grouped into a common time domain

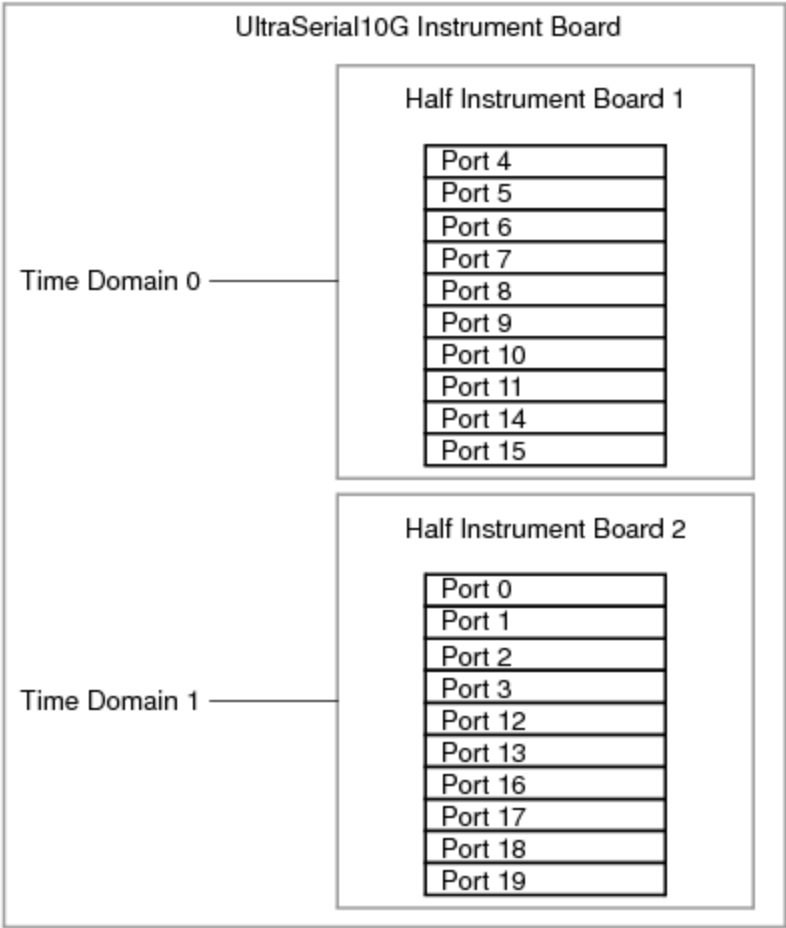
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UltraSerial10G fully supports flow domains and asynchronous pattern starts

•

Serial timing and jitter setup can be programmed using `.ApplyLevelsTiming` per flow domain.
Note that only one jitter measurement can be active at a time.

The following diagram shows the concurrent testing architecture for the UltraSerial10G.



UltraSerial10G Concurrent Testing Architecture

Note that you cannot datalog the UltraSerial10G when using the instrument as part of concurrent test.

For more information, see [About Concurrent Test](#).

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UltraSerial10G Theory of Operation

UltraSerial10G Theory of Operation

This section provides an overview of high speed serial buses that the UltraSerial10G tests, describes the functional components of the UltraSerial10G instrument. The following topics are available:

- [High Speed Serial Buses](#)
- [UltraSerial10G Functional Overview](#)
- [UltraSerial10G Time Domains](#)
- [UltraSerial10G Timing Groups](#)
- [Differential Driver](#)
- [Drive Midband](#)
- [Differential Receiver](#)
- [UltraSerial10G Driver Pin Levels](#)
- [UltraSerial10G Timing](#)
- [DC Coupling](#)

- [Vector Memory \(VM\) and Capture Memory \(CMEM\)](#)
- [PRBS Generator](#)
- [Receive Align Trigger](#)
- [Digital Loopback](#)
- [Drive-only Site Copy](#)
- [UltraSerial10G PPMU](#)
- [UltraSerial10G Scan Testing](#)
- [UltraSerial10G Jitter Insertion and Measurement](#)

For more information about the UltraSerial10G, see:

- [About the UltraSerial10G](#)
- [UltraSerial10G Programming](#)
- [UltraSerial10G Specifications](#)

For information on UltraSerial10G instrument performance specifications and standards, see [UltraSerial10G Specifications](#).

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UltraSerial10G Theory of Operation : High Speed Serial Buses

High Speed Serial Buses

The UltraSerial10G instrument tests high speed serial buses. High speed serial buses receive and transmit packets of information. A packet is a formatted stream of bits that represent control commands or data. An important characteristic of high speed serial buses is that serial clocks are encoded into the pattern data and must be "recovered." Another important characteristic is the occurrence of nondeterministic data (NDD) in the device transmit stream.

Nondeterministic DUT behavior occurs when the response of a device complies with the device specification and the bus protocol, but the order and content of the information on the bus is different than original simulation. Common causes of NDD are:

- Unknown or random initial states
- Unknown or random lock time for internal PLLs or other dynamic circuits on the DUT
- Different time domains on the DUT leading to race conditions at the DUT transmitter
- The physical layer encoding or processing steps that inherently add nondeterminism

Types of nondeterminism include:

Cycle slipping at initialization

Random inactive periods that occur at the beginning of bus transmission.

Cycle slipping in the middle of transmission	Random inactive periods that occur in the middle of a transmission, in between control and data packets.
Out-of-order of packets	The order of transmitted control or data packets differs from one run to the next.
Running disparity	Due to other types of nondeterminism, the disparity of the transmitted 10-bit symbol may vary from one run to the next.

8b10b Encoding and Disparity

8b/10b encoding is used for many high speed serial bus devices. During data serialization, eight lanes of data are serialized and encoded into 10 bits according to the 8b10b standard. Each 8-bit byte can be translated into two 10-bit symbols, one for positive disparity and one for negative disparity. The transmitter decides which of the two 10-bit symbols to send next based on the running disparity resulting from the prior symbol.

The 8b10b encoding scheme limits the number of 1’s and 0’s in the data to be no more than five in a row and addresses the following issues with serial data transmission:

- Guarantees enough bit transitions for Clock Data Recovery circuits to extract the clock out of the data.
- Defines a finite number of patterns (each with a small, deterministic length), which simplifies error detection and correction.
- Produces a transmitted signal that does not contain a DC component
- Decreases intersymbol interference and ensures signal amplitude by providing limitations on the maximum spacing of transitions

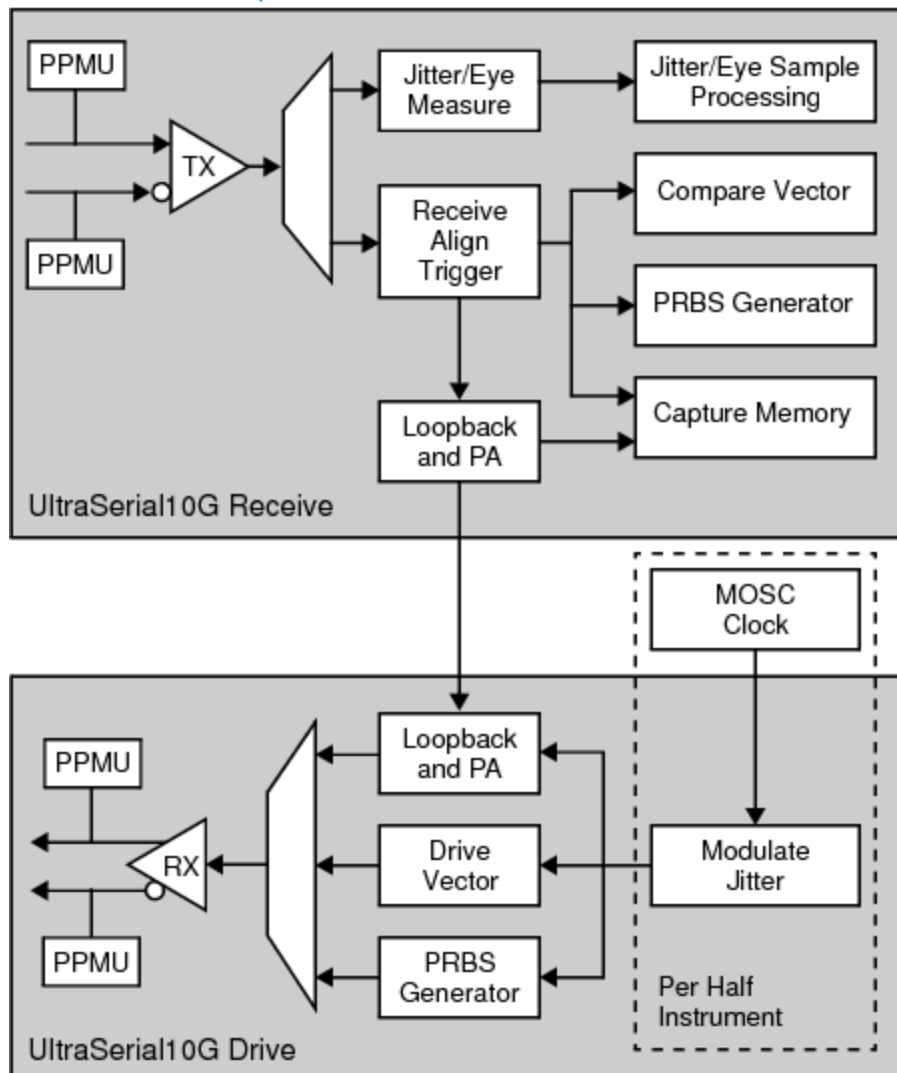
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[UltraSerial10G Theory of Operation](#) : UltraSerial10G Functional Overview

UltraSerial10G Functional Overview

UltraSerial10G drive hardware drives differential signals to the DUT RX pins. UltraSerial10G receive hardware receives differential signals from the DUT TX pins. The figure below shows a differential drive and receive pair on the UltraSerial10G.



UltraSerial10G Block Diagram

On the differential driver, the MOSC clock provides the timing for the drive signals for each half instrument. You can also enable jitter modulation on the drive signals to test DUT receiver clock data recovery (CDR) tolerances. Drive signal timing is routed to drive vector memory, the pseudo-random binary sequence (PRBS) generator, digital loopback hardware, or the Protocol Aware (PA) hardware to generate the DUT receiver (RX) drive signal into the DUT receiver. The PPMUs can be used to perform DC parametric testing.

The differential receiver takes the DUT transmitter output (TX) and routes it to either the jitter and eye measurement hardware or the receive align trigger (RAT) hardware. Jitter and eye measurement hardware samples and analyzes DUT output to return random jitter (Rj) and data dependent jitter (DDj), generate a six-point eye, and return rise and fall times. RAT consists of CDR, 10b align, and 20b match hardware that recovers the embedded clock and performs a series of pattern matches used to align DUT output to the expect pattern.

After the RAT completes its alignment, the DUT output can be routed to several locations:

- Compare vector memory, where DUT output is compared to stored expect patterns
- PRBS generator, where DUT output is compared to one of four available sequences (7, 15, 23, or 31)
- Capture memory, where DUT output can be stored for use in additional analysis or debugging
- Digital loopback hardware, where DUT output for a finite or infinite number of vectors can be fed back into the differential driver digital loopback hardware
- PA hardware that supports the PCI Express (PCIe) protocol

As with the differential driver, the receiver PPMUs can be used to perform DC parametric testing.

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UltraSerial10G Theory of Operation : UltraSerial10G Time Domains

UltraSerial10G Time Domains

UltraSerial10G can have up to two time domains, and each time domain can contain up to 10 ports on each half board for a total of 20 ports on each instrument board. Note that the UltraSerial10G ports are not divided so that the first time domain includes the first 10 ports and second time domain includes the last 10 ports. Instead, the UltraSerial10G time domains include these ports:

- | | |
|---|--|
| • | First Time Domain (0) Ports: 4, 5, 6, 7, 8, 9, 10, 11, 14, 15 |
| • | Second Time Domain (1) Ports: 0, 1, 2, 3, 12, 13, 16, 17, 18, 19 |

For more information, see these topics:

- | | |
|---|--|
| • | UltraSerial10G Timing Groups |
| • | UltraSerial10G Timing |
| • | Using UltraSerial10G in Concurrent Testing |

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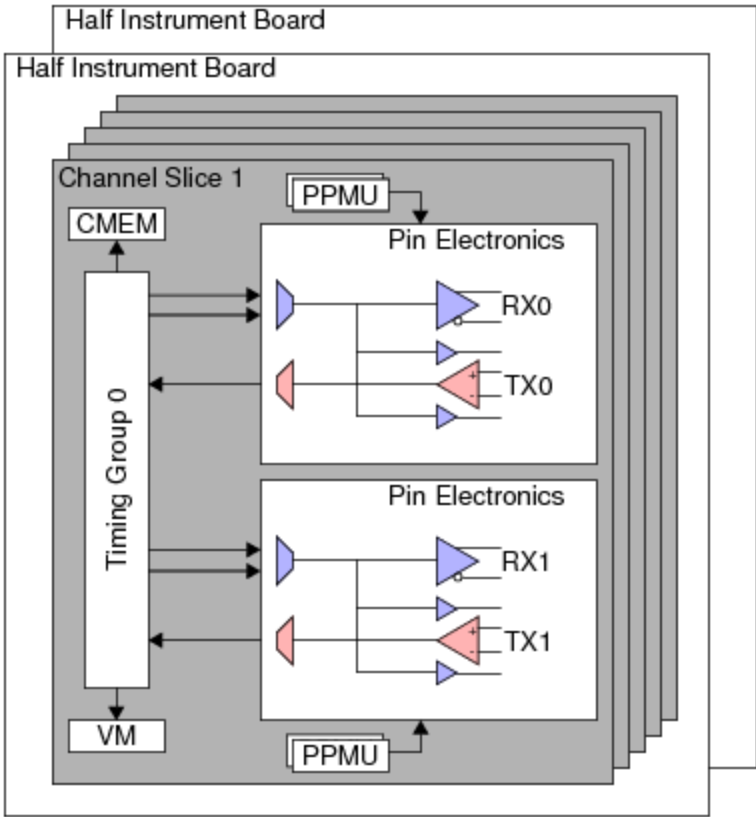
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UltraSerial10G Theory of Operation : UltraSerial10G Timing Groups

UltraSerial10G Timing Groups

UltraSerial10G has 10 timing groups, five on each half instrument board. Each timing group controls two pin electronics (PE) blocks. Each PE has one TX and one RX port. Both PE blocks in each timing group must operate in only one of these modes:

- Base serial: This mode includes all serial operations, including PRBS, but not loopback, site copy, or Protocol Aware.
- Loopback: This mode provides digital loopback signals through the instrument hardware. See Digital Loopback.
- Drive-only site copy: This mode allows you to reuse receive channels as drive channels, which doubles the number of available drive channels. See Drive-only Site Copy.
- PCIe Protocol Aware (PA): This mode allows you to test PCIe ports at a protocol level. See PCIe Protocol Aware.



UltraSerial10G Timing Groups

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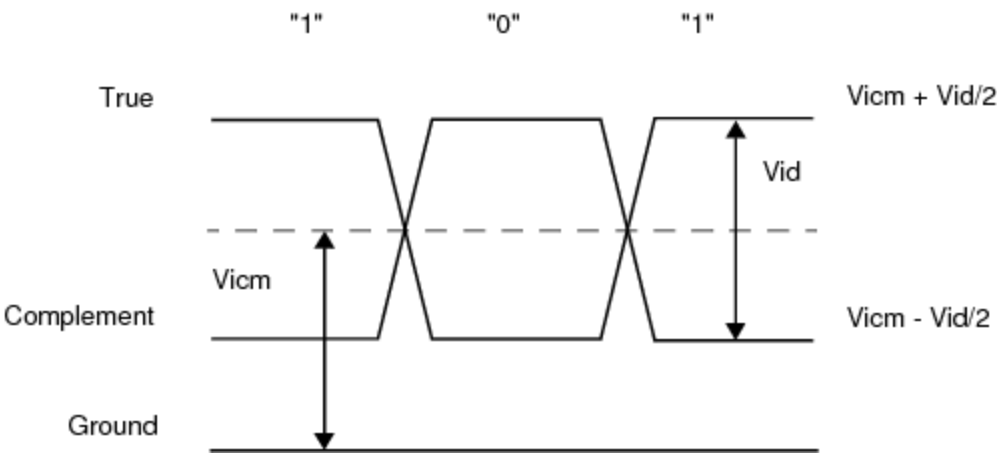
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UltraSerial10G Theory of Operation : Differential Driver

Differential Driver

Each UltraSerial10G has 20 true differential driver channels that are formed by pairing 40 digital channels. Each driver channel can be programmed to high, low, or midband. You can also program the driver’s initial and start states. UltraSerial10G only supports differential mode. The driver has two programmable levels: differential voltage (V_{id}) and common mode voltage (V_{icm}).



Differential Driver Levels

The programmable range of the drivers is:

- V_{id} : +100mV to +1.8V
- V_{icm} : 0V to +1.8V

☒ **Note:**

V_{id} is programmed assuming an open connection (no termination). When V_{id} is terminated to 50 ohms, the terminated range is: +50mV to +900mV.

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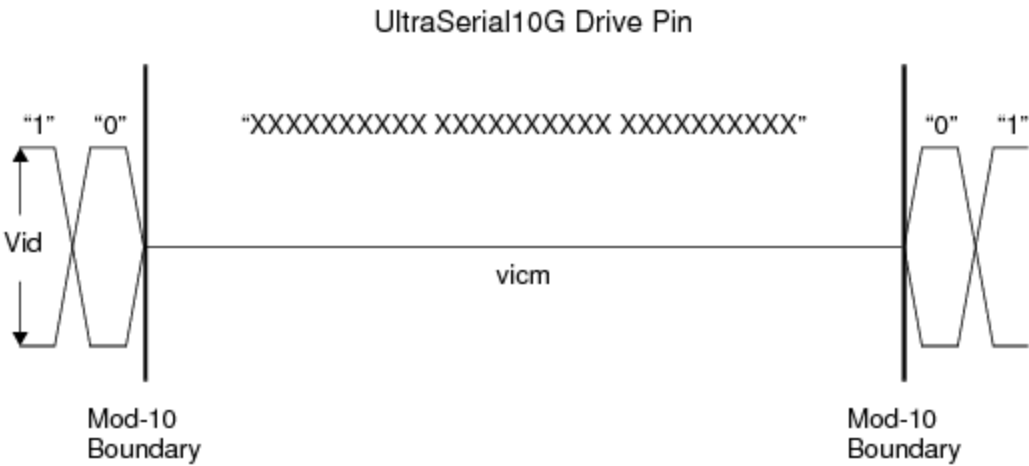
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UltraSerial10G Theory of Operation : Drive Midband

Drive Midband

In addition to driving 0 or 1, the UltraSerial10G can also drive midband which is represented by the X data state. Drive midband must start on a modulo-10 boundary of VM and occur in blocks of 10. When an UltraSerial10G drive pin is in midband, both single-ended pins of the differential pair drive the Vicm level. The diagram below illustrates a waveform during drive to midband.



Drive Midband

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UltraSerial10G Theory of Operation : Differential Receiver

Differential Receiver

The UltraSerial10G differential receiver takes the difference of the true and complement pins to determine the high or low signal. In effect, the differential receiver tracks the crossing point of the differential signal.

Vod_Main is the standard setting used for functional test. When performing functional tests, you must set Vod to 0V to ensure that CDR locks on an edge. IG-XL uses Vod_Main, Vod_Alt1, and Vod_Alt2 for jitter eye and rise time/fall time tests. For these tests:

- Vod_Alt1 is the minimum Vod level, which is typically set to 20% of the output voltage swing.
- Vod_Main is the mid Vod level, which is typically set to 50% of the output voltage swing.
- Vod_Alt2 is the maximum Vod level, which is typically set to 80% of the output voltage swing.

You can program Vod_Main, Vod_Alt1, and Vod_Alt2 from -0.8V to +0.8V. Each receiver also has a programmable termination level, Vt. The programmable range of Vt is from 0V to +1.7V.

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UltraSerial10G Theory of Operation : UltraSerial10G Driver Pin Levels

UltraSerial10G Driver Pin Levels

Pin Levels sheets define voltage and current levels for digital, analog, and power pins and pin groups. The values on the sheet can be expressed as absolute values or as formulas using symbols previously defined on a DC Specs sheet or the Global Specs sheet.

You must specify levels for every digital, analog, and power pin, either individually or as part of a pin group. For valid level settings, see:

- [UltraSerial10G Pin Voltage Levels](#)
- [UltraSerial10G Level Ranges and Swings](#)
- [UltraSerial10G Level Peaking Modes](#)

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UltraSerial10G Theory of Operation : UltraSerial10G Driver Pin Levels : UltraSerial10G Pin Voltage Levels

UltraSerial10G Pin Voltage Levels

The table below describes the UltraSerial10G pin voltage levels.

Vid	Differential drive voltage swing. Half of this voltage is above Vicm, the other half is below. See UltraSerial10G Level Ranges and Swings.
Vicm	Common mode drive voltage. This is at the center point of the two halves of a differential signal.
Vod_Main	Output voltage difference between a pair of signals which comprise a differential pair. For jitter testing, this level is the mid Vod level which is typically set to 50% of the output voltage swing.
Vod_Alt1	For six-point eye and rise time/fall time testing, this level is the minimum Vod level which is typically set to 20% of the output voltage swing.
Vod_Alt2	For six-point eye and rise time/fall time testing, this level is the maximum Vod level which is typically set to 80% of the output voltage swing.
Vod_Oob	The out of band (OOB) Vod detection threshold level.
Vt	The termination input voltage level for the driver
VodTyp	Typical differential output voltage of the device at the test system comparator. Used for calibration.
VocmTyp	Typical common mode output voltage of the device at the test system comparator. Used for calibration.

Peaking

The voltage levels setting used to compensate for skin effect loss on the drive or comparator signal. See [UltraSerial10G Level Peaking Modes](#).

For more information, see [Programming Pin Levels](#).

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UltraSerial10G Theory of Operation : [UltraSerial10G Driver Pin Levels](#) : UltraSerial10G Level Ranges and Swings

UltraSerial10G Level Ranges and Swings

The level ranges and swings for the UltraSerial10G instrument is dependent on the termination voltage at the DUT, as shown in the table below.

UltraSerial10G Level Ranges and Swings		
Termination Voltage at DUT	Vid Range	Maximum Swing
0V	0 to 900mV	900mV
Midpoint of DUT output voltage swing	0 to 1765mV	1215mV - (0.68 * VIL)
1.5V	0 to 1650mV	1200mV
Open	0 to 1800mV	1800mV

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UltraSerial10G Theory of Operation : [UltraSerial10G Driver Pin Levels](#) : [UltraSerial10G Level Peaking Modes](#)

UltraSerial10G Level Peaking Modes

The [voltage level setting](#), Peaking, allows you to compensate for skin effect loss on the drive and comparator signals.

The table below lists driver ranges (Vid) and output amplitude peak-to-peak as a function of termination condition and peaking setting. Note that the Vid programmed range does not change appreciably when using the peaking settings, but the programmed output amplitude (Vid) decreases at higher peaking settings. This amplitude reduction with peaking is equivalent to the cable losses if peaking is not used, but with a better rise/fall time at the DUT.

UltraSerial10G Level Ranges and Swings for Vt Modes

Termination Voltage at DUT	Driver Specification	Peaking None	Peaking Mid	Peaking High
0V	Vid Range	0 to 900mV	0 to 895mV	0 to 890mV
	Output Amplitude (peak- to-peak)	50 to 900mV	50 to 775mV	50 to 650mV
Midpoint of DUT output voltage swing	Vid Range	0 to 1775mV	0 to 1770mV	0 to 1765mV
	Output Amplitude (peak- to-peak)	50 to (1215mV - (0.68 * VIL))	50 to (1130mV - (0.68 * VIL))	50 to (1048mV - (0.68 * VIL))
1.5V	Vid Range	0 to 1650mV	0 to 1645mV	0 to 1640mV

	Output Amplitude (peak-to-peak)	50 to 1200mV	50 to 1030mV	50 to 870mV
Open	Vid Range	0 to 1.8V	N/A	N/A
	Output Amplitude (peak-to-peak)	100 to 1800mV	N/A	N/A

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UltraSerial10G Theory of Operation : UltraSerial10G Timing

UltraSerial10G Timing

Each UltraSerial10G instrument uses the test system’s 100MHz reference to generate its own clock, called the Master Oscillator (MOSC). The MOSC is the time base that runs between 5.35GHz and 10.7GHz in 100Hz increments.

The DUT period is generated from the MOSC and a MOSC divider, called the MOSC per vector (MPV). The MOSC runs between 5.35GHz to 10.7GHz and provides a MOSC period of 93ps to 187ps.

MPV is the number of MOSC cycles to elapse before transitioning to the next vector. The valid values for the MOSC per vector (MPV) are: 1, 2, 4, 8, 16, 32, 64, and 128.

DUT period = MOSC period * MOSC Per Vector (MPV)

Based on the MOSC frequency and MPV values, the minimum frequency is 41.8Mbps (5.35GHz/128), and the maximum frequency is 10.7Gbps (10.7GHz/1).

For more information, see Programming the Serial Timing.

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UltraSerial10G Theory of Operation : DC Coupling

DC Coupling

DC coupling is always enabled on all channels. For the UltraSerial10G driver, DC coupling adds a common mode component (Vicm) to the waveform. See Differential Driver Levels. For the UltraSerial10G receiver, DC coupling adds a DC load to the signal being output by the DUT. Note that the UltraSerial10G does not support AC coupling.

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[UltraSerial10G Theory of Operation](#) : Vector Memory (VM) and Capture Memory (CMEM)

Vector Memory (VM) and Capture Memory (CMEM)

The UltraSerial10G has 512M (426M available) X 1 bit of VM per drive pin and 512M (426M available) x 2 bits of VM per receive pin. The UltraSerial10G driver can drive 0 or 1 data, and the UltraSerial10G receiver can compare L, H, or X (don't care) data. VM has these length requirements:

- [Minimum 2560 vectors](#)
- [Maximum 256M vectors](#)
- [Modulo 2560 vectors](#)

For a pattern set with only one pattern or the last module in a pattern set, you can use modulo 40 vector length.

☒ **Note:**

When masking cycles in a serial receive pattern, the don't care (X) data starts on a modulo 10 cycle boundary and occurs in blocks of 10 cycles.

The UltraSerial10G also has 1G X 1 bit of CMEM per receive pin. You can use CMEM to capture device data. In capture mode, CMEM collects the DUT actual response data and stores 1 bit of data (0 or 1). When running above 8Gbps, you can only capture CMEM data using one lane at a time per timing group.

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UltraSerial10G Theory of Operation : PRBS Generator

PRBS Generator

Pseudo-random binary sequence (PRBS) hardware autogenerates DUT stimulus and response data. The UltraSerial10G PRBS generator supports four sequences: $2^7 - 1$, $2^{15} - 1$, $2^{23} - 1$, and $2^{31} - 1$. All other sequences ($2^9 - 1$ and so on) must be programmed using pattern files (VM).

The drive PRBS generator uses an linear function shift register (LFSR) to determine where to begin in the PRBS algorithm. You must specify this seed value in hex format. The receive side PRBS generators are seeded automatically by the incoming data stream after the Receive Align Trigger (see Receive Align Trigger). The first 32 bits from the data stream are input to the PRBS register, and this initial value is used to generate the expect data sequence.

The length of PRBS patterns can be either infinite or a finite length. Infinite PRBS requires you to manually stop the PatGen when complete. Finite lengths must obey modulo rules for pattern length and minimum pattern size (both are 2560 or 40).

The UltraSerial10G has fail and cycle counters that are used to calculate the bit error rate (BER) value.

For more information on programming the PRBS seed value, see Programming Serial Pattern Sets.

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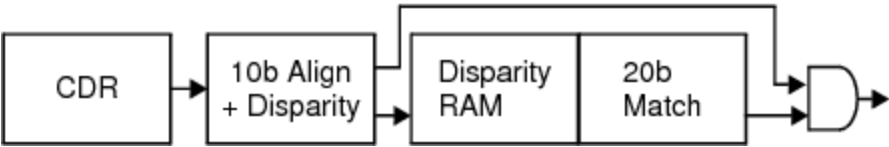
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UltraSerial10G Theory of Operation : Receive Align Trigger

Receive Align Trigger

The Receive Align Trigger (RAT) recovers the embedded clock and performs a series of pattern matches to align to DUT functional data. RAT handles uncertain timing and start of data for nondeterministic devices. The following are the stages of the RAT:

- Clock Data Recovery (CDR)
- 10b Align
- 20b Match



Stages of the Receive Align Trigger (RAT)

The RAT has a timeout capability that allows you to specify how long it should take for the RAT to occur. The RAT times out If it does not assert in the specified amount of time. This timeout should include time for all stages you have set up for the RAT (CDR, 10b, 20b).

Clock Data Recovery

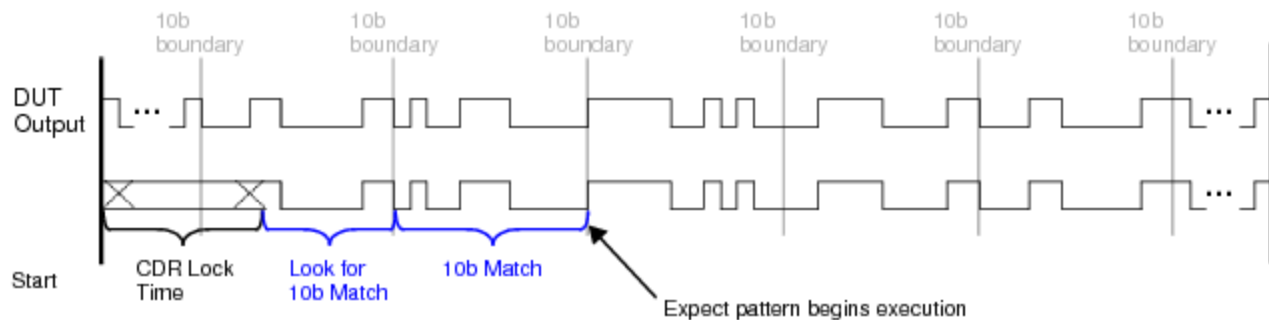
Clock Data Recovery (CDR) recovers the clock that is embedded in the DUT response data, places the strobe at 50% of the anticipated DUT eye, and then tracks changes in the phase offset. If CDR is disabled, you can program the strobe using the Start Delays Receive Delay column on the Serial Timing sheet.

The UltraSerial10G receive channel requires a training interval of 80s (nominal) to allow the CDR to find the center of the data eye. The receive pattern data need not be padded to account for these

training vectors.

10b Align

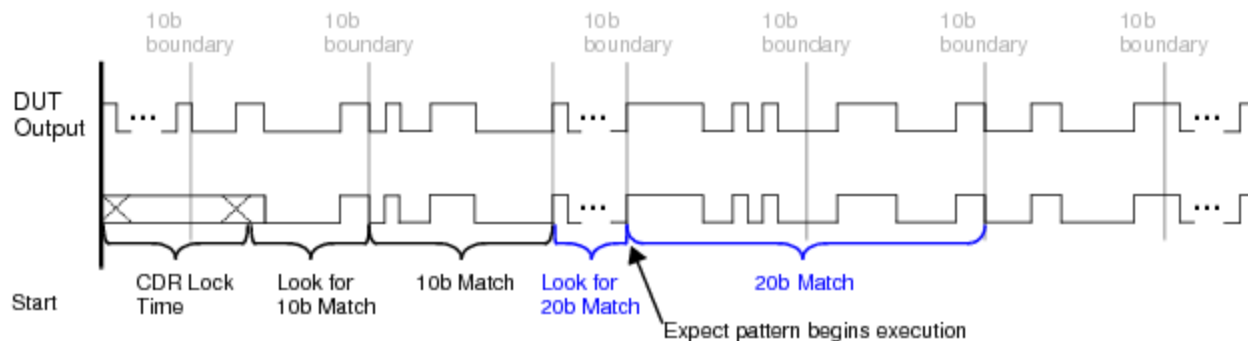
10b align is used to align to the 10b boundary of the device. To gain alignment, you specify on a per pin basis a 10b symbol to match. The 10b symbol must include disparity (positive or negative) and should be chosen such that aliasing does not occur. Aliasing means that the 10b symbol could be matched from a combination of two other symbols. Once the symbol is matched, the RAT asserts and expect pattern execution begins. When the 10b disparity is enabled for the 10b align, the 10b alignment will be true on either positive or negative disparity.



10b Align

10b Align and 20b Match

Once the RAT is aligned to the 10b boundary of the device, 20b match is used to align the beginning of the DUT data. The 20b symbol is specified on a per-pin basis using language or from the first 20 bits of the expect pattern (Vector Memory). When the Disparity RAM is enabled for the 20 bit symbol, both negative and positive disparity are mapped to positive disparity. When the Disparity RAM is disabled, there is a 1-to-1 mapping. The RAT holds off until the first 20 bits of the expect pattern match the data coming out of the DUT. Pattern execution continues after the 20 bits have been matched.



10b Align and 20b Match

Immediate Mode

In addition to the 10b Align mode and 10b Align and 20b Match mode, the RAT also has an immediate mode. When the RAT mode is set to immediate, the CDR, 10 bit match, and 20 bit match is not performed, and the expect pattern is started immediately. You must use the start delay to place the strobe for the DUT expect data.

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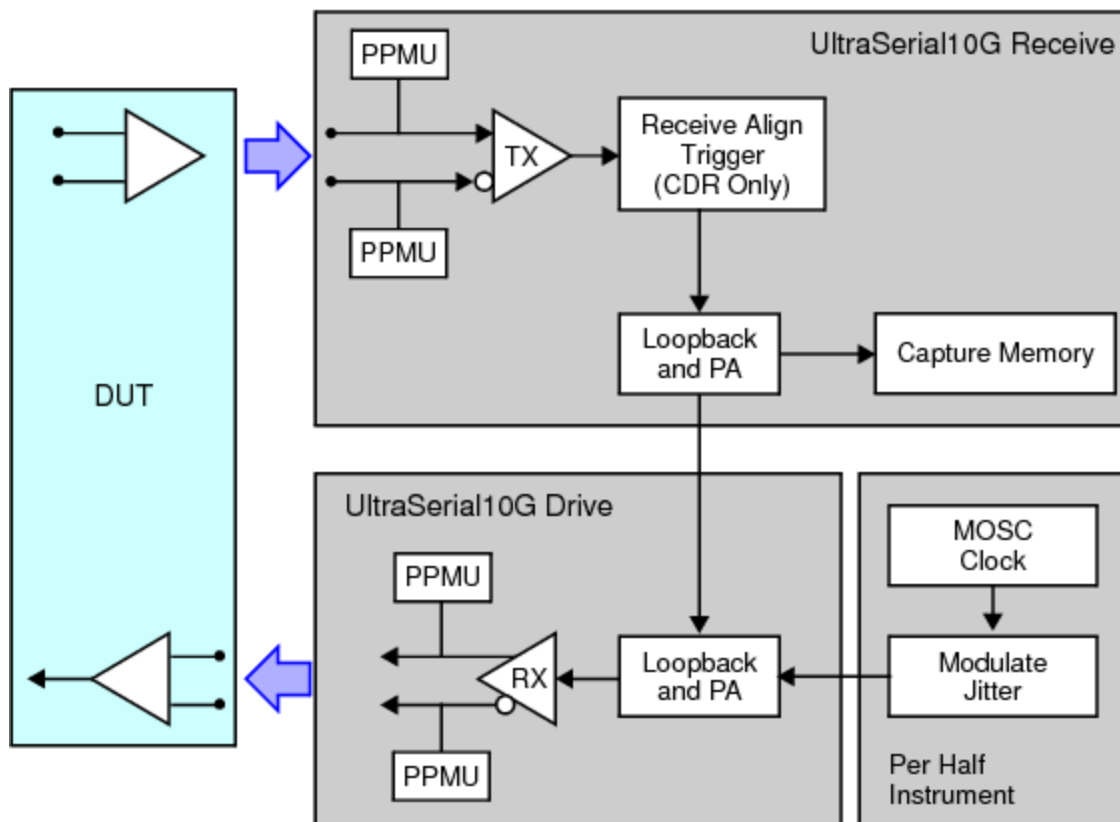
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UltraSerial10G Theory of Operation : Digital Loopback

Digital Loopback

UltraSerial10G has a digital loopback mode that takes the DUT transmitter output (TX) on the UltraSerial10G receiver and loops it back into the UltraSerial10G driver, which sends it into the DUT receiver input (RX). While in this mode, you can still capture the DUT output in CMEM for debugging. You can also enable jitter insertion on the input returning to the DUT receiver (RX). This mode is enabled per differential timing controller, each of which controls two differential drive and two differential receive channels.



UltraSerial10G Digital Loopback

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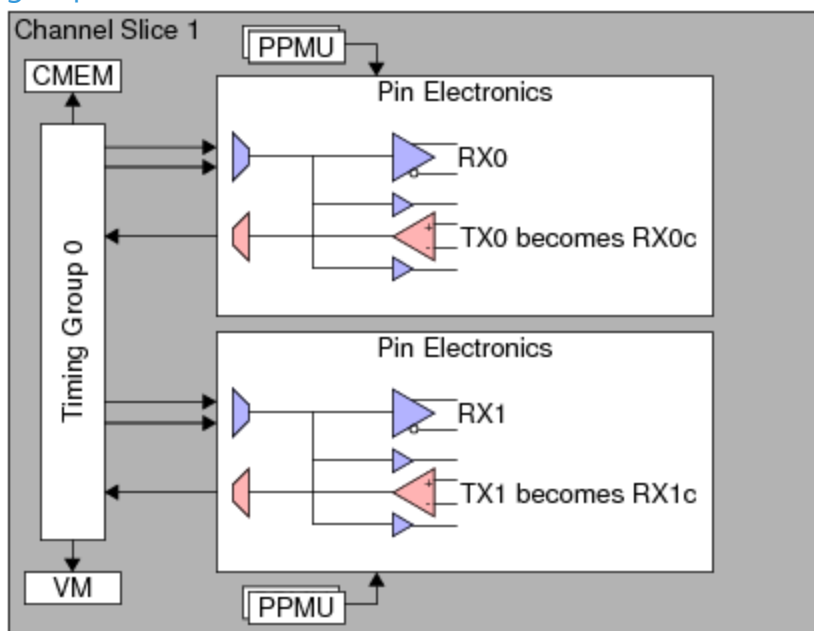
UltraSerial10G Theory of Operation : Drive-only Site Copy

Drive-only Site Copy

The UltraSerial10G site copy feature allows you to reuse all of the receive channels as drive-only channels on a different site. These channels become site copies of the primary drive channels within the same timing group. Typically, this mode is used for HDMI receiver testing, which requires the use of drive channels only.

Site copy is enabled by timing group. So, for instance, you can use site copy on timing group 0, normal serial mode on timing group 1, loopback mode on timing group 2, site copy mode again on timing group 3, and so on.

The figure below shows the how the receive channel becomes a drive-only site copy for one timing group across two sites.



Drive-only Site Copy

For more information, see [Site Copy Guidelines](#) in the UltraSerial10G DIB Design Guidelines and [Site Copy and Receive Channel Reuse](#) in the UltraSerial10G Channel Map Programming.

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UltraSerial10G Theory of Operation : UltraSerial10G PPMU

UltraSerial10G PPMU

Each UltraSerial10G digital channel has a four quadrant pin parametric measurement unit (PPMU), which is used to:

- Force voltage while measuring current (for leakage measurements)
- Force current while measuring voltage (for open/shorts tests)

This section contains the following topics:

- UltraSerial10G PPMU Ranges and VI Characteristics
- Voltage Clamps
- UltraSerial10G PPMU Operation
- Improving Digital PPMU Performance
- Controlling PPMU Voltage Glitches

For more information, see:

- About the PPMU

- Pin Parametric Measurement Unit (PPMU) in the UltraSerial10G Hardware Specification



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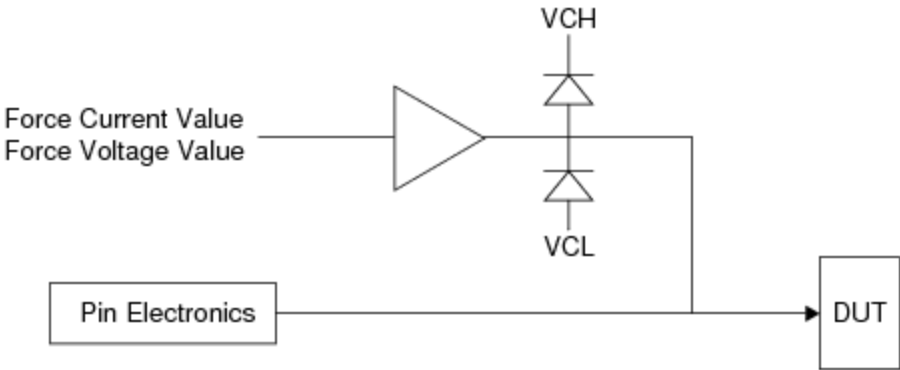
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UltraSerial10G Theory of Operation : [UltraSerial10G PPMU](#) : UltraSerial10G PPMU Ranges and VI Characteristics

UltraSerial10G PPMU Ranges and VI Characteristics

The UltraSerial10G has two modes: Force Current Measure Voltage (FIMV) and Force Voltage Measure Current (FVMI).



UltraSerial10G PPMU

The table below lists the available ranges:

UltraSerial10G PPMU Ranges

Description	Range
Force Voltage	-1.0V to 1.5V (for 2mA and 200A ranges) 0V to 1.5V (for 30mA range¹)
Measure Voltage	-1.0V to 1.5V

Force and Measure Current

- -2mA to +2mA (for 2mA range)
- -200A to +200A (for the 200A range)
- -30mA to +30mA (for 30mA range²)

Force Current Ranges

- 30mA
- 2mA
- 200A

Measure Current Ranges

- 30mA
- 2mA
- 200A

Voltage Clamp Range (VCL/VCH)

-0.8 to 1.5V³

¹ For the 30mA measure current range, the full voltage range is only valid when the load current into the DUT is 0. As the load current increases, this range narrows as described below.

² For the 30mA force and measure current range, the full current range is only valid when the voltage at the DUT is 0. As the voltage at the DUT increases, this range narrows as described below.

³ These voltage clamps are not programmable.

The PPMU voltage and current (V/I) characteristics are summarized in the following figures.

PPMU 30mA Operating Range

For the 30mA force and measure current range, the PPMU uses the UltraSerial10G driver. This driver consists of a VT amplifier and a programmable current source. The VT amplifier can provide a

maximum voltage of 1.5V and supply current until this voltage limit is reached. The following formulas define the operating region shown in the figure below.

- Force Voltage Range (I_{DUT} sinking current) = $-50\text{mV/mA} * I_{DUT}$

- Force Voltage Range (I_{DUT} sourcing current) = $1.5\text{V} - 50\text{mV/mA} * I_{DUT}$

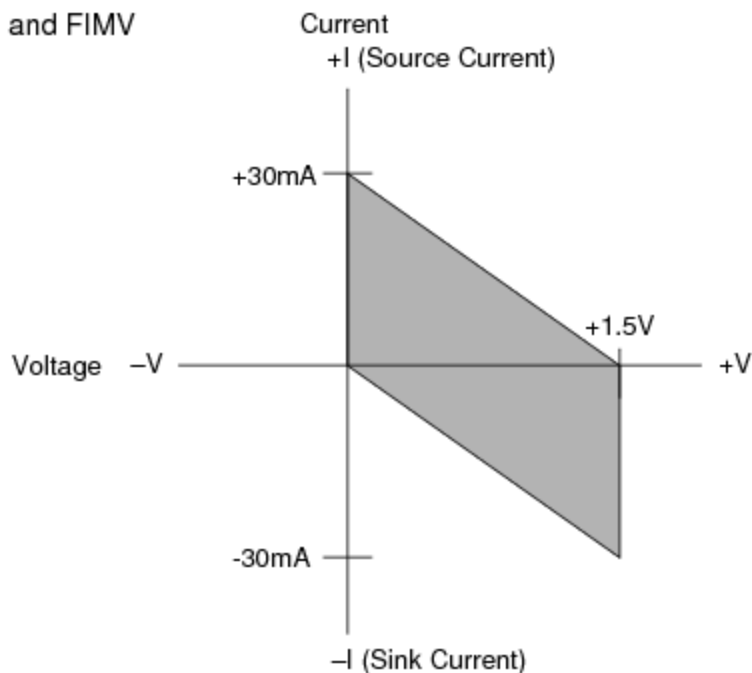
where I_{DUT} is the load current going into or coming from the DUT

- Measure Current Range (I_{DUT} sinking current) = $-20\text{mA/V} * V_{DUT}$

- Measure Current Range (I_{DUT} sourcing current) = $30\text{mA} - 20\text{mA/V} * V_{DUT}$

where V_{DUT} is the voltage at the DUT

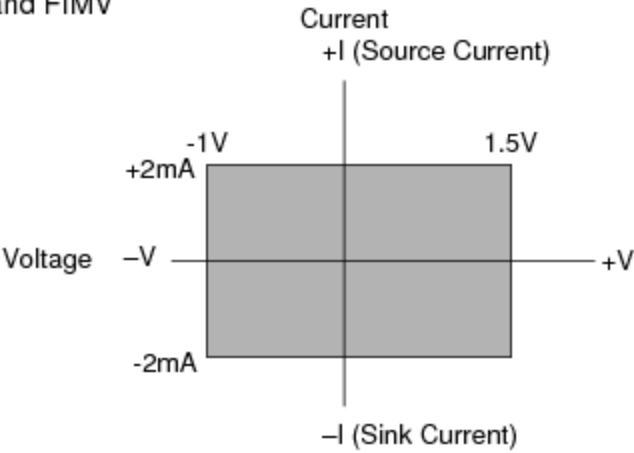
FVMI and FIMV



UltraSerial10G PPMU V/I Curve, 30mA Range (not to scale)

PPMU 2mA Operating Range

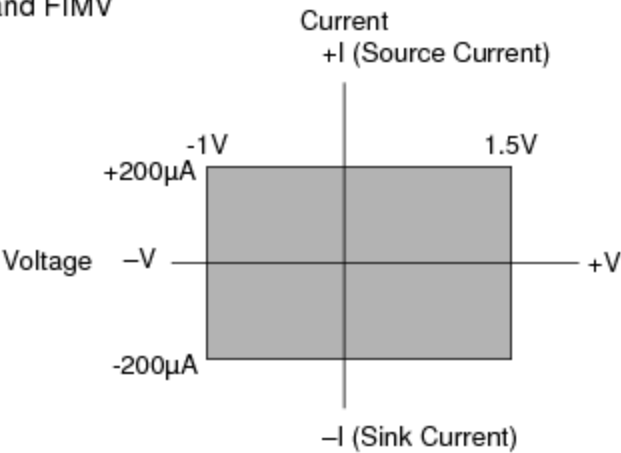
FVMI and FIMV



UltraSerial10G PPMU V/I Curve, 2mA Range (not to scale)

PPMU 200A Operating Range

FVMI and FIMV



UltraSerial10G PPMU V/I Curve, 200A Range (not to scale)

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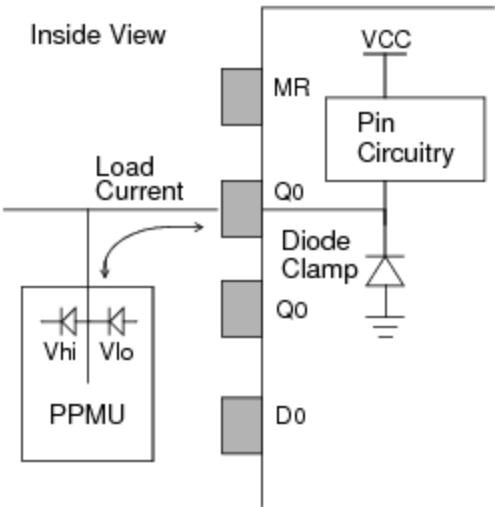
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UltraSerial10G Theory of Operation : UltraSerial10G PPMU : Voltage Clamps

Voltage Clamps

The UltraSerial10G PPMU provides individual voltage clamps on each pin to protect the DUT from high voltage, especially when the PPMU is in force current mode. In this mode the PPMU attempts to force the programmed current and if the output has a high resistance, the PPMU output could reach the maximum voltage limit, which is greater than 1.5V. These clamps are not programmable and are set to keep the voltage in the -0.8V to 1.5V force voltage range.



PPMU Clamps

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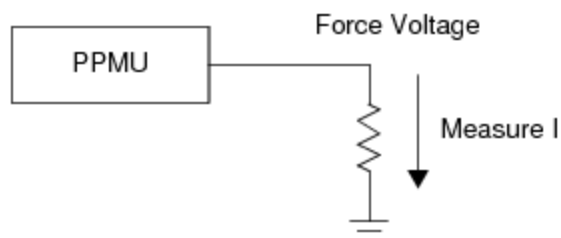
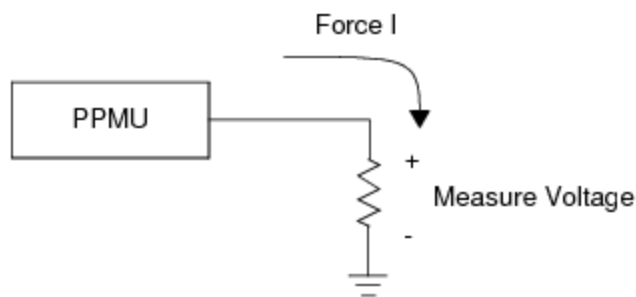
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[UltraSerial10G Theory of Operation](#) : [UltraSerial10G PPMU](#) : [UltraSerial10G PPMU Operation](#)

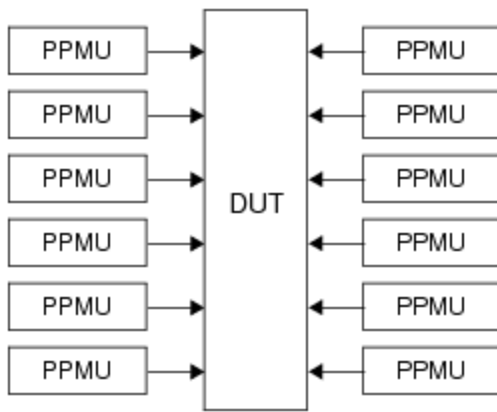
UltraSerial10G PPMU Operation

The Pin Parametric Measurement Unit (PPMU) is a device which measures a current for an applied voltage or a voltage for an applied current.



PPMU Usage

For digital circuitry, the PPMU provides a way of measuring the DC characteristics of a DUT pin. Two common types of testing which the PPMU is used for are leakage and open/shorts testing. For leakage or open and shorts testing, one PPMU is connected to each DUT pin to be tested.



Digital Channel PPMU Connections

To make parametric tests, you normally program all pins to one forcing setup and then switch each pin under test to a different setup just before measuring. For example, to test for open or shorts, you would force all pins to some voltage and then switch each pin to force current and measure the resulting voltage to check for opens or shorts.

Voltage forcing and measuring are done on a single voltage range.

Current forcing and measuring can be done on one of these ranges: 30mA, 2mA, or 200A. The current range also acts as a current limit when forcing voltage.

For leakage testing, each PPMU is set up to measure current while forcing the same voltage. For example, if all pins are high impedance input pins, we would expect all the leakage currents to be very small. If the current on any pin is too large, then the part is bad and the test fails. If the leakage current is too small, then perhaps the pin is not connected, in which case it could be a bad part or it could be that the part was not inserted into the socket correctly.

Open-short testing determines that a pin is connected correctly, not open or shorted to an adjacent pin. This type of test is typically done on a DUT input pin. It measures a voltage on one pin at a time while pulling a current from it. All other DUT pins are held at a default voltage. This test usually measures the input protection diode. By pulling a current, the test forward biases the diode and expects to see a diode drop across it.

- If the diode is bad, it could be shorted to ground.
- If the diode is open, the test might go into compliance.
- If the DUT pin is shorted to an adjacent pin, the voltage will be driven by that pin not the diode drop.

In the first and the last case the part is bad and the test fails. The middle case can be caused by a bad part or by the part not correctly inserted in the test socket. In either case, the part cannot be tested further and you can bin the part for retest.

In the preceding tests, note the two different ways the PPMU is used:

- | | |
|----|---|
| 1. | Set up all PPMUs the same way and measure all pins in parallel. |
| 2. | Set up all PPMUs the same way and measure one pin at a time while changing the setup on that PPMU only. |

In the preceding tests, the test compares the results against set limits. The PPMU can also be used as a general VI source and measurement instrument.

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UltraSerial10G Theory of Operation : UltraSerial10G PPMU : Improving Digital PPMU Performance

Improving Digital PPMU Performance

Each UltraSerial10G channel board contains a Pin Slice Processor (PSP), which is used for level and calibration management. For PPMU measurement however, this processor performs the test for the measured value against the limits in cases where only the result must be logged and not the measured value itself. This improves the performance on PPMU measurements since individual values need not be transferred back to the tester computer. Examples include continuity and leakage tests.

Effect of Datalog Settings on PPMUs

If you disable the datalog and set the number of samples to one, a single pass/fail result is read back, giving you the best test time performance.

If you enable datalog, the individual results and measured values for the pins must be read back, even for the short format. If the number of samples is greater than one, you must use software to perform calculations such as averaging.

If you use the enhanced datalog, you do not get the throughput gain of reading back a single pass/fail measurement.

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UltraSerial10G Theory of Operation : UltraSerial10G PPMU : Controlling PPMU Voltage Glitches

Controlling PPMU Voltage Glitches

All measurements made by the PPMU are assumed to be DC because the bandwidth of the PPMU is limited and the PPMU takes some time to respond to instantaneous or sudden changes. PPMU response time ranges from tens of microseconds to a few milliseconds depending on the PPMU mode, range, and its load. IG-XL allows enough time for the PPMU to respond to these changes. However, you must be aware of transient glitches that can be caused by changing the PPMU range or mode.

Glitches are caused when voltages or currents in the PPMU are switched and because the PPMU has a finite bandwidth. For example, a glitch can occur when the PPMU is in Force Voltage, Measure Current mode (200A) range while drawing 100A, and you suddenly change its range to 2mA. In this case the current sensing resistor must be switched from the old range to the new range, and a glitch of 100A multiplied by R_{sense_new} appears for a short period of time until the PPMU closes the loop and catches up with this change. Glitches can occur in other combinations of PPMU settings. In all cases the magnitude, direction, and duration of these glitches depend on the PPMU mode, range, and loading.

The PPMU has a built-in clamp to help suppress these glitches. It has two clamp levels: VCL (clamp low) and VCH (clamp high), which are set to -0.8V and 1.5V, respectively. These levels should help avoid any damage to a DUT while using the PPMU to make any DUT measurements. Another way to suppress glitches is to gate the instrument off when switching the mode or range.

Note that even when using these methods, you may not be able to suppress all PPMU voltage glitches. In most cases, however, these transient glitches should not affect your PPMU measurements.

	
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UltraSerial10G Theory of Operation : UltraSerial10G Scan Testing

UltraSerial10G Scan Testing

Scan testing on UltraSerial10G is done using standard UltraSerial10G drive and receive patterns (VM). The UltraSerial10G only operates as a differential pin pair and cannot generate single-ended scan signals. Scan tests must operate differentially.

The UltraSerial10G scan data must be aligned to the DUT source clock to meet the device scan timing requirements. You disable CDR circuitry for scan testing and use the UltraSerial10G drive and receive offsets to position the drive and strobe edges.

The minimum data rate for scan testing on the UltraSerial10G is 42Mbps.

Note that you may need to expand vectors to create a slower running data rate. For example, to run the scan pattern at 20Mbps, you must replicate each vector by 3x and program the instrument to 60Mbps, which makes the effective data rate 20Mbps.

✔ Note:

When masking cycles in a scan pattern, the don't care (X) data starts on a modulo 10 cycle boundary and occurs in blocks of 10 cycles. For low-speed scan patterns that do not have a 10-bit boundary and require masking control on a per-bit basis, you can either mask more bits than desired with the possibility of losing some coverage or expand the pattern by 10X which allows you to mask on a per-bit basis.

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UltraSerial10G Theory of Operation : UltraSerial10G Jitter Insertion and Measurement

UltraSerial10G Jitter Insertion and Measurement

The UltraSerial10G can insert jitter on UltraSerial10G drive pins. You can enable and disable jitter insertion on a per channel basis. The jitter insertion waveform is a sine wave. You can program the period and amplitude of the jitter. The maximum jitter you can insert is a function of data rate and jitter frequency. You can control the amplitude and frequency on a per half board basis.

For more information, see [Timing](#) in the UltraSerial10G specification.

The UltraSerial10G can measure jitter and perform a six-point eye measurement. Jitter measurement returns random jitter (Rj) and data dependent jitter (DDj) at a 50% threshold level. The six-point eye measurement returns the eye opening of the DUT output at the 20, 50, and 80% threshold levels. It also returns rise and fall time results at the 20% and 80% threshold levels.

For more information, see:

- [Programming UltraSerial10G Serial Jitter](#)
- [UltraSerial10G Jitter and Eye Measurement Test Programming](#)

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UltraSerial10GDefTopic

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About the UltraSerial10G

About the UltraSerial10G

The UltraSerial10G (US10G) is the next generation of the high speed serial bus test instruments for the UltraFLEX test system. The earlier SB6G instrument tests serial buses up to 6.4Gbps by replicating the high-speed device operation environment. The UltraSerial10G has most of the earlier SB6G features, and it now supports new features including:

- Data rate up to 10.7Gbps using PRBS and Loopback and up to 8.0Gbps using VM
- Digital loopback
- Protocol Aware (PA) functionality to test PCIe Generation 1 and 2 protocols
- PA functionality to test sWire protocols such as SATA (Generations 1, 2, and 3)

Depending on the test system configuration, each UltraFLEX test system can have the following number of UltraSerial10G boards and ports:

UltraSerial10G Board and Channel Counts			
UltraFLEX Configuration	Total Available Slots	Maximum UltraSerial10G Boards	Maximum UltraSerial10G Ports
UltraFLEX-HD	12	10	200
UltraFLEX-SC (Lower Power PDU)	24	12	240

UltraFLEX-SC (Higher Power PDU) ¹	24	14	280
UltraFLEX-HC ²	36	24	480

- ¹ This PDU was manufactured after January 2010.
- ² UltraSerial10G instrument boards can be placed only in slots 0-24 of the 36-slot test system.

This section includes the following topics:

- [Features](#)
- [Safety Information](#)
- [UltraSerial10G Software Licensing Requirements](#)
- [Key Terms](#)

UltraSerial10G information includes the following sections:

- [UltraSerial10G Theory of Operation](#)
- [UltraSerial10G Programming](#)
- [UltraSerial10G Debug Displays](#)
- [UltraSerial10G Examples](#)

Related Topics

- [UltraSerial10G Specifications](#)
- [About Protocol Aware](#)
- [Protocol](#) (VBT syntax)

- [Serial](#) (VBT syntax)
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- [UltraSerial10G DIB Design Guidelines](#)

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